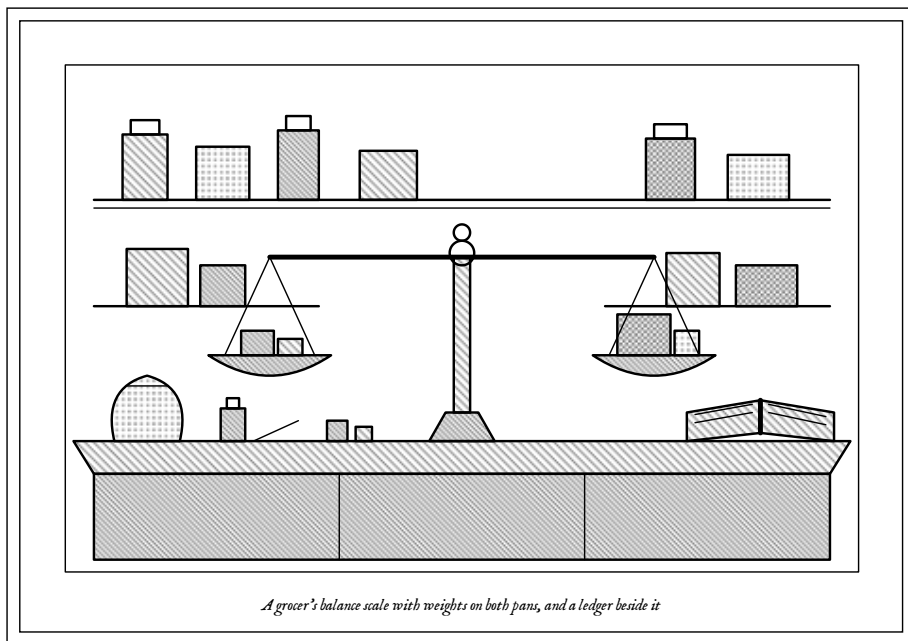




CHAPTER III

The Accounting Equation

Why must what a business holds always match the claims on it?



This chapter teaches assets, liabilities and owner's equity, the three building blocks of every business record. It shows why they always balance, adds revenue, expenses and drawings, and tests the idea on seven January events in a small binding workshop.

THE ELEMENTS OF ACCOUNTANCY



Being a Plain Account of Recording, Classifying, Summarising and Reporting

A. JAIRA, N. TIMI

one-person shop, because accounting treats the business as separate whatever the law says.

A fourth confusion is to think that historical cost shows what an item is worth today. It does not. It shows the price paid, which can be proved. Present value may differ, and that is why the rule is a choice of reliability over currency.

Historical cost is not always the best measure, but it is the most reliable.

IN HISTORY

By the early 1970s, businesses in many countries sold goods and borrowed money across borders, yet each country kept its own accounting rules. A bank in one country could not easily compare a borrower's reports with those of a borrower in another. On 29 June 1973, professional accounting bodies from several countries formed the International Accounting Standards Committee in London. The idea came from Sir Henry Benson, a leading British accountant and former president of the Institute of Chartered Accountants in England and Wales. The bodies came from Australia, Canada, France, Germany, Japan, Mexico, the Netherlands, the United Kingdom and Ireland, and the United States.

The committee of 1973 first tried to give the world one set of accounting rules.

Its members worked part-time as standard setters while holding other jobs. They aimed to write standards in the public interest and to win acceptance for them across the world. The committee worked until 2000, and on 1 April 2001 a new body, the International Accounting Standards Board, replaced it and continued its work. The story shows what this chapter has argued: accounting is trustworthy when many people agree on the rules first.

The committee's standards were later taken over by a board.

REVIEW QUESTIONS

- 2.1 Match each reader of Meghna and Sons to the question it asks: Nurul, the bank, the publishers, the tax authority and the staff.
- 2.2 Suppose the bank asks the shop for its records, but only the weekly book-sales report is shown. Which branch of accounting does the weekly report belong to, and why is it not enough for the bank?
- 2.3 A village tailor, who sells stitching, keeps her family's rent and food costs in her business book. Which rule is broken, and who is misled?
- 2.4 A shop is told it will close in three months. Explain what happens to the going concern assumption and to the way its shelves are recorded under historical cost.

Suppose the shop buys a display shelf. A year later, a neighbour says the shelf would sell for more than the shop paid. Which number should the records show? The rule of **historical cost** is the rule of recording an asset at the price paid for it, not at its present value. The word *asset* means something of value that the business owns, such as a shelf or a book; Chapter 3, 'The Accounting Equation', studies assets properly.

The reason for the rule is trust. The price paid is a fact that an invoice or a receipt can prove. A present value is an opinion, and two honest people may hold different opinions. A bank would prefer the fact. There is an exception: some accounting standards allow certain items to be shown at present value, and later chapters show how the cost of an item is lowered over time. This book first teaches the plain rule of historical cost, because it is the base on which the exceptions stand.

Before moving on. Business entity keeps the owner's money apart from the shop's. Going concern assumes the shop will continue. Historical cost records items at the price paid. You should now be able to say: each idea protects the reader from a misleading number.

The reports that result from these rules are called financial statements. A **financial statement** is a formal report of a business's results or position, prepared at the end of a period. A period may be a month, a quarter or a year. The yearly reports of Meghna and Sons for 2025 are financial statements. They carry the profit of Tk 750,000 that Nurul, the bank and the tax authority read.

Three statements are most common: the income statement, which shows profit; the statement of owner's equity, which shows what the owner's share has become; and the balance sheet, which lists what the business owns and owes on one day.² Chapter 9, 'Financial Statements and Closing Entries', teaches them. Notice how the whole process now closes into a circle. Events are recorded, recording follows the standards and the three ideas, and the statements carry the result to the readers.

IN SHORT

Accounting records, sorts and reports money events. Standards, the business entity rule, going concern and historical cost make the reports trustworthy for their readers.

Before moving on. Accounting standards make businesses report alike. Three ideas decide what is recorded: separate entity, continued operation and price paid. Financial statements deliver the result. You should now be able to say: readers can trust accounting reports because rules, not moods, decide what the reports contain.

* * *

WORKED CASE: MATCHING USERS TO

² Under the International Financial Reporting Standards, the balance sheet is called the statement of financial position. This book uses the name balance sheet.

The third basic idea records items at the price actually paid.

Historical cost. The rule of recording an asset at the price paid for it, not at its present value.

Historical cost is reliable because the price paid can be proved.

A short review of the three basic ideas.

Financial Statements are the formal reports at the end of the process.

Financial statement. A formal report of a business's results or position, prepared at the end of a period.

The main Statements are previewed here and taught in a later chapter.

A short review of the section.

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First edition

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WHAT EACH READER OF THE SHOP'S RECORDS ASKS

READER	GROUP	QUESTION ASKED	FIGURE FROM THE CASE
Nurul	Internal	Did the shop earn enough profit?	Profit Tk 750,000
The bank	External	Can the loan be repaid?	Loan Tk 600,000
The publishers	External	Will the shop pay for the books?	Owed Tk 1,150,000
The tax authority	External	Is the profit honestly reported?	Profit return
Staff	External (see text)	Are wages and jobs safe?	Wages Tk 2,940,000

Now the puzzle from the start of the chapter can be partly answered. The five questions differ, but the facts behind them are the same: what the shop earned, what it owes, what it holds. One set of records can serve all five readers when every reader receives reports built from the same events and the same rules. Section 2.3 explains the rules. Notice also the viewpoint of the owner, who is both a reader and a person who must answer to the others: Nurul needs honest reports precisely because the bank, the publishers and the tax authority will test them.

Remember that Meghna and Sons is a merchandising business: it buys books and sells them again. A service business, such as a tailor who sells stitching, and a manufacturing business, such as a printing works that makes books, have the same kinds of reader. The figures differ, but the questions are alike. A lender always asks about repayment, a supplier about payment, and a tax authority about honest profit.

Before moving on. Internal users run the business and use information to plan and control. External users stand outside it and ask about repayment, payment, honest profit, return on money invested, or security of work. One set of records serves them all when the reports share the same facts and rules.

You should now be able to say: an internal user helps to run the business, an external user does not, and each external user asks a different question of the same records.

* * *

2.3 RULES AND CONCEPTS

Suppose two bookshops each report a profit. The first records its books at the price it paid for them. The second records them at the price it hopes to receive one day. The two profits cannot be compared, and a bank reading both reports would not know which to believe. To prevent this, accountants agree on rules before the records are made. This section presents the rules that give accounting its trustworthiness: written standards, then three basic ideas about what is being recorded, then the reports that result.

One set of records can answer many questions because the questions share the same facts.

The same five kinds of reader exist whatever kind of business is studied.

Recall: merchandising business, service business, manufacturing business, Chapter 1, section 1.1.

A short review of the readers and their questions.

A sentence the reader can test against the section.

Reports can be trusted only if every business follows the same rules.

You may know the word *account* from a bank, where it means a record of the money you hold there. Accountants use the word in a related but different way. In their sense, an account is a named record of one kind of item, such as Cash or Sales. Chapter 4, 'Accounts, Debits and Credits', teaches accounts in detail. For now, remember only that accounting is built from such named records.

The figure that follows shows the whole process on one page. Read the top row from left to right. Each box is a stage, and each arrow shows the direction in which information moves. The first box holds the events. The second holds the records made from them. The third holds the reports built from the records. The lines that leave the third box lead to the readers in the bottom row, who are the subject of the next section.

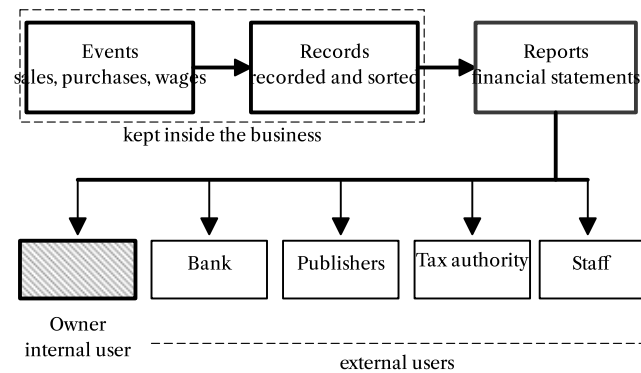


FIG. 2.1. Accounting as a flow from events to reports to users

People often use the words accounting and bookkeeping as if they meant the same thing, but they do not. **Bookkeeping** is the recording part of accounting: writing down each transaction in the right book. A transaction is one event that can be measured in money, such as the sale of one novel. The bookkeeper records. The accountant also sorts, checks and reports, and decides how a doubtful item should be treated. An everyday bridge helps here. The person who writes each shopping payment in the family notebook is keeping books. The person who studies the notebook to plan next month's spending is doing accounting.

Accounting divides into branches, and the readers decide the division. Think first of Nurul, who wants a weekly report to decide which books to order. Think next of the bank, which wants a formal report once a year. These two readers need different reports, prepared in different ways. That difference gives us the two main branches of accounting.

Financial accounting is accounting that prepares reports for outsiders, such as lenders, owners and tax authorities. An owner who takes no part in managing the business is also an outsider. The bank's yearly report and the tax authority's profit return come from financial accounting. Its reports follow written rules, which

The word account has an everyday meaning and a special accounting meaning.

Account. Word in use. In daily life, a record of money in a bank. In accounting, a named record of one kind of item, such as Cash or Sales.

A flow chart shows events becoming records, then reports, then decisions by readers.

Bookkeeping is the recording stage; accounting is the whole process.

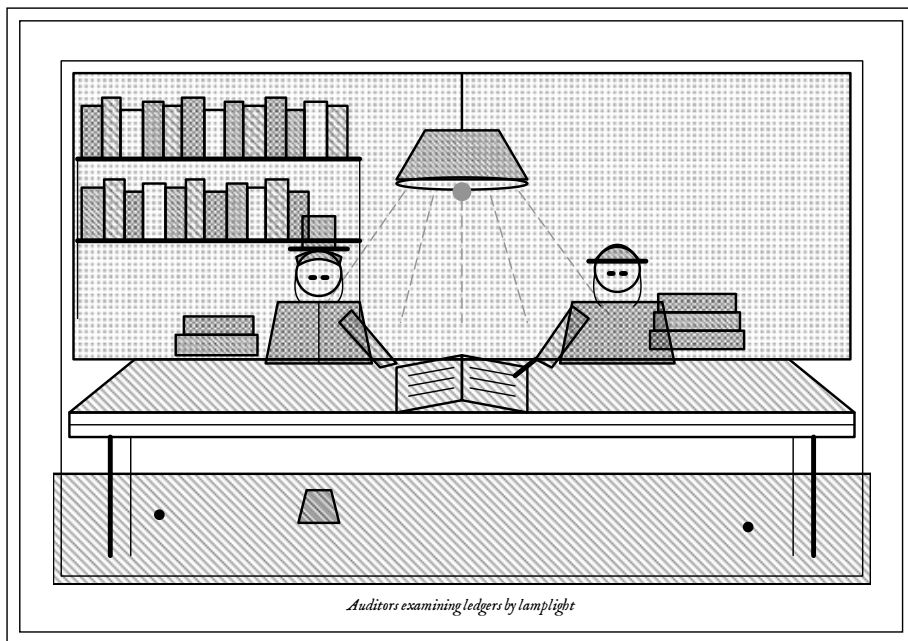
Bookkeeping. The recording part of accounting: writing down each transaction in the right book.

Accounting divides into branches according to the readers it serves.

Financial accounting serves outsiders, and its reports follow written rules.

Financial accounting. Accounting that prepares reports for outsiders, such as

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CHAPTER II

What Accounting Is

Who needs to know how a shop is doing, and why should they believe the answer?



Accounting turns the daily events of a business into reports that other people can read and trust. This chapter explains what accounting is, which people use its reports, and which basic rules keep its numbers fair. It is the map for the rest of the book.

records. The later political history of the Company, which includes the governing of large parts of South Asia, belongs to history books and is not told here.

REVIEW QUESTIONS

- 1.1 A tailor stitches clothes from cloth that customers bring. A grocer buys rice and sells it again. A furniture workshop builds chairs from timber and sells them. State the kind of business each one is.
- 1.2 List four people or groups who would want to see the records of a bakery, and say what each wants to learn.
- 1.3 In the proposal for Meghna and Sons Limited, there are 410,000 shares of Tk 10 each. Calculate their total face price, and say how many shares would be one tenth of the company.
- 1.4 In one sentence each, explain what a shareholder is and how a merchandising business differs from a manufacturing business.
- 1.5 Meghna Binding Service began on 1 January 2026 as a separate set of books from the shop. Name its kind of business, and give one reason why the owner may want the shop and the workshop recorded apart.

YOU WILL NEED

Nothing from earlier pages, because this is the first chapter. You need only your own experience of buying, selling and saving money.

A shop can be busy all day and still leave its owner unsure of the result.

The cash in the drawer does not say whether the day was profitable.

This chapter asks what businesses do, who owns them, and why they keep records.

The book marks its key words, notes and questions in the same way every time.

Picture a bookshop on a busy street in Dhaka. From morning until evening, customers enter, choose a book, pay, and leave. At closing time the owner empties the cash drawer onto the desk and counts the notes. The pile looks large. Was it a good day? The owner hesitates, because the cash alone cannot answer the question.

Consider what the owner would need to know. Some of the cash may be payment for a book that has not yet been delivered. Some books on the shelves may have been bought from publishers who have not yet been paid. The rent for the month must still be found from somewhere. Even part of the pile may be borrowed from a relative last week. A drawer of cash is a fact. A good day is a judgement, and a judgement needs more facts.

This chapter begins to teach you how such judgements are made. First we ask what kinds of work businesses perform, because a bookshop, a tailor and a bakery have different things to count. Next we ask who owns a business, because one owner and a thousand owners need different things from the same records. Last we ask why records are kept at all. The answer gives you the reason for the rest of the book.

A word about the way this book is built. A new key word appears in bold, and its meaning follows at once. A short note in the side column repeats that meaning. Each section ends with a paragraph that begins *Before moving on*. Where an idea is new and hard, a flag tells you to go slowly, and a *Pause and check* question lets you test yourself before you continue. Many students find the first chapters slow, because every word is new. That is normal, and the pace is meant to help you.

* * *

1.1 WHAT BUSINESSES DO

Begin with something you may have seen yourself. A woman sells tea from a small stall beside a bus stop. She buys leaves and milk, boils them, pours the cups, and takes payment. Each action belongs to one pattern: she gives people something they want, and she receives money in return. We call that pattern business activity, the work an organisation performs to supply goods or services to customers for money. The word *business* has two meanings, so notice which one this book uses.¹

Look closer at the tea seller, because her day shows three parts that every business activity contains. First, she obtains what she

¹ The word *business* is also used for any private matter or occupation. This book uses it only for an organisation that supplies goods or services to earn a profit.

GO SLOWLY HERE.

A business supplies something people want and receives money in return.

Business. Word in use. In daily life, it can mean any occupation or concern. In accounting, it is an organisation that supplies goods or services to earn a profit.

Every business activity has three parts: obtain, work and deliver.

reason the later chapters teach not only how to write records but how to check them.

Before moving on. A record is a written note of an event in the business, kept so that facts do not depend on memory. Owners, lenders, tax officers and shareholders each make decisions that need those records. You should now be able to say why a busy owner cannot rely on memory, and name two people who need the records.

* * *

WORKED CASE: CLASSIFYING THE MEGHNA BUSINESSES

Case file, illustrative. Meghna and Sons is a family-run bookshop in Dhanmondi, Dhaka, owned by Nurul Meghna and trading since 4 April 1996. In 2025 it sold books worth Tk 14,400,000, which later chapters will call its net sales (Chapter 10, 'Accounts of a Merchandising Company'). The shop buys books and sells them. On 1 January 2026 a separate workshop, Meghna Binding Service, began to repair and bind books for schools, with its own set of books. A proposal exists to form a company in 2026 with 410,000 shares of Tk 10 each, a total of Tk 4,100,000.

Question. What kind of business is each of the two Meghna businesses? Who owns the shop today? If the company is formed, what is the total value of the shares at their face price of Tk 10? Use only the facts in the case file.

Reasoning. Begin with the shop. The case file says it buys books and sells them. It does not change the books, and it is not paid for work. This matches the definition of a merchandising business. Next, the workshop repairs and binds books for schools. The schools pay for work, and the workshop does not resell goods it bought. This matches the definition of a service business.

Now consider ownership. The case file names one owner, Nurul Meghna, so the shop today has the shape of a sole proprietor. The proposal would change the shape, because a company has shares and shareholders. To find the total value, multiply the number of shares by the price of one share: 410,000 shares × Tk 10 = Tk 4,100,000. A buyer of any of these shares would become a shareholder.

Conclusion. The shop is a merchandising business owned by one person, Nurul Meghna. The workshop is a service business. If the company is formed, its 410,000 shares have a total face price of Tk 4,100,000.

Check. Multiply again from the other direction: Tk 4,100,000 ÷ Tk 10 = 410,000 shares, which agrees with the proposal. Test each kind against its definition: goods bought and resold give merchandising, and work done for payment gives service. Notice that the sales figure of Tk 14,400,000 was not needed to decide the kind, because the kind depends on what the business does, not on how much it earns.

The recap gathers the section into three sentences.

The case file gives the facts about three Meghna businesses and one proposal.

The question asks for the kind, the shape of ownership and the sum.

The shop buys finished books and resells them, so it is a merchandising business.

One named owner makes the shop a sole proprietor today; the proposal would change that.

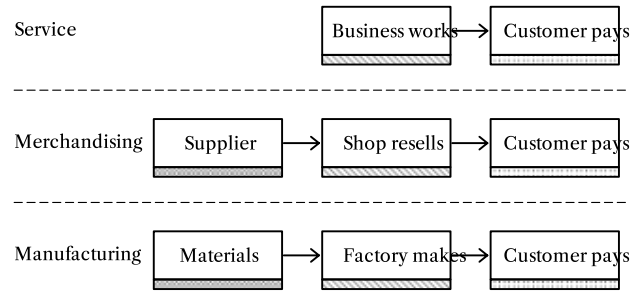
The conclusion gives the three answers in plain sentences.

The check repeats the arithmetic and the definitions.

THE THREE KINDS OF BUSINESS

KIND	THE CUSTOMER RECEIVES	THE BUSINESS MUST COUNT	EXAMPLE
Service	Work	Fees earned and the cost of doing the work	Meghna Binding Service
Merchandising	Goods bought from others	Goods bought, goods sold, and the price gap	Meghna and Sons, a bookshop
Manufacturing	Goods the business made	Materials, goods half finished, goods finished	A bakery that bakes its own bread

The figure shows each kind as a chain. Read each row from left to right. A box names who holds the work or the goods, and an arrow shows the direction in which they travel toward the customer. A service chain is short, a merchandising chain has a supplier before the shop, and a manufacturing chain has the factory in the middle.



Strip under each box: lines = the business, dots = the customer, crosses = the

FIG. 1.1. Three kinds of business drawn as chains from source to customer

Why does the kind matter? Because it tells the owner where to look. In a service business, the owner watches the fees earned and the cost of the workers' time. In a merchandising business, the owner watches how many goods were bought, how many were sold and what remains on the shelves. In a manufacturing business, the owner watches all of these and also the materials and goods still being made. More things to count means more records.

Real businesses often combine kinds. A bakery that also sells tea bought from a wholesaler is manufacturing and merchandising together. A shop that repairs what it sells is merchandising and service together. The three groups are tools for thinking, not boxes into which every firm must fit. When a firm mixes kinds, we ask which kind brings in most of its money, and we describe the other kinds as additions.

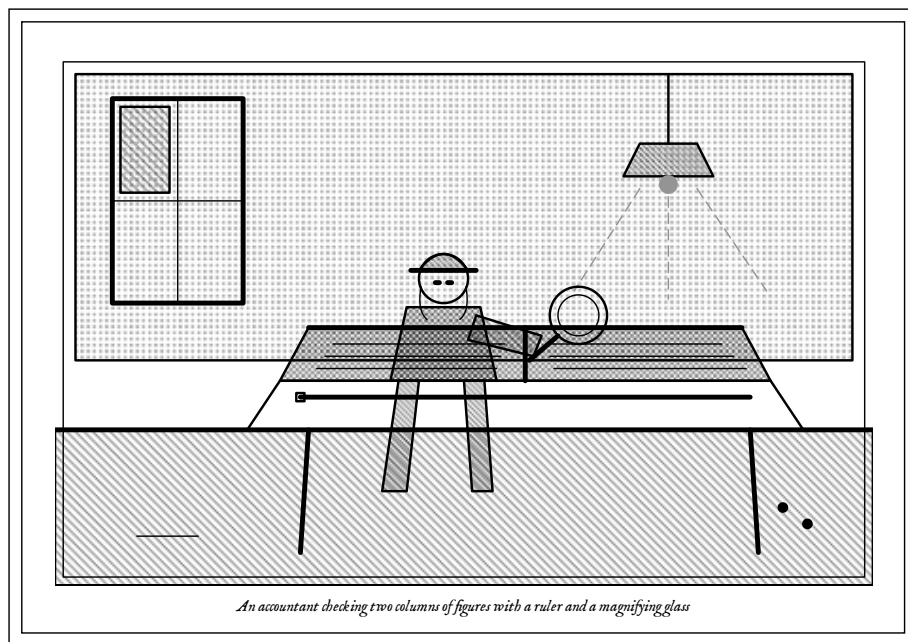
Before moving on. Business activity is work done to supply goods or services for money, and profit is what remains after costs. A ser-

The figure shows each kind as a chain from the source to the customer.

The kind of business decides which facts its owner must watch.

Many real businesses combine two kinds, and that is allowed.

The recap gathers the section into three sentences.



CHAPTER VI

The Trial Balance

If every fact is written twice, can the books be tested against themselves?



Every entry in the books has equal debits and credits, so the whole ledger must carry the same equality. This chapter shows how to list the ledger balances, test the equality, and find and fix a difference, using the January 2026 books of Meghna Binding Service.

YOU WILL NEED

Chapter 2, 'What Accounting Is', section 2.1 'The Idea of Accounting', for the meaning of accounting and bookkeeping; Chapter 2, section 2.3 'Rules and Concepts', for historical cost; and Appendix A1, 'Maths You Will Use', for moving a term across an equals sign.

A market stall holds Tk 100,000 of value, and someone must have a claim on all of it.

The uncle's claim and Rahim's claim together equal the whole value of the stall.

This chapter explains why what a business holds always equals the claims on it.

Picture a market stall on a busy street in Dhaka. Its owner, Rahim, counts what the stall holds: a folding table, a rack of umbrellas for sale, and cash in a tin. A careful friend values everything together at Tk 100,000. "Whose is all this?" the friend asks. Rahim answers: "Mostly mine. But my uncle lent me Tk 30,000 to buy the umbrellas, and he expects it back."

Now the friend has an answer. The uncle has a claim of Tk 30,000. The rest, Tk 70,000, belongs to Rahim. Add the two claims and the total is Tk 100,000 again. Try to find one taka of value in the stall that nobody has a claim on. You cannot. Every taka entered the stall in one of two ways: someone lent it, or Rahim put it in or earned it.

This small discovery is the subject of the whole chapter. A business holds things of value. The people who lent money, and the owner, hold claims on those things. The two totals must agree, in every business, in every country, on every day. Is that a law of nature, or only a habit of bookkeepers? By the end of the chapter you will see that it is a rule of arithmetic with a plain reason, and you will watch seven events in a small workshop obey it, one after another.

* * *

3.1 ASSETS, LIABILITIES AND EQUITY

Bookkeeping, as Chapter 2 explained, means writing down what happens to the money of a business.¹ But which events deserve a place in the records? A business meets many events in a day: a telephone call, a promise, a visit from a customer. Only some of them change its money. A **transaction** is an event that has a money effect on a business and must be recorded. Buying shelving for Tk 120,000 is a transaction. A customer who says she may buy a book next month has not yet created one.

Case file, illustrative. Meghna Binding Service is a book-binding workshop that repairs and binds books for schools. It began on 1 January 2026 and keeps a separate set of books from the bookshop Meghna and Sons of Dhanmondi, Dhaka. This chapter uses the workshop's records for January 2026. Amounts are in taka (Tk), with international digit grouping.²

¹ Recall: accounting and bookkeeping, Chapter 2, 'What Accounting Is', section 2.1 'The Idea of Accounting'.

² In Bangladesh you may also see lakh and crore. Tk 500,000 is five lakh. This book writes every amount with commas after each three digits.

A third slip is to place a debit on the credit side of the account because the other account's name was written there. The Particulars column names the other account; the amount goes on the side of the account whose page you are on.

A fourth confusion treats the balance as the sum of both sides. It is the difference between them, placed on the larger side.

IN HISTORY

Francesco Datini was a merchant of Prato, a town near Florence in Tuscany, Italy. He was born around 1335 and died on 16 August 1410. He traded in cloth and many other goods, and kept offices in several countries. After his death, his papers stayed in his house in Prato. In 1870, workers altering the house found them stored in sacks in a disused stairwell.

The find included some 150,000 letters and more than 500 account books and ledgers. Historians still use them to study how merchants of the fourteenth century recorded debts, payments and goods. The survival of so many books, six centuries later, shows the aim of this chapter: a transaction written once in an orderly book can be found again by someone who never met the writer.

REVIEW QUESTIONS

- 5.1 A workshop owner invests Tk 100,000 in cash. A week later the workshop pays Tk 20,000 in cash for shelving. Journalise both transactions, with narrations.
- 5.2 Post the two entries of question 5.1 to the Cash account and balance it.
- 5.3 A Cash account has debits of Tk 125,500 and Tk 48,250 and credits of Tk 30,750 and Tk 61,400. Find its balance and say on which side it falls.
- 5.4 On 1 February 2026 the Cash account of Meghna Binding Service starts with the balance of Tk 245,000 brought down. During February it receives Tk 52,000 and pays out Tk 118,500. Find the balance at 28 February 2026.
- 5.5 A different workshop buys a press for Tk 30,000, paying Tk 10,000 in cash and owing the rest to the dealer. Journalise the compound entry, post it to the three accounts, and state the balance of each.

A debit goes to the debit side of its own account, not to the other account's side.

A balance is a difference, not a total.

A medieval merchant's surviving books show that recorded transactions endure.

The Datini archive shows how long a well-kept record can last.

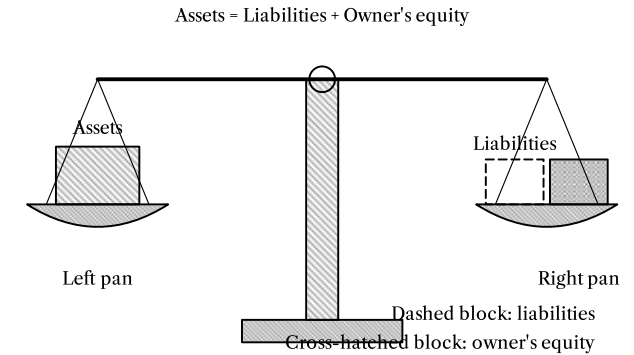


FIG. 3.1. An equation as a balanced scale: assets on one pan, liabilities and owner's equity on the other

Read the scale as an equals sign: the two pans carry the same total.

The equation can be rearranged to find any one of its three parts from the other two.

Worked example: owner's equity from assets and liabilities, with small numbers.

The equation measures recorded amounts, and equity can fall below zero.

Each person reads the same equation with a different question in mind.

The picture shows the idea a second time. Read it from left to right. The left pan carries the assets. The right pan carries two blocks: the liabilities and the owner's equity. The beam is level because the totals are equal. If a block is added to one pan, a block of the same weight must be added to the other, or the scale tips.

Often you know two parts and need the third. Rearranging the equation gives $E = A - L$, and also $L = A - E$ (Appendix A1, section A1.3 'Rearranging an Equation', shows the move). Here is the first calculation, set out in the order used throughout this book.

Given. A business has assets of Tk 50,000 and liabilities of Tk 20,000. *Find.* The owner's equity. *Formula.* In words: owner's equity equals assets minus liabilities. In symbols: $E = A - L$. *Steps.* $E = 50,000 - 20,000 = 30,000$. *Answer.* The owner's equity is Tk 30,000. *Check.* Liabilities plus equity should equal assets: $20,000 + 30,000 = 50,000$. It does.

Two limits are worth knowing. First, the equation uses the prices written in the records. Chapter 2, section 2.3 'Rules and Concepts', taught historical cost: an asset is recorded at the price paid.³ Shelving bought for Tk 120,000 stays at Tk 120,000, even if a dealer would offer a different price today. Second, if liabilities exceed assets, then $E = A - L$ gives a negative number. The owner then has no claim left, and the business owes more than it holds. This is a warning sign, not an error in the equation.

Different people read the equation differently. A lender looks at A and L and asks whether the assets are large enough to repay the loan. The owner looks at E and asks how large the personal claim has become. A supplier looks at the account payable and asks whether the business can pay it.

PAUSE AND CHECK 3.P1 A workshop has assets of Tk 80,000 and liabilities of Tk 25,000. What is its owner's equity, and how do you check the result?

³ Recall: historical cost, Chapter 2, 'What Accounting Is', section 2.3 'Rules and Concepts'.

JOURNAL ENTRIES FOR 15 TO 31 JANUARY 2026

DATE	PARTICULARS	FOLIO	DEBIT (TK)	CREDIT (TK)
15 January 2026	Rent	L7	60,000	
	Cash	L1		60,000
	Narration: Paid the month's rent.			
20 January 2026	Publisher Payable	L4	80,000	
	Cash	L1		80,000
	Narration: Paid part of the amount owed to the publisher.			
31 January 2026	Wages	L8	35,000	
	Cash	L1		35,000
	Narration: Paid wages for January.			

Step 2: post. Each of the fourteen lines above is copied to its account. The Cash account receives six postings, and the table that follows shows them with the totals of each side.

THE CASH ACCOUNT AFTER POSTING (LEDGER PAGE L1)

DATE	PARTICULARS	FOLIO	DEBIT (TK)	CREDIT (TK)
1 January 2026	Owner's Equity	J1	500,000	
2 January 2026	Shelving	J1		120,000
10 January 2026	Service Revenue	J1	40,000	
15 January 2026	Rent	J1		60,000
20 January 2026	Publisher Payable	J1		80,000
31 January 2026	Wages	J1		35,000
Totals			540,000	295,000

Step 3: balance Cash. Debit total: $500,000 + 40,000 = \text{Tk } 540,000$. Credit total: $120,000 + 60,000 + 80,000 + 35,000 = \text{Tk } 295,000$. The larger total is the debit side, so $B = 540,000 - 295,000 = \text{Tk } 245,000$, a debit balance. The other accounts are balanced in the same way, and the table that follows lists every balance.

Step 3 posts every line, and the Cash account shows six postings.

Step 4 balances Cash line by line, then the other accounts.

PAUSE AND CHECK 3.P2 The owner invested Tk 40,000. Revenue was Tk 15,000, expenses Tk 9,000 and drawings Tk 2,000. What is the owner's equity?

Before moving on. Revenue raises owner's equity, expenses lower it, and drawings lower it without being an expense. The expanded form is $E = I + R - X - D$. You should now be able to say which of the four movements raises the owner's claim and which three lower it.

3.3 TRANSACTIONS AND THE EQUATION

How can a business record hundreds of events and still keep both sides equal? The answer lies in how a transaction behaves. Each transaction changes the items of the equation, but always in matching ways: when one side changes, the other side changes by the same amount, or the change happens inside one side only.

Take any transaction and ask three questions in order. First, which items change, and by how much? Second, is each item an asset, a liability, owner's equity, revenue, an expense or drawings? Third, after the change, does $A = L + E$ still hold? If the third answer is no, an effect is missing or a sign is wrong. Begin with the smallest case. On 1 January 2026 the owner puts Tk 500,000 into the workshop. The asset cash rises by Tk 500,000, and owner's equity rises by Tk 500,000. Both sides are Tk 500,000. The equation holds.

The table below groups the effects into five patterns. Read a row from left to right: the first cell names the pattern, the second states what happens to the two sides, and the third gives an example from the workshop.

THE FIVE PATTERNS OF CHANGE IN THE EQUATION

PATTERN	EFFECT ON THE EQUATION	EXAMPLE FROM JANUARY 2026
One asset rises, another falls	Total assets unchanged	Shelving bought for cash
Asset rises, liability rises	Both sides rise equally	Inventory bought on credit
Asset rises, owner's equity rises	Both sides rise equally	Owner invests; binding fee earned
Asset falls, owner's equity falls	Both sides fall equally	Rent paid; wages paid
Asset falls, liability falls	Both sides fall equally	Publisher paid

Notice that every row contains at least two effects. No transaction changes only one item and leaves the rest alone. This fact is the seed of double entry, which Chapter 4, 'Accounts, Debits and Credits', section 4.3 'Debit and Credit' builds into a full method of recording. For now, keep the simpler tool: test $A = L + E$ after every transaction. A transaction that changes only the asset side, such as the shelving purchase, still balances, because one asset rises exactly as another falls.

Before moving on. Analyse any transaction with these questions:

Three new items and one longer formula have been added to the equation.

Every transaction has matching effects, so the equation stays balanced.

Three questions analyse any transaction in the same order.
GO SLOWLY HERE.

Five patterns cover all seven events of the month.

Every row has two effects, which is the seed of a larger system.

Three questions: five patterns

The method has four steps, and the same four serve every entry. First, find the account that was debited and enter on its debit side the date, the name of the other account, the journal page, and the amount. Second, find the account that was credited and enter on its credit side the date, the name of the other account, the journal page and the amount. Third, repeat for any further lines of the entry. Fourth, write the ledger page of each account in the Folio column of the journal. The tables that follow show the postings of the 1 January entry, with the journal on page 1 and the accounts on pages L1 and L5.

THE 1 JANUARY POSTING IN THE CASH ACCOUNT (LEDGER PAGE L1)

DATE	PARTICULARS	FOLIO	DEBIT (TK)	CREDIT (TK)
1 January 2026	Owner's Equity	J1	500,000	

THE 1 JANUARY POSTING IN THE OWNER'S EQUITY ACCOUNT (LEDGER PAGE L5)

DATE	PARTICULARS	FOLIO	DEBIT (TK)	CREDIT (TK)
1 January 2026	Cash	J1		500,000

Folio references are small numbers that link the two books. In the ledger, "J1" says that the amount came from journal page 1. In the journal, "L1" written beside Cash says that the amount was posted to ledger page L1. With these links a reader can start from any figure in either book and find the other half of the record in seconds. A folio entry in the journal also shows that the line has been posted, which prevents posting the same line twice.²

Posting leaves each account with many amounts on its two sides. To learn how much the account holds, we find its balance. Balancing an account means totalling both sides and finding the difference. Many students find this step confusing, because the arithmetic is easy but the layout is not. The method is as follows. Total the debit side. Total the credit side. Subtract the smaller total from the larger. The result is the balance, and it belongs to the side with the larger total.

Formula. In words: the balance equals the larger side total minus the smaller side total. In symbols: $B = L - S$, where B is the balance, L is the larger total and S is the smaller total. For a tiny example with made-up figures, suppose an account has debits of Tk 200 and Tk 100, and a credit of Tk 120. Then $L = 300$ on the debit side and $S = 120$, so $B = 300 - 120 = \text{Tk } 180$, a debit balance. **Check.** Add the balance to the smaller side: $120 + 180 = 300$, which equals the larger total, so the two sides now agree.

Before moving on. The ledger collects the journal's entries under the name of each account. Posting copies each debit and each credit to its account and records the folio. A balance is the larger side total less the smaller, placed on the larger side.

* * *

² Accounting software posts automatically, but the logic is the same: each journal line reaches the account it names, once.

Posting follows four short steps for every journal entry.

The folio columns link each journal line to its ledger account and back.

Folio. Word in use. In daily life, a page number or a large sheet of paper. In accounting, a reference number linking a journal line to a ledger page, or the reverse.

After posting, an account is balanced to show how much it now holds.

The formula for a balance is the larger total less the smaller total.

A short review of the ledger, posting and balancing.

Read the figure by rows, each showing the three totals after one event.

The answer is stated in a sentence, and two independent checks confirm it.

Three slips are common in a worked list like this one.

A long list is solved one row at a time and then totalled.

The chapter in a few lines: three items, one equation, and what moves it.

Owner's equity is a claim, and it is not the same as the owner's cash.

To read the second figure, take one row at a time. The three number columns show the totals of assets, liabilities and owner's equity after that event. Look at the row for 3 January: assets rise to 700,000, and liabilities to 200,000, with owner's equity still at 500,000. In every row, the first number equals the sum of the other two.

Answer. At 31 January 2026 the workshop has assets of Tk 565,000, liabilities of Tk 120,000 and owner's equity of Tk 445,000. The assets are cash Tk 245,000, shelving Tk 120,000 and inventory Tk 200,000. **Check.** Liabilities plus equity: $120,000 + 445,000 = 565,000$, which equals the assets. The cash can be tested alone: $500,000 - 120,000 + 40,000 - 60,000 - 80,000 - 35,000 = 245,000$. The expanded form agrees: $500,000 + 40,000 - 95,000 = 445,000$.

Students often match a rent payment to the liability column, when it should lower owner's equity. They also forget that the shelving appears twice, once rising and once falling. And they sometimes change owner's equity when the publisher is paid. Test each row against the three questions before totalling.

Before moving on. A long list of events is solved one row at a time, then totalled. Two checks, $A = L + E$ and the expanded form, give confidence. You should now be able to analyse a new set of events in the same way.

* * *

SUMMARY

Assets are what a business owns, liabilities are what it owes, and owner's equity is the owner's claim after debts. The accounting equation, $A = L + E$, holds for every business at every date. Revenue raises owner's equity, expenses lower it, and drawings lower it without being an expense, giving $E = I + R - X - D$. Every transaction has matching effects, so the equation stays balanced. At 31 January 2026, Meghna Binding Service has assets of Tk 565,000, liabilities of Tk 120,000 and owner's equity of Tk 445,000.

THINK IT THROUGH

3.T1 A transaction changes only the asset side of the equation. Give one example from this chapter and explain why both sides stay equal.

3.T2 A student says that paying the publisher Tk 80,000 on 20 January 2026 was an expense. Decide whether she is right, and explain what changed.

WHERE PEOPLE GO WRONG

Many readers think owner's equity is the cash the owner has in the business. It is not. Owner's equity is a claim, not an item. At 31 January 2026 the workshop's owner's equity is Tk 445,000, but its

amount. Both facts belong to one event, so both must be written together in one place.

A reminder from Chapter 4 helps here. Each type of account increases on one side and decreases on the other. The table that follows lists the five types met in this chapter. Read a row from left to right: the type, the side on which it increases, and an example from the case. A decrease is always entered on the opposite side.

Recall: debit and credit, Chapter 4, section 4.3; account types, Chapter 4, section 4.2.

The table reminds the reader which side increases each type of account.

WHICH SIDE INCREASES EACH TYPE OF ACCOUNT

TYPE OF ACCOUNT	INCREASES WITH	EXAMPLE IN THE CASE
Asset	Debit	Cash
Liability	Credit	Publisher Payable
Owner's equity	Credit	Owner's Equity
Revenue	Credit	Service Revenue
Expense	Debit	Rent

The first book is called the journal. The **journal** is the book of first entry, in which transactions are recorded in date order with debits and credits. It is called the book of first entry because every transaction is written there before it is written anywhere else. The record of one transaction in the journal is a journal entry. A journal may be a bound volume, a ledger-style sheet or a computer file; the rules are the same.¹

A word with two meanings appears in the name of the first book.

The journal is the book of first entry, kept in date order.

Book. Word in use. In daily life, a set of printed pages that you read. In accounting, a record in which transactions are written, such as the journal or the ledger.

Journal. The book of first entry, in which transactions are recorded in date order with debits and credits.

A journal entry has five columns and a fixed order of lines.

A journal entry follows a fixed pattern. The columns are Date, Particulars, Folio, Debit and Credit. The date goes on the first line. The account to be debited is written first, with its amount in the Debit column. The account to be credited is written on the next line, indented a little to the right, with its amount in the Credit column. The Folio column stays empty until the entry is posted, a step that section 5.3 explains. The table that follows shows the entry for 1 January.

THE JOURNAL ENTRY FOR 1 JANUARY 2026

DATE	PARTICULARS	FOLIO	DEBIT (TK)	CREDIT (TK)
1 January 2026	Cash		500,000	
	Owner's Equity			500,000
	Narration: Owner invested cash to begin the workshop.			

The last line of the entry is a narration. A **narration** is a short note

A narration records the reason for the entry in one line.

¹ Some businesses keep several journals, one for each frequent kind of transaction. Chapter 11, 'Special Journals', teaches them. This chapter uses the single general journal.

THE EIGHT STEPS OF THE ACCOUNTING CYCLE

STEP	WHAT HAPPENS	WHERE TAUGHT
1	Collect and check source documents	This chapter
2	Record each transaction in the journal	This chapter
3	Post the journal to the ledger	This chapter
4	List the ledger balances in a trial balance	Chapter 6, 'The Trial Balance'
5	Adjust the accounts for changes not yet recorded	Chapter 7, 'Adjusting the Accounts'
6	Collect the figures on a worksheet	Chapter 8, 'The Worksheet'
7	Prepare the financial statements	Chapter 9, 'Financial Statements and Closing Entries'
8	Prepare the closing records	Chapter 9, 'Financial Statements and Closing Entries'

Every cycle begins with proof. A **source document** is paper or electronic proof of a transaction, such as an invoice, receipt or cheque stub. When the workshop buys books on credit from a publisher, the publisher's invoice is the source document. When the workshop pays rent, the landlord's receipt is the source document. Records begin from the document and never from memory, because a document can be shown to a bank, a tax authority or an auditor, who is a person that checks records for accuracy.

The idea has limits. Some events have no paper from an outsider: the owner's own investment may be proved only by a bank slip or a signed note. Other entries, which Chapter 7 describes, rest on a calculation made inside the business. A careful bookkeeper therefore asks one question of each source document before recording it: does this prove that an event happened, and does it show the date and the amount? If it does not, the entry waits.

The next figure shows the path that one transaction takes. Read it from left to right. The three boxes are the source document, the journal and the ledger. The arrow between the first two is labelled "recorded", and the arrow between the last two is labelled "posted". Below each box is a short note saying how that record is arranged. The example in the bottom row follows the publisher's invoice of 3 January.

A source document is the first proof that a transaction happened.

Source document. Paper or electronic proof of a transaction, such as an invoice, receipt or cheque stub.

Not every event has outside paper, and the bookkeeper must still act with care.

The figure that follows shows the path of one transaction through the first three steps.

YOU WILL NEED

Chapter 3, 'The Accounting Equation', section 3.1 'Assets, Liabilities and Equity'; section 3.2 'Revenue, Expenses and Drawings'; and section 3.3 'Transactions and the Equation'. Also Appendix A1, 'Maths You Will Use', for adding and subtracting columns of figures.

A diary of events makes the simplest question hard to answer.

The equation of Chapter 3 works for a few events but not for thousands.

The chapter teaches the place, the six kinds and the rule that catches mistakes.

Imagine that you keep a diary for a small business. Each evening you write one line for every event: money received, a bill paid, books bought. After a month the diary holds dozens of lines. Then a lender asks one question: how much cash does the business hold today? To answer, you must read every line, decide which ones touched cash, and add and subtract them in the right order. If your answer is wrong, you cannot see where the mistake lies.

Chapter 3 gave one answer: change the accounting equation after each transaction. That method works well for seven events. It fails for seven thousand, because the table would need a column for every kind of item and a row for every event. A real business needs two things. It needs to keep each kind of item in one place, with every increase and decrease together. It also needs a rule that catches mistakes at once.

This chapter teaches both. First we meet the account, which is the place. Next we sort accounts into six types. Then we learn the rule of debit and credit, which links every entry to a second entry and so exposes many errors. The rule was written in a printed book more than five hundred years ago, and it has hardly changed since. Few ideas in business have lasted so long.

* * *

4.1 THE ACCOUNT

GO SLOWLY HERE.

A household notebook with one page for each item shows the idea of an account.

An account holds all the changes to one kind of item.

Account. A record that collects all the increases and decreases of one kind of item, such as Cash.

A chart of accounts is the list that stops duplicate or missing accounts.

Chart of accounts. A list of every account a business uses, with a code for each.

Think of a household exercise book. One page is headed *Rice*, another *Electricity*, a third *Money lent to a neighbour*. Each time something happens, you write it on the page where it belongs. To learn how much was spent on rice this month, you read one page and not the whole book. A business uses the same idea, with many more pages and with strict rules for writing on them.

An **account** is a record that collects all the increases and decreases of one kind of item. For example, the Cash account collects every receipt and every payment of cash, so it shows how much cash the business holds. The business keeps one account for each kind of item it must follow: one for Cash, one for shelving, one for money owed to a publisher, and so on.

As the number of accounts grows, the business needs a list, so that nobody opens a second Cash account by mistake. A **chart of accounts** is a list of every account a business uses, with a code for each. For example, a business might give the code 101 to Cash and the code 201 to Accounts Payable. Meghna Binding Service, the running case, uses only eight accounts, so its chart is short. A large

Asset accounts record what is owned; liability accounts record what is owed.

The owner's capital account records money put in; drawings records money taken out.

Revenue accounts record what is earned; expense accounts record what is used up.

The recap gathers the section into three sentences.

An asset account records something of value that the business owns or controls, such as Cash, Shelving or Inventory. Here *inventory* means the books held for sale, a subject that Chapter 13, 'Inventory: Meaning and Measurement', teaches in full. A liability account records an amount owed to others. Accounts Payable, for example, holds the amounts owed to publishers. The owner can look at the first and see what the business has, and look at the second and see what it owes.

The owner's capital account records the value the owner has put into the business, and it is part of owner's equity. The drawings account records what the owner takes out for personal use. Why keep them apart? Because the owner should see both how much was put in and how much was taken out. A single combined account would hide the second figure behind the first.

A revenue account records money earned or due from selling goods or doing work. Service Revenue is the revenue account of Meghna Binding Service. An expense account records a cost used up in earning revenue, such as Rent or Wages. The business keeps a separate expense account for each cost, because an owner wishes to know the rent and the wages separately and not as one lump.

Before moving on. The six types are asset, liability, owner's capital, drawings, revenue and expense. Each account belongs to exactly one type, and its type decides how it behaves. You should now be able to name the type of Cash, Accounts Payable, Rent and Service Revenue.

* * *

CHAPTER V

The Journal and the Ledger

If every event is written down once, how can any fact be found again?



Every transaction leaves a paper trail: a source document, a journal entry and a ledger account. This chapter follows that trail step by step, names the steps of the accounting cycle, and works through the seven January 2026 transactions of Meghna Binding Service.

GO SLOWLY HERE.

Bank Statements use the two words, but with a meaning that does not fit here.

Debit. Word in use. In daily life, a bank debits an account when it takes money out. In accounting, a debit is an entry on the left side of an account, whether the balance rises or falls.

Credit. Word in use. In daily life, credit means buying now and paying later, or a bank adding money to an account. In accounting, a credit is an entry on the right side of an account.

A debit is an entry on the left side of an account.

Debit. An entry on the left side of an account; it increases assets and expenses and decreases liabilities and equity.

A credit is an entry on the right side of an account.

4.3 DEBIT AND CREDIT

You have probably seen the words *debit* and *credit* on a bank statement. The bank says it has debited your account when it takes money out and credited it when it adds money. Shops also use *credit* for buying now and paying later. In accounting, the two words have a fixed and different meaning, and the bank's habit can mislead you. Put it aside for this chapter. The words come from Venice: *debit* from the Latin *debere*, to owe, and *credit* from *credere*, to believe.

A **debit** is an entry on the left side of an account. That is the whole meaning of the word: the left side. It does not mean good or bad, and it does not mean more or less. In the Cash T-account, a figure written on the left side is a debit to Cash. Whether a debit raises or lowers the balance depends on the type of account, as the next paragraphs show.

A **credit** is an entry on the right side of an account. Again, the

Common slips. Students often treat debit as meaning decrease, which is wrong for assets. They record an expense on the credit side because the cash leaves. They forget the second entry, so the totals differ. Or they read a liability as an asset and debit it when the business borrows. When an entry looks strange, return to the four questions.

Before moving on. The four questions are: which accounts, what types, increase or decrease, and which side. Each transaction gets one debit and one credit of equal amount. You should now be able to record any simple transaction without memorising it.

* * *

SUMMARY

An account collects all the changes to one kind of item, and the chart of accounts lists them all. There are six types: asset, liability, owner's capital, drawings, revenue and expense. A debit is the left side and a credit is the right side. Assets, drawings and expenses increase by debit, and liabilities, owner's capital and revenue increase by credit. Double entry records every transaction with an equal debit and credit, which keeps the equation in balance.

THINK IT THROUGH

4.T1 Explain, using the accounting equation, why Cash increases on the debit side and not on the credit side.

4.T2 The owner takes Tk 10,000 in cash from the business for personal use. Name the two accounts, say which is debited and which credited, and explain why the business does not record this as an expense.

WHERE PEOPLE GO WRONG

A mistaken belief: "Debit means less and credit means more." **Correction:** the words name the left side and the right side. Debits raise assets, expenses and drawings, but they lower liabilities, owner's capital and revenue.

A mistaken belief: "A credit is good news and a debit is bad news." **Correction:** neither word has a moral meaning. A debit to Cash is good news for the business, and a credit to Accounts Payable means the business owes more.

A mistaken belief: "Money going out is a credit, so every payment is a credit to the expense." **Correction:** the credit goes to Cash, which falls. The expense account increases, so it is debited.

A mistaken belief: "If debits equal credits, the books are correct." **Correction:** equal totals show agreement only. A wrong account, a missing transaction or a wrong amount on both sides still leaves the totals equal.

IN HISTORY

Four common slips cause most wrong entries.

The recap gathers the section into three sentences.

The summary restates the chapter in a few lines.

Two problems test whether the rule can be explained and applied again.

Debit does not mean decrease, and credit does not mean increase.

Debit is not bad and credit is not good.

Expenses are debited even though money leaves the business.

Equal totals do not prove that every entry is right.

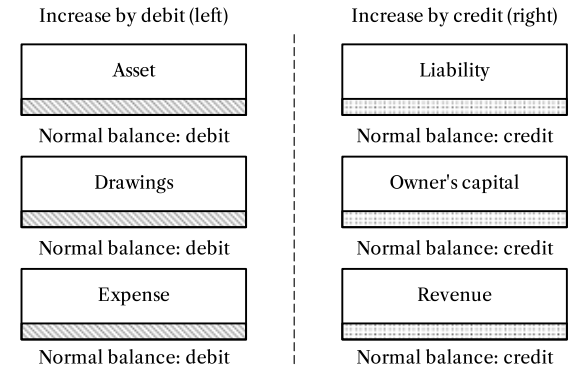


FIG. 4.2. Six account types, each with its side of increase and normal balance

Double entry means every transaction has two equal sides.

Double entry records each transaction twice, with equal debit and credit.

Double entry. The rule that every transaction is recorded twice, once as a debit and once as a credit of equal amount.

Equal entries keep the equation in balance and expose many mistakes.

The owner and the lender both gain from the check of equal totals.

Double entry has limits: equal totals do not prove that every entry is right.

Many students find this the hardest step, because it asks you to hold two ideas together. Start with a question. If all changes to an item sit in one account, how can we know that none was forgotten? Chapter 3 gives the clue. Every transaction left the accounting equation in balance, so every transaction had at least two effects. The method of recording should show both effects each time. That method is called double entry.

Double entry is the rule that every transaction is recorded twice, once as a debit and once as a credit of equal amount. Take the first transaction of Meghna Binding Service. On 1 January 2026 the owner invests cash of Tk 500,000. Cash is an asset and it increases, so Cash is debited Tk 500,000. Owner's Capital increases, so Owner's Capital is credited Tk 500,000. One fact, two entries, one amount.

PAUSE AND CHECK 4.P2 A business buys shelving for cash. Name the two accounts, and say which one is debited and which one credited.

Why does the rule help? Each entry changes the equation by the same amount on each side, so the equation stays in balance after every transaction. It follows that, at any moment, the total of all debit entries must equal the total of all credit entries. If they differ, an error exists, and the business knows it at once. This check is far better than the diary of the opening puzzle. The Venetian method was described in print by Luca Pacioli in 1494, and its essential rule is the one you have now learned.

Consider the viewpoints. The owner gains a quick way to find mistakes before they grow. A lender, who cannot watch the business, gains confidence that the figures were kept under a rule that exposes many errors. An employee who keeps the records gains protection, because a difference points to the mistake and not to the person.

Where does the rule fail? Equal totals show that debits and credits agree. They do not show that the right accounts were chosen. If rent is debited to Wages by mistake, the totals still agree. If a trans-

8.4 Combined problem. The unadjusted balances are, in taka: Cash 9,000; Equipment 12,000; Prepaid Insurance 1,800; Wages 3,000 (debits); Capital 17,400; Unearned Revenue 2,400; Fees Revenue 6,000 (credits). Adjustments: (a) insurance of 900 has expired; (b) fees of 1,500 in Unearned Revenue are now earned; (c) depreciation is 1,200; (d) wages of 400 are owed. Complete the worksheet and state the net income.

8.5 Name two jobs that a worksheet does and a ledger alone does not. Then explain why the worksheet does not change any account.

YOU WILL NEED

Chapter 5 'The Journal and the Ledger', section 5.3 'The Ledger and Posting'; Chapter 5 'The Journal and the Ledger', section 5.1 'The Accounting Cycle'; Appendix A1 'Maths You Will Use'.

A second, independent count tests the first, and the ledger offers such a pair.

After a month of posting, the bookkeeper needs a test that avoids rereading every line.

The ledger should keep the equality of debits and credits that every entry has.

Recall: ledger and posting, Chapter 5, section 5.3.

A tiny example with made-up figures shows the equality surviving the balancing.

Two everyday words take a special meaning in this chapter.

Balance. Word in use. In daily life, to keep steady, or the amount left in a bank account. In accounting, the difference between the two sides of an account.

Trial. Word in use. In daily life, a hearing in a court of law or a test of something.

A shopkeeper closes for the evening and counts the coins in the till. Then the shopkeeper adds the till roll, the paper strip on which each sale was printed. If the two answers agree, the count is probably right. If they differ, something is wrong, and the shopkeeper knows to search. Neither answer proves anything alone, but two independent answers that agree give real comfort.

The books of a business offer a similar pair of answers. Every transaction was written with debits equal to credits. After a month of journalising and posting, a bookkeeper has hundreds of lines in many accounts. How can the bookkeeper learn, without rereading every line, whether a slip has occurred? And if a test says "wrong", how can the slip be found without starting again? This chapter answers both questions, and it also explains an uncomfortable truth: a test that says "right" does not prove that the books are perfect.

* * *

6.1 PREPARING A TRIAL BALANCE

Recall that every journal entry has debits equal to credits, and that posting copies each debit and each credit to its account. If both jobs are done correctly, one result must follow: the total of all debits in the ledger equals the total of all credits. Let D be the total of all debit amounts posted and C the total of all credit amounts. Because each entry has equal sides, $D = C$. Balancing an account removes the same amount from both of its sides, so the equality survives. The total of the debit balances therefore equals the total of the credit balances.

Here is a tiny example with made-up figures. Entry one debits Cash and credits Owner's Equity for Tk 100. Entry two debits Rent and credits Cash for Tk 30. The debit total D is $100 + 30 = 130$, and the credit total C is $100 + 30 = 130$. After balancing, Cash is a debit of $100 - 30 = 70$, Rent is a debit of 30, and Owner's Equity is a credit of 100. The debit balances total $70 + 30 = 100$, and the credit balance is 100. The equality holds.

Two words need care. In everyday speech, a *balance* may be the amount left in a bank account. In accounting, the balance of an account is the difference between its two sides, as Chapter 5 showed. The word *trial* in everyday speech may mean a hearing in court. Here it means only a test. The test asks one question and gives a plain answer, but it is not a verdict on the books.

IN SHORT

A worksheet moves figures through five pairs of columns: trial balance, adjustments, adjusted trial balance, income statement and balance sheet. Net income is revenue minus expenses, and it is entered on the debit side of the income columns and the credit side of the balance sheet columns.

Many students find the last step confusing, because one amount appears in two places. These are the common slips. Some students extend a debit balance into a credit column. Some treat Drawings as an expense and extend it to the income columns. Some forget an account that came from an adjustment, such as Wages Payable. Some place a net loss on the same sides as a net income, which reverses the sign. Finally, some mix units, for example by writing 3,500 for a figure that is in thousands of taka. Name each slip before you begin, and you will notice it when it occurs.

Before moving on. To extend a balance, copy it to one statement column on its own side. Net income is revenue minus expenses, and net loss is expenses minus revenue. The balancing figure appears on opposite sides in the income columns and the balance sheet columns. Three checks test the worksheet, and none of them finds an amount in the wrong account. *You should now be able to say* where net income is entered, and why it is entered twice.

* * *

SUMMARY

A worksheet is a working paper with columns. It moves the figures of the trial balance through the adjustments, the adjusted trial balance and the statement columns, without changing any ledger account. Net income is total revenue minus total expenses, and net loss is the reverse. The balancing figure is entered twice, on opposite sides. Equal totals check the arithmetic, but they cannot show that an amount is in the right account.

THINK IT THROUGH

8.T1 A clerk extends the stall's balances. The balance sheet columns total Tk 11,400 (debit) and Tk 8,500 (credit) before net income. Net income of Tk 2,600 is added to the credit side. Find the difference between the two totals, name the stall account with the same balance, and say what you would check first.

8.T2 The stall's adjusted trial balance shows a debit total of Tk 14,530 and a credit total of Tk 14,800. Find the difference. Is it divisible by 9? What kind of error does this suggest, and which balance is the likely cause?

WHERE PEOPLE GO WRONG

Common slips: a wrong column, a wrong side, a forgotten account and a mixed-up sign.

The recap restates extending, finding net income or loss, and checking.

The chapter in a few lines: five pairs of columns, one result, three checks.

The formula is a plain addition of each column, followed by a comparison.

Formula. In words: the debit total is the sum of all debit balances, the credit total is the sum of all credit balances, and the test is passed when the two are equal. In symbols: DT = sum of debit balances; CT = sum of credit balances; the test is DT = CT. *Steps.* DT = 245,000 + 120,000 + 200,000 + 60,000 + 35,000 = Tk 660,000. CT = 120,000 + 500,000 + 40,000 = Tk 660,000. The table that follows sets out the list.

TRIAL BALANCE OF MEGHNA BINDING SERVICE AS AT 31 JANUARY 2026

ACCOUNT	DEBIT (TK)	CREDIT (TK)
Cash	245,000	
Shelving	120,000	
Inventory	200,000	
Publisher Payable		120,000
Owner's Equity		500,000
Service Revenue		40,000
Rent	60,000	
Wages	35,000	
Totals	660,000	660,000

A trial balance is taken often, because later steps depend on it.

The answer and the check show a passed test that still proves less than it seems.

Common slips in preparing the list are easy to avoid with a tick for each account.

Different readers use the trial balance in different ways.

A short review of the trial balance and its preparation.

The test has a place in the cycle. Steps 5 to 8 of the accounting cycle build on the ledger balances, so an undetected slip at this point would travel into every later report. Taking the trial balance at the end of each month, or even each week, keeps the search for a slip short. In a small bookshop with few transactions, one list a month is enough; a busy business may take one daily.

Answer. Both columns total Tk 660,000, so the test is passed. **Check.** A second sum in a different order helps: debits, 35,000 + 60,000 + 200,000 + 120,000 + 245,000 = 660,000. The same total appears. From the owner's viewpoint, the result is reassuring. From the viewpoint of a bank, it is a first sign of care, though not a proof that every figure is right, as section 6.2 explains.

Common slips are few. A balance may be written in the wrong column. An account that has a balance may be left out. A digit may be added wrongly. Tick each ledger account as it is listed, and compare the number of accounts on the list with the number in the ledger.

The trial balance serves several people inside the business. The bookkeeper uses it to learn whether posting went well. The owner, Nurul, uses it as a quick picture of what the accounts hold, though it is not a formal report. An employee who prepares the later steps of the cycle uses it as the starting list, because adjustments and statements are built from its figures. A bank usually asks for the formal reports rather than for this working paper.

PAUSE AND CHECK 6.P1 Explain in your own words why the debit total of a trial balance must equal the credit total when every journal entry and every posting is correct.

Before moving on. A trial balance lists all ledger balances in two columns and tests that the totals are equal. The test works because every entry has equal debits and credits. The five steps are: bal-

Now we total columns 7 and 8. The difference between them is the result of the year, and it has a name. **Net income** is the amount by which total revenue exceeds total expenses for a period. For example, if the stall earns Tk 6,000 and its expenses are Tk 3,400, its net income is Tk 2,600.

Formula. In words: net income equals total revenue minus total expenses. In symbols, $NI = R - E$. Here NI stands for net income, R stands for total revenue and E stands for total expenses.

Steps. Column 7 total: $E = \text{Tk } 2,300 + \text{Tk } 600 + \text{Tk } 500 = \text{Tk } 3,400$. Column 8 total: $R = \text{Tk } 6,000$. Then $NI = \text{Tk } 6,000 - \text{Tk } 3,400 = \text{Tk } 2,600$. **Answer.** The stall's net income is Tk 2,600. **Check.** Enter Tk 2,600 under column 7, on the debit side, so that column 7 totals Tk 6,000 like column 8. Enter Tk 2,600 under column 10, on the credit side. Column 9 totals Tk 11,400, and column 10 totals Tk 8,800 + Tk 2,600 = Tk 11,400.

Why does Tk 2,600 appear twice, on opposite sides? In the income columns, the debit side is short by Tk 2,600, so the figure balances that pair. In the balance sheet columns, net income increases owner's equity, and owner's equity has a credit balance, so the figure belongs on the credit side. One amount, two columns, opposite sides.

PAUSE AND CHECK 8.P2 The stall's income columns total Tk 3,400 (debit) and Tk 6,000 (credit). On which side of the income columns is the balancing figure placed, and why?

A period can also end badly. A **net loss** is the amount by which total expenses exceed total revenue for a period. For example, if the stall earned Tk 3,000 in a slow quarter and its expenses were Tk 3,400, its net loss would be Tk 400. The formula reverses: $NL = E - R$. The balancing figure now goes to the credit side of the income columns and to the debit side of the balance sheet columns, because a loss reduces owner's equity.

A worksheet is checked three times. First, the totals of columns 1 and 2, of columns 3 and 4, and of columns 5 and 6 must agree. Second, after the balancing figure is added, columns 7 and 8 must agree, and columns 9 and 10 must agree. Third, the balancing figure under the income columns must equal the one under the balance sheet columns, on opposite sides. These checks catch arithmetic slips and extension errors. They do not catch an amount placed in the wrong account, because the totals still agree.

Case file, illustrative. Meghna and Sons uses the dataset of the Chapter 7 case. After the adjustments, the balances are, in taka: Cash 150,000; Accounts Receivable 110,000; Supplies 15,000; Prepaid Insurance 27,000; Equipment 360,000; Accumulated Depreciation 72,000; Accounts Payable 60,000; Wages Payable 14,000; Unearned Revenue 12,000; Capital 400,000; Drawings 80,000; Service Revenue 788,000; Wages 344,000; Rent 120,000; Utilities 26,000; Supplies Expense 33,000; Insurance Expense 9,000; Depreciation Expense 72,000. The file records total expenses of 604,000 and net income of $788,000 - 604,000 = \text{Tk } 184,000$.

GO SLOWLY HERE.

Net income is the amount by which revenue exceeds expenses.

Net income. The amount by which total revenue exceeds total expenses for a period.

The formula for net income, in words and in symbols.

Steps, answer and check for the stall: net income is Tk 2,600.

The balancing figure goes to opposite sides because net income increases owner's equity.

Net loss is the opposite case, with expenses above revenue.

Net loss. The amount by which total expenses exceed total revenue for a period.

Three checks show whether a worksheet is sound, and each has limits.

The case file gives the adjusted balances of the running case.

where it began. Such luck is rare, but it shows that equal totals cannot rule out several errors.

An auditor who reads a balanced trial balance treats it as a starting point and not as an end. The auditor tests source documents against the journal to find the errors that the totals cannot show. An owner who sees equal totals may feel safe, but a missing sale or a payment posted to the wrong account still spoils the reports. A careful bookkeeper therefore combines the test with other checks, such as confirming that every source document has been recorded.

Before moving on. The trial balance finds errors that change one side only. It cannot find a missing transaction, an equal error on both sides, a wrong account of the same type, or errors that cancel. A transposition error on one side gives a difference divisible by 9.

* * *

6.3 FINDING AND FIXING DIFFERENCES

Suppose the wages of 31 January were posted to the debit side of the Wages account as Tk 53,000, while the credit to Cash was posted correctly as Tk 35,000. The trial balance now shows a debit total of Tk 678,000 and a credit total of Tk 660,000. The difference is Tk 18,000. The bookkeeper holds one number and faces a long ledger. Where should the search begin?

Locating a difference follows a fixed order, from the cheapest check to the most costly. First, add both columns again, in a different order. Second, test whether the difference is divisible by 9, which points to a transposition error. Third, test whether the difference equals one account balance, which points to a posting that was missed, and whether half of the difference equals a balance, which points to a balance listed in the wrong column. Fourth, compare every posting with the journal and tick each line that agrees. Fifth, recalculate each balance. Stop as soon as the cause is found.

The checks have examples in the January ledger. Suppose the Rent debit of Tk 60,000 was never posted. The debit total would be Tk 600,000 against a credit total of Tk 660,000. The difference of Tk 60,000 equals one amount in the journal, so the bookkeeper would look for a missed posting of that size. Suppose instead that Service Revenue, a credit balance of Tk 40,000, were listed in the Debit column. The debit total would become Tk 700,000 and the credit total Tk 620,000. The difference of Tk 80,000 is twice Tk 40,000, so half of it points to a balance in the wrong column.

The best search is the one that is never needed. Careful bookkeepers check each entry before posting, tick each journal line when it has been posted, and take a trial balance often, for example at the end of each week. A difference found among a week of entries is far easier to trace than a difference found among a year of them.

When time runs short, the bookkeeper may open a suspense account. A **suspense account** is a temporary account that holds a difference until its cause is found. If the credits fall short, the difference is entered as a credit balance in the suspense account. If

A balanced list is a starting point for the auditor and a warning for the owner.

A short review of what the test finds and what it misses.

A difference of Tk 18,000 shows where a search for a slip begins.

The search moves from the cheapest check to the most costly.

Two further differences show how the second and third checks work.

Careful habits reduce the search.

Sometimes the search cannot finish before the reports are due.

Suspense account. A temporary account that

The rule for step 3 has two parts. If the balance and the adjustment stand on the same side, add them. Wages is a debit of Tk 2,000, and adjustment (b) adds a debit of Tk 300, so the adjusted balance is a debit of Tk 2,300. If they stand on opposite sides, subtract the smaller from the larger, and place the answer on the larger side. Supplies is a debit of Tk 1,000 and a credit of Tk 600, so the answer is a debit of Tk 400. A row with only an adjustment, such as Wages Payable, keeps that figure on its own side.

The table below shows the stall's worksheet through column 6. Read each row from left to right. For Supplies, Tk 1,000 is a debit in column 1. The letter (a) beside Tk 600 marks a credit in column 4. The result, Tk 400, is a debit in column 5. The bottom row shows that every pair of columns has equal totals, and the adjusted trial balance totals Tk 14,800 on each side.

STALL WORKSHEET, COLUMNS 1 TO 6 (TAKA)

ACCOUNT	TRIAL BALANCE DR	TRIAL BALANCE CR	ADJUSTMENTS DR	ADJUSTMENTS CR	ADJUSTED DR	ADJUSTED CR
Cash	5,000				5,000	
Supplies	1,000			(a) 600	400	
Equipment	6,000				6,000	
Capital		8,000				8,000
Repair Revenue		6,000				6,000
Wages	2,000		(b) 300		2,300	
Supplies Expense			(a) 600		600	
Depreciation Expense			(c) 500		500	
Wages Payable				(b) 300		300
Accumulated Depreciation				(c) 500		500
Totals	14,000	14,000	1,400	1,400	14,800	14,800

PAUSE AND CHECK 8.P1 In the stall's worksheet, what is the adjusted balance of Supplies, and on which side does it stand?

In one line. The adjusted trial balance combines columns 1 to 4 row by row, and its two totals must agree before we move on.

The fourth and fifth pairs are the statement columns. In step 4, every adjusted balance is copied to one column only, and the kind of account decides which. Revenue and expense accounts go to the income statement columns, which are 7 and 8. Asset, liability and owner's equity accounts go to the balance sheet columns, which are 9 and 10. Within each pair, a debit balance stays a debit and a credit balance stays a credit.

The table below shows the destination of each kind of account. Read a row from left to right: the kind of account, the side on

The combining rule: add figures on the same side, subtract figures on opposite sides.

The table shows columns 1 to 6 for the stall; read each row from left to right.

In one line: the adjusted trial balance is the trial balance after the changes, with equal totals.

GO SLOWLY HERE.

Step 4 sends each adjusted balance to exactly one statement column.

The table shows where each kind of account is entered.

TRIAL BALANCE AS AT 31 JANUARY 2026 WITH THE ERROR AND A SUSPENSE ACCOUNT

ACCOUNT	DEBIT (TK)	CREDIT (TK)
Cash	245,000	
Shelving	120,000	
Inventory	200,000	
Rent	60,000	
Wages	53,000	
Publisher Payable		120,000
Owner's Equity		500,000
Service Revenue		40,000
Suspense		18,000
Totals	678,000	678,000

Step 6 corrects the books with one journal entry.

Step 6: correct. Wages is too high by Tk 18,000, so Wages is credited by that amount. The suspense account is cleared by a debit of the same amount. The table that follows shows the journal entry.

THE CORRECTING ENTRY

DATE	PARTICULARS	FOLIO	DEBIT (TK)	CREDIT (TK)
31 January 2026	Suspense		18,000	
	Wages			18,000
	Narration: Corrected wages posted as 53,000 instead of 35,000, and cleared the suspense account.			

The answer and the check agree with the case file.

Common slips arise when the sign, the column or the unit is mixed up.

Answer. After posting, Wages = 53,000 - 18,000 = Tk 35,000, and the suspense account = 18,000 - 18,000 = nil. The totals of the correct trial balance are Tk 660,000 on each side. *Check.* Wages now equals the Tk 35,000 of the source document. The suspense account has no balance. The debit total and the credit total both equal the Tk 660,000 of section 6.1.

Common slips occur at this stage. A student may place the suspense balance on the wrong side, which doubles the difference instead of removing it. A student may correct the error by overwriting the figure in the Wages account, which hides the history. A student may forget to clear the suspense account, so that it stays in the books. Check the side, the entry and the final balance of Suspense, in that order.

* * *

SUMMARY

A trial balance lists all ledger balances in two columns and tests that the debit total equals the credit total. It is step 4 of the accounting

The chapter's main ideas in a few lines.

and the last columns show the result of the period before any statement is drafted.

Here is the idea in the smallest possible numbers. Supplies show Tk 1,000 in the trial balance, and Tk 600 of them have been used. On the worksheet we enter Tk 600 as a credit to Supplies, and we see at once that Tk 400 remain. Nothing in the ledger has changed. If we had made an arithmetic slip, we would correct it on the worksheet and begin again at no cost.

Different people value these jobs in different ways. The owner wants to know the result of the year early, before the statements are ready. The clerk wants to avoid rewriting, because a worksheet error is corrected on the sheet itself, while a ledger error needs a formal correction. A bank or a customer never sees the worksheet at all, because it is private working paper.

A worksheet is not a statement. This difference is easy to forget, because the worksheet holds the same figures that later appear in the statements. A financial statement is a formal report for outside readers. In this book we prepare two of them, and Chapter 9, 'Financial Statements and Closing Entries', teaches them fully. For now, here are short previews in plain words. The income statement lists the revenue and the expenses of a period. The balance sheet lists what the business owns and owes on one date. The worksheet only prepares the figures that these two statements need.

Because the worksheet is working paper, it changes no account. The ledger keeps its old balances until the adjusting entries are recorded in the journal and posted, as Chapter 7, section 7.4 'Recording Adjusting Entries' explains. A worksheet is also optional: a small business may omit it, and a computer program may perform the same task without showing it. Finally, a worksheet has limits. Equal totals show that debits equal credits, but they cannot show that an amount sits in the right account. A cost entered in the wrong account leaves the totals equal. A clerk must never hide a difference in a suspense account merely to make the totals agree.

Before moving on. A worksheet is a working paper with columns. It gathers balances, tests the adjustments and checks the arithmetic, and it is private. It is not a statement, and it changes no ledger account. Equal totals do not prove that every amount is in the right account. *You should now be able to say* why an accountant uses a worksheet before the statements, and why it changes no record.

8.2 BUILDING THE TEN COLUMNS

We now build a worksheet one pair of columns at a time. To keep the arithmetic small, we use a repair stall that exists only for teaching; it is not part of the running case. At the end of its year, the stall's trial balance shows these debit balances: Cash Tk 5,000, Supplies Tk 1,000, Equipment Tk 6,000 and Wages Tk 2,000. It shows these credit balances: Capital Tk 8,000 and Repair Revenue Tk 6,000. Each side totals Tk 14,000. Three changes are needed. Supplies worth Tk 600 have been used, wages of Tk 300 are owed, and the equipment has lost Tk 500 of its value.

The same idea in numbers: a Tk 600 change is tested without touching the ledger.

Owner, manager, clerk and outsider value the worksheet in different ways.

GO SLOWLY HERE.

The worksheet is private working paper; it is not a financial statement.

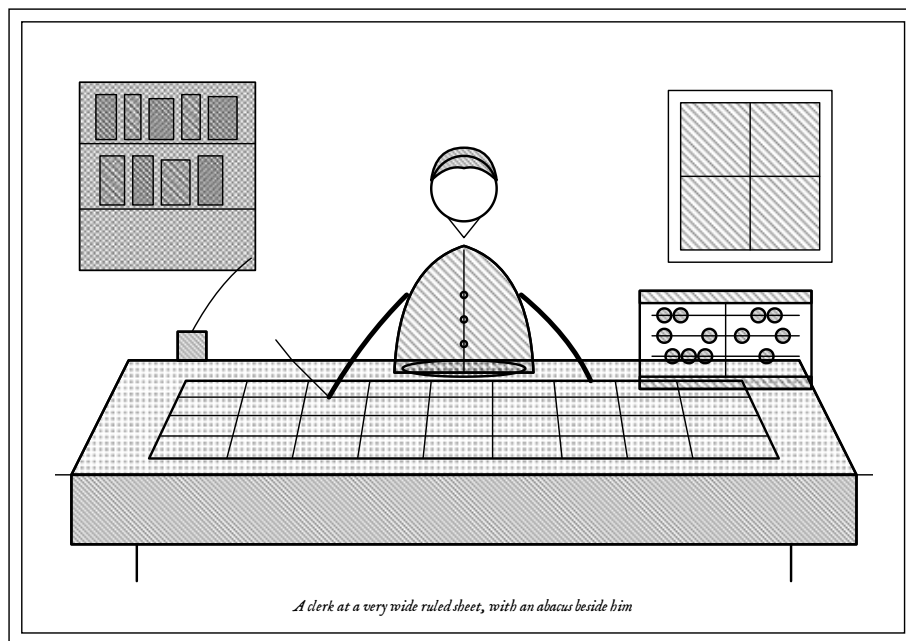
The worksheet changes no record, and it has limits.

The recap restates the three jobs, the private status and the limits.

A tiny repair stall gives clean numbers for learning the method.

6.4 The credit total of a trial balance is Tk 7,200 larger than the debit total. State the suspense account balance that would be added and on which side.

6.5 Rent of Tk 18,000 was posted to the debit side of the Rent account as Tk 81,000, and the credit to Cash was correct. Find the difference, say on which side the suspense balance goes, and write the correcting entry with its narration.



CHAPTER VIII

The Worksheet

How can an accountant test every change on paper before touching a single record?



A large ruled sheet carries figures from the trial balance through the adjustments to the statements. This chapter builds its ten columns one pair at a time, extends each balance, and teaches the checks that reveal a hidden mistake.

Paying for a year of use on one day raises the question of how much belongs to each month.

A business meets the same puzzle, and its records must choose one answer.

The records are true as a diary of events, but incomplete as a measure of the period.

Reports cover a fixed span of time, which must be named before anything can be measured.

GO SLOWLY HERE.

Period. Word in use. In daily life, a lesson at school or the full stop at the end of a sentence. In accounting, a span of time such as a month or a year.

Accounting period. The span of time, such as a month or year, for which results are measured and reported.

YOU WILL NEED

Chapter 6, 'The Trial Balance', section 6.1 'Preparing a Trial Balance'; Chapter 4, 'Accounts, Debits and Credits', section 4.3 'Debit and Credit'; and Appendix A1, 'Maths You Will Use', for percentages and simple division.

Imagine that you pay Tk 12,000 on 1 October for a bus pass that lasts twelve months. By 31 December you have used it for three months. A friend asks: "How much did the pass cost you this year?" Your wallet says Tk 12,000, because that is the amount that left it. Your journeys say Tk 3,000, because that is the amount you have used. Both answers are honest. They answer two different questions.

Now give the puzzle to a business. On 1 October 2026 the book-binding workshop Meghna Binding Service pays Tk 36,000 for twelve months of insurance. On 31 December 2026 its records still show the whole Tk 36,000 as paid. In the same records nothing appears for the wages of the last days of December, which the workers have earned but not yet received. The first item makes the year look dearer than it was. The second makes it look cheaper. A lender who read such records would form a wrong picture in both directions.

The records are not false. Each payment did happen, and each was written down correctly on its day. They are incomplete as a measure of one period of time. Closing that gap is the work of this chapter: a small set of corrections, made once at the end of each period, each needing only a few lines of arithmetic. Many students find this chapter difficult at first, because a ledger that balances looks finished. It is not finished, and you will soon see why.

* * *

7.1 WHY ADJUST

A school reports on your work term by term, not once for your whole life. A business reports in the same way. An **accounting period** is the span of time, such as a month or year, for which results are measured and reported. Meghna Binding Service uses the calendar year 2026, from 1 January to 31 December. Some businesses also report every month or every quarter. The shorter the period, the more often the records need correction, because more items fall across the edge of the period.¹

¹ Reporting periods of different lengths are taught in Chapter 20, 'The Whole Cycle and Reporting Periods', section 20.1 'Fiscal Year and Interim Periods'.

In 2002 the United States passed the Sarbanes-Oxley Act, which tightened the rules on company reporting and auditing. It applies to companies in the United States; other countries have their own laws. The story shows why correct and complete accounts matter: a report that leaves out what a business owes gives its readers a false picture.

A new United States law followed in 2002.

REVIEW QUESTIONS

7.1 On 1 March a workshop pays Tk 6,000 for six months of insurance. Find the insurance expense and the prepaid expense at 30 June of the same year.

7.2 The Supplies account shows Tk 52,500 before adjustment. A count on the last day of the year shows supplies on hand of Tk 18,200. Find the supplies expense and write the adjusting entry.

7.3 On 1 November a workshop receives Tk 18,000 for a school's binding work, to be done evenly over six months. Find the revenue earned and the unearned revenue at 31 December.

7.4 Equipment cost Tk 240,000, has a life of 8 years and no residual value. At year end wages of Tk 9,000 are owed, and work worth Tk 15,000 has been done but not billed. Write the three adjusting entries and state the effect on profit.

UNADJUSTED TRIAL BALANCE, MEGHNA BINDING SERVICE, 31 DECEMBER 2026 (TK)

ACCOUNT	DEBIT	CREDIT
Cash	150,000	
Accounts Receivable	90,000	
Supplies	48,000	
Prepaid Insurance	36,000	
Equipment	360,000	
Drawings	80,000	
Wages	330,000	
Rent	120,000	
Utilities	26,000	
Accounts Payable		60,000
Unearned Revenue		24,000
Capital		400,000
Service Revenue		756,000
Total	1,240,000	1,240,000

The uncorrected figures give a profit that is too high by Tk 96,000.

Different readers of the records are harmed in different ways by a wrong profit.

The accrual basis is the basis used here, though other bases exist.

The section's four ideas are the period, the accrual basis, matching, and the need to correct.

Case file, illustrative. Meghna Binding Service, 31 December 2026. Before correction, revenue is Tk 756,000 and the expenses are wages Tk 330,000, rent Tk 120,000 and utilities Tk 26,000, a total of Tk 476,000. The profit, meaning revenue minus expenses, appears to be Tk 280,000. After the six corrections of this chapter, revenue is Tk 788,000 and expenses are Tk 604,000, so profit is Tk 184,000. Chapter 8, 'The Worksheet', section 8.3 'Worked Worksheet', calls this amount net income.

Each reader of the records would be misled by the first figure. The owner might withdraw money in the belief that Tk 280,000 was earned. A lender might judge the business safer than it is. A tax authority would assess tax on the wrong amount. An employee might read the high profit as a reason to expect higher pay. Correct timing protects all of these people.

The accrual basis is not the only method in use. Some very small businesses, in some countries, keep records on the cash basis and may be allowed to report tax that way. Rules differ from country to country, and this book describes the accrual basis because it is the basis behind the statements studied here. Even on the accrual basis, judgement is needed: a very small amount may be left uncorrected if leaving it would mislead nobody.

Before moving on. An accounting period is the span of time being reported. The accrual basis places revenue and expenses in the period of the activity, and matching places each cost beside the revenue it helped to earn. Because ordinary entries follow cash and events, corrections are needed at period end. You should now be able to say why a ledger that balances can still give the wrong profit.

7.2 DEFERRALS

Before moving on. An adjusting entry is dated on the last day of the period, carries a narration and involves no cash. The workshop's six entries cover supplies, insurance, depreciation, advance fees, wages owed and unbilled work. You should now be able to write any of the six from its data.

WORKED EXAMPLE: FROM UNADJUSTED TO ADJUSTED BALANCES

Case file, illustrative. *Given.* The unadjusted trial balance of Meghna Binding Service at 31 December 2026 (see the table in the first section), totalling Tk 1,240,000 on each side, and the six adjustments: (a) supplies on hand Tk 15,000, so supplies expense Tk 33,000; (b) insurance for 12 months paid on 1 October 2026, Tk 3,000 a month, expired Tk 9,000; (c) equipment depreciation Tk 72,000 (5 years, no residual value); (d) unearned revenue earned Tk 12,000 (Tk 4,000 a month from 1 October); (e) wages owed Tk 14,000; (f) revenue earned but not billed Tk 20,000.

Find. The adjusted balance of every account that changes, the profit after adjustment, and a check that debits equal credits. *Formula.* Adjusted balance equals unadjusted balance plus or minus the adjustment. Profit equals revenue minus expenses. *Steps.* Apply each of the six entries to its two accounts, as in the next table. An asset or expense rises on a debit and falls on a credit; a liability or revenue rises on a credit and falls on a debit.

BALANCES OF THE ACCOUNTS CHANGED BY THE SIX ADJUSTMENTS (TK)

ACCOUNT	SIDE	BEFORE	CHANGE	AFTER
Accounts Receivable	Debit	90,000	+20,000	110,000
Supplies	Debit	48,000	-33,000	15,000
Prepaid Insurance	Debit	36,000	-9,000	27,000
Accumulated Depreciation	Credit	0	+72,000	72,000
Wages Payable	Credit	0	+14,000	14,000
Unearned Revenue	Credit	24,000	-12,000	12,000
Service Revenue	Credit	756,000	+32,000	788,000
Wages	Debit	330,000	+14,000	344,000
Supplies Expense	Debit	0	+33,000	33,000
Insurance Expense	Debit	0	+9,000	9,000
Depreciation Expense	Debit	0	+72,000	72,000

Service Revenue rises by $12,000 + 20,000 = 32,000$, from 756,000 to 788,000. The expenses are wages 344,000, rent 120,000, utilities 26,000, supplies expense 33,000, insurance expense 9,000 and depreciation expense 72,000, which total 604,000. The profit is therefore $788,000 - 604,000 = 184,000$.

Answer. After the six adjustments the profit for 2026 is Tk 184,000, not the Tk 280,000 that the unadjusted figures suggested. Tk 96,000 of the apparent profit had not really been earned. *Check.* The unchanged accounts are Cash 150,000, Equipment 360,000, Drawings 80,000, Rent 120,000 and Utilities 26,000 on

Six entries, one date and no cash complete the section.

The unadjusted balances and the six adjustments are the data, and the task is to find the new balances.

Find, formula and steps: apply each entry to its two accounts, and total the effect on profit.

The expenses and revenue after adjustment give the true profit of the year.

The answer and the two checks show that the books still balance.

Worked example: supplies used during 2026.

Money received in advance is a debt of work until the work is done.

Unearned revenue. Money received before work is done, held as a liability until the work is performed.

Worked example: the part of the advance that has been earned.

Equipment is paid for once but serves for years, so its cost is spread.

Worked example: one year of depreciation on the equipment.

Four corrections have been made; the pattern is to find the used part and move it.

Given. The Supplies account shows Tk 48,000 before adjustment; a count on 31 December 2026 values the supplies on hand at Tk 15,000. *Find.* The supplies used during 2026. *Formula.* In words: supplies used equal the supplies account balance minus supplies on hand. *Steps.* $48,000 - 15,000 = 33,000$. Entry: debit Supplies Expense 33,000; credit Supplies 33,000. *Answer.* Supplies used in 2026 are Tk 33,000, and the Supplies account is left at Tk 15,000. *Check.* $33,000 + 15,000 = 48,000$, the original balance.

Now turn the idea around. Suppose a school pays a magazine publisher for a year of issues in advance. The publisher has the money, but it still owes twelve issues. The money is not yet earned. For the workshop, **unearned revenue** is money received before work is done, held as a liability until the work is performed. On 1 October 2026 a school pays Tk 24,000 in advance for binding work; the trial balance shows this as a liability of Tk 24,000. The work is done evenly at Tk 4,000 a month from 1 October. By 31 December three months of the work are complete, so part of the liability has become revenue.

Given. Unearned Revenue of Tk 24,000; work worth Tk 4,000 a month, started on 1 October 2026; period ends on 31 December 2026. *Find.* The revenue earned in 2026 and the liability that remains. *Formula.* In words: revenue earned equals monthly work times months done; liability remaining equals the advance minus revenue earned. *Steps.* $4,000 \times 3 = 12,000$. $24,000 - 12,000 = 12,000$. Entry: debit Unearned Revenue 12,000; credit Service Revenue 12,000. *Answer.* Tk 12,000 is earned in 2026, and Tk 12,000 stays as unearned revenue. *Check.* The whole advance covers $24,000 \div 4,000 = 6$ months, and 3 months are done, so half remains.

Equipment follows the same spirit. The workshop paid Tk 360,000 for equipment that will serve for about five years. It would be unfair to charge the whole cost to the first year. The method that spreads the cost is called depreciation, taught in Chapter 16, 'Depreciation', section 16.1 'Why Depreciate'. For now, use the simplest rule: divide the cost, less any amount expected at the end, by the years of service. The charge is recorded by debiting Depreciation Expense and crediting an account called Accumulated Depreciation, explained in Chapter 16, section 16.4 'Recording and Disposal'.

Given. Equipment cost Tk 360,000; life 5 years; no residual value, which means nothing is expected to be received at the end. *Find.* The depreciation for 2026. *Formula.* In words: yearly charge equals cost divided by years of life. In symbols: $D = C \div N$. *Steps.* $D = 360,000 \div 5 = 72,000$. Entry: debit Depreciation Expense 72,000; credit Accumulated Depreciation 72,000. *Answer.* The depreciation for 2026 is Tk 72,000. *Check.* $72,000 \times 5 = 360,000$, the whole cost after five years.

Before moving on. A prepaid expense is an asset until used, and the used part moves to an expense. Supplies are counted, and the difference is the amount used. Unearned revenue is a liability until the work is done, and the earned part moves to revenue. Depreciation spreads the cost of equipment over its life. You

one count for each kind of fact. Notice that this is the two-stage method of this chapter: record each item on its own line or card, then total each kind once. In 1911 Hollerith's firm joined three other firms to form the Computing-Tabulating-Recording Company, which later became IBM.

A tabulating machine is not a special journal, but the two share one idea: group records of the same kind and total each group at once. In 1896 Hollerith incorporated the Tabulating Machine Company. In later decades, similar machines were adapted to bookkeeping, and the company became part of the business that grew into IBM. Mechanised records, like special journals, show that large numbers of similar transactions reward the effort of organising them.

The link with accounting is the idea of grouping like records and totalling them quickly.

REVIEW QUESTIONS

11.1 A tea stall makes three credit sales: Tk 100, Tk 200 and Tk 300. Enter them in a sales journal, total the amount column, and state which two main accounts receive the total and on which side.

11.2 A different bookshop makes these credit sales in December 2026: 2 December, invoice 801, Greenfield School, Tk 42,500; 9 December, invoice 802, Padma College, Tk 18,750; 15 December, invoice 803, Greenfield School, Tk 31,200. Its accounts receivable control account opens at Tk 150,000. Enter the sales, total them, and find the closing balance of the control account.

11.3 In one week a sales journal totals Tk 126,400. The control account opened at Tk 310,000. After posting, the customer accounts show balances of Tk 120,000, Tk 95,400, Tk 150,000 and Tk 61,000. Do the totals agree? If not, state the difference.

11.4 A shop records three credit purchases: Tk 36,000, Tk 54,500 and Tk 29,500. The accounts payable control account opened at Tk 200,000. Total the purchases journal, state the debit and credit posted, and find the closing balance of the control account.

The same method applies to suppliers. Meghna and Sons posts each line of the purchases journal to the supplier named on it: Tk 150,000 to Ananda Publishers, Tk 96,000 to Prothom Print and Tk 54,000 to Jonaki Books. These are credits, because they increase debts owed. The credit side of the accounts payable control account receives the total of Tk 300,000. The credits in the supplier accounts also total Tk 300,000, since $Tk\ 150,000 + Tk\ 96,000 = Tk\ 246,000$, and $Tk\ 246,000 + Tk\ 54,000 = Tk\ 300,000$. Only the new purchases are counted here, because the case gives no opening balance for suppliers.

Before moving on. A subsidiary ledger holds individual accounts. A control account in the main ledger sums them. To agree the totals, find the closing balance of the control account and compare it with the sum of the individual balances. You should now be able to say what a difference between the two totals tells you.

* * *

WORKED CASE: RECORDING AND AGREEING ONE MONTH

Situation. It is 30 November 2026. Meghna and Sons has recorded its credit sales on invoices 701, 702 and 703, with a total of Tk 269,000, and its credit purchases from three suppliers, with a total of Tk 300,000. The accounts receivable control account opened the month at Tk 280,000. **Question.** Which totals reach the main ledger, and do the control account and the customer accounts agree at month end?

Reasoning. First, total each journal. The sales journal total is $Tk\ 85,000 + Tk\ 120,000 + Tk\ 64,000$, which is Tk 269,000. The purchases journal total is $Tk\ 150,000 + Tk\ 96,000 + Tk\ 54,000$, which is Tk 300,000. Second, post each total once. For sales, debit the accounts receivable control account Tk 269,000 and credit Sales Tk 269,000. For purchases, debit Purchases Tk 300,000 and credit the accounts payable control account Tk 300,000. Third, post each line to the customer or supplier account named on it. Fourth, compare the control account with the sum of the customer accounts.

Conclusion. The accounts receivable control account is $Tk\ 280,000 + Tk\ 269,000$, which is Tk 549,000. The three customer accounts have received debits of Tk 85,000, Tk 120,000 and Tk 64,000, and, as assumed, the brought-forward customer balances total Tk 280,000. Their sum is also Tk 549,000. The totals agree. **Check.** Debits and credits posted from the two journals are Tk 569,000 each, as $Tk\ 269,000$ plus $Tk\ 300,000$ gives the same sum on both sides.

Posting the lines. The three customer accounts each receive a debit: Dhanmondi Girls' School Tk 85,000, Rose Valley School Tk 120,000 and Lalmatia College Tk 64,000. The three supplier accounts each receive a credit: Ananda Publishers Tk 150,000, Prothom Print Tk 96,000 and Jonaki Books Tk 54,000. Each posting names the journal that it came from, so any figure can be traced to its source document. This trail is the reason a business records a

The same method applies to suppliers, with the control account on the credit side.

Recap: one control account summarises one subsidiary ledger, and the two totals must agree.

This worked case joins the sales journal, the purchases journal and the control account in one month.

The reasoning starts with the totals and then moves to the check.

The conclusion is that the control account stands at Tk 549,000, matching the customer accounts.

Posting the lines to individual accounts is the step that makes later checking possible.

This chapter rests on three earlier pieces of work.

Recall: worksheet, Chapter 8, 'The Worksheet', section 8.3 'Worked Worksheet'.

Recall: deferrals, Chapter 7, 'Adjusting the Accounts', section 7.2 'Deferrals'.

A market stall owner must answer two different questions on the last evening of the month.

The earnings belong to the month that ended; the tin and the debts continue.

Meghna Binding Service meets the same puzzle at the end of its first year.

The chapter answers the puzzle in three parts, then applies them to the workshop.

The word Statement means a formal report of figures, not a spoken claim.

Statement. Word in use. In daily life, something said or written, such as a bank statement. In accounting, a formal report of figures for a period or for a date.

Here are the stall's small, round figures for the month.

You will need three earlier pieces of work. Chapter 8, 'The Worksheet', section 8.3 'Worked Worksheet', showed how the adjusted figures are collected on one sheet. Chapter 7, 'Adjusting the Accounts', section 7.2 'Deferrals', explained the entries that move amounts between periods. Appendix A1, 'Maths You Will Use', holds the arithmetic.

A woman keeps a tea stall in a market. On the last evening of the month she empties the cash tin onto the table. She counts the takings, remembers what she paid for leaves, milk and sugar, and keeps apart the money she took home. Then she faces a question that every business must answer sooner or later. How much did the stall earn this month? And what does the stall own and owe tonight?

These are two different questions, and their answers behave differently. The earnings belong to the month that has ended. The cash tin, the debts and the owner's stake continue into the next month. On the first morning of the new month, the earnings must start again at nothing, while the cash tin must keep what it holds. How can one set of records do both?

Case file, illustrative. Meghna Binding Service, the separate set of books for the book-binding workshop that began on 1 January 2026, ends its first year on 31 December 2026. Its Revenue account then holds Tk 788,000. If nothing is done, the first sale of 2027 would be added to that balance, and the account could never show what the workshop earned in one year alone.

This chapter answers the puzzle in three parts. Section 9.1 teaches the three reports that present the results of a period. Section 9.2 teaches the closing entries, which reset the accounts that measure one period. Section 9.3 teaches reversing entries, a step that makes the first payment of a new period easier to record. A worked example with Meghna Binding Service follows, and the chapter ends with the history of the law that first required companies to publish a balance sheet.

* * *

9.1 THE THREE STATEMENTS

Return to the tea stall. At the end of the month the owner wants three answers. What did the stall earn after paying its costs? How did her own stake in the stall change? What does the stall own and owe tonight? Each answer needs a separate page, because each answers a different question. Accountants call each such page a statement, and this section teaches the three pages in the order in which they are prepared.

Here are her figures. She sold tea for Tk 5,000. She spent Tk 3,000 on leaves, milk, sugar and fuel. She took Tk 500 home for her family. When she started the stall, she had put in Tk 2,500. At the month's end the stall holds cash of Tk 3,700 and equipment worth Tk 1,300, and it owes its supplier Tk 1,000. We will use these figures for every statement in this section.

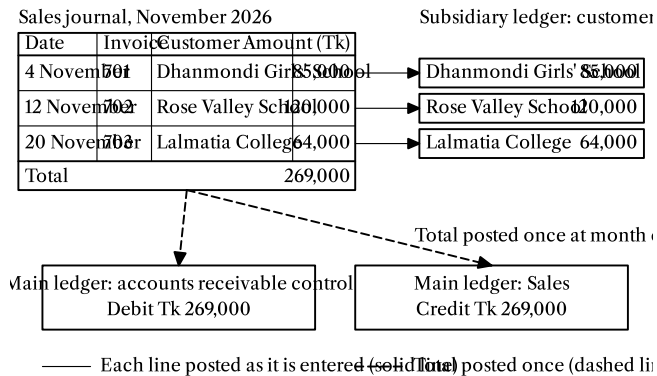


FIG. 11.1. Sales journal columns and posting to ledgers

The **purchases journal** is a special journal that records only purchases of goods on credit. Its source document is the invoice received from the supplier. The columns are the date, the name of the supplier and the amount. In November 2026, Meghna and Sons bought on credit as the next case paragraph records.

Case file, illustrative. On 3 November 2026, Meghna and Sons bought goods on credit from Ananda Publishers for Tk 150,000. On 9 November it bought goods on credit from Prothom Print for Tk 96,000. On 17 November it bought goods on credit from Jonaki Books for Tk 54,000. The total of credit purchases for the month is Tk 300,000.

The posting rule is the same in reverse. Each line is posted to the account of the supplier named on it. The total of Tk 300,000 is posted once to two main accounts: a debit to Purchases, an account that collects the cost of goods bought for resale, and a credit to the account payable. The debit and credit follow the rules of Chapter 4 'Accounts, Debits and Credits': a purchase of goods is a debit, and an increase in a debt owed is a credit. Debits and credits for the month are equal, at Tk 300,000 each.

Exceptions apply here too. A purchase paid in cash does not belong in the purchases journal, because no debt to a supplier arises. A purchase of equipment on credit, such as a new shelving unit, does not belong there either, because the business buys it to use and not to resell. Both go elsewhere: the first in the cash book of Chapter 12 'The Cash Book and Bank Reconciliation', the second in the general journal. A clerk therefore asks two questions of each invoice received. Was it bought on credit? Was it goods for resale? Only two answers of yes place it in the purchases journal.

Each posting should carry a reference. The journal line records the account to which it was posted, and the account records the journal and the date from which the figure came. This two-way trail costs a few seconds for each posting. It repays that time whenever a customer queries a bill or an auditor who is a person

The purchases journal is the matching book for goods bought on credit from suppliers.

Purchases journal. A special journal that records only purchases of goods on credit.

The case file lists three credit purchases, which total Tk 300,000.

Posting the totals gives each main account one entry for the month, and each supplier's account one entry per line.

Only credit purchases of goods for resale belong in the purchases journal.

A posting reference lets anyone trace a figure back to its source.

The balance sheet shows one date, so its heading says at, not for.

The three statements are prepared in a fixed order because each feeds the next.

Notice the difference. The balance sheet shows a single date, so its third line reads *At 31 December 2026*. The other two statements cover a period.

The order matters, because each statement supplies a number to the next. The income statement is prepared first, because it produces net income. The statement of owner's equity comes second, because it needs net income and produces ending capital. The balance sheet comes last, because it needs ending capital. The figure below shows the links. Each box is one statement. Each arrow shows a number moving from one box to the next, and the words on the arrow name that number.

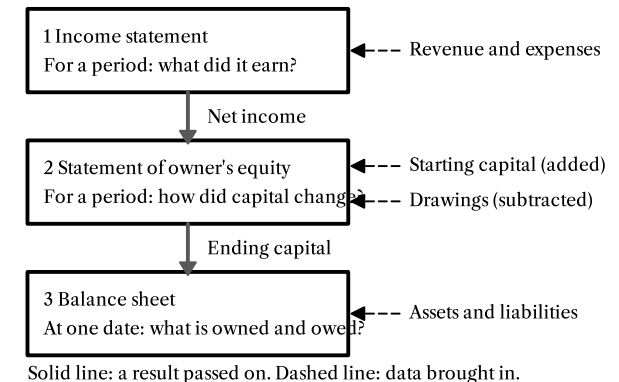


FIG. 9.1. The three statements and the numbers that link them

The owner reads the income statement to judge the work. A bank that lends money reads the balance sheet to see whether assets cover liabilities.

Before moving on. The income statement reports revenue, expenses and net income for a period. The statement of owner's equity reports how capital changed, using net income and drawings. The balance sheet reports assets, liabilities and owner's equity at one date. Prepare them in that order, and give each the right third line of heading.

* * *

9.2 CLOSING THE BOOKS

Look again at the tea stall's books after the month's results are known. Its Revenue account holds Tk 5,000, its Expenses account holds Tk 3,000, and its Drawings account holds Tk 500. Next month the owner wants the results of the new month only. If the old figures stay, each new sale is added to Tk 5,000, and no one can separate the two months. The books must be reset, but not every account may be reset. The Cash account must keep its Tk 3,700. Which accounts may be emptied, and which must not?

Owners, lenders and managers read the three statements for different reasons.

Before moving on: each statement answers one question, and each feeds the next.

Some accounts must reset to zero at period end and others must not.

The first limit is repetition. Each credit sale needs its own entry, and each entry names the same two accounts. The second limit is that only one person can write in one book at a time, so a second clerk must wait. The third limit is posting. Each entry is posted separately to two ledger accounts, so the number of postings is twice the number of entries. Errors also grow, and an error among hundreds of similar entries is hard to find.

The table compares the work in the general journal with the work in a book kept for credit sales alone. The first row counts the ledger postings to the two main accounts, the receivable account and the sales account. A row is read from left to right: the task, the work for three sales, then the work for two hundred sales. For the general journal the count is twice the number of sales, because each sale touches two accounts. For a book of credit sales alone the count is two, because only the total is posted.

POSTINGS TO THE TWO MAIN ACCOUNTS WHEN SALES ARE RECORDED ON CREDIT

METHOD	THREE SALES	TWO HUNDRED SALES
General journal, posting every entry	6 postings	400 postings
Book for credit sales, posting the total	2 postings	2 postings

A **special journal** is a journal used for only one kind of repeated transaction, such as credit sales. It names the two accounts once, in its heading, and records only the changing parts on each line. Return to the tea stall. Its three credit sales of Tk 100, Tk 200 and Tk 300 take three short lines in a special journal. In the general journal they would take three full entries, each with two account names and an explanation.

Consider the benefits from several viewpoints. The owner receives records sooner and at lower cost. A manager can give the book of credit sales to one clerk and the book of credit purchases to another, and both clerks work at the same time. An employee writes fewer words and makes fewer slips. A customer, who may query a bill, is helped because every sale sits on a line that is easy to find. The plain rule is this: when many transactions of one kind occur, give them a book of their own.

Special journals also have exceptions. A special journal for credit sales does not record cash sales, because no account receivable arises when the customer pays at once; Chapter 12 'The Cash Book and Bank Reconciliation' describes the book used for cash. It does not record the sale of a delivery van, because a van is not goods held for sale. Adjusting and correcting entries also stay in the general journal. Finally, a very small business may have too few transactions to justify extra books, and one journal is then enough.

Before moving on. A special journal is a book for one repeated kind of transaction. It saves writing, shares work and reduces postings. The general journal remains for everything that does not fit a special journal. You should now be able to say why two hundred credit sales need two postings in a special journal and four hun-

The general journal has three limits, and each one grows with the number of transactions.

A small count shows how quickly the work grows, so read the table row by row.

A special journal groups transactions of one kind in one book, with columns for the parts that change.

Special journal. A journal used for only one kind of repeated transaction, such as credit sales, to speed up recording.

Special journals give four benefits: less writing, shared work, fewer postings and easier checking.

A special journal has limits, because it accepts only its own kind of transaction.

Recap: one book per repeated kind of transaction, with the rest staying in the general journal.

THE FOUR CLOSING ENTRIES OF THE TEA STALL

STEP	ACCOUNT	DEBIT (TK)	CREDIT (TK)
1	Revenue	5,000	
1	Income Summary		5,000
2	Income Summary	3,000	
2	Expenses (all accounts)		3,000
3	Income Summary	2,000	
3	Owner's Capital		2,000
4	Owner's Capital	500	
4	Drawings		500

After Step 2, the income summary holds the net result: credits of revenue less debits of expenses.

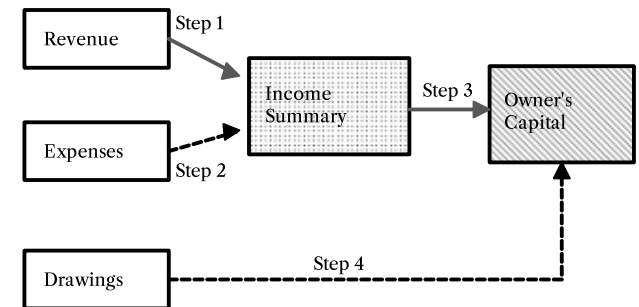
Follow the income summary through the steps. After step 1 it holds a credit of Tk 5,000. After step 2 it also holds a debit of Tk 3,000, so its balance is a credit of Tk 2,000. That balance is the net income. Step 3 debits Income Summary for Tk 2,000 and moves the amount to capital, and the account is left at zero. The rule is: a credit balance is closed by a debit of the same amount, and a debit balance is closed by a credit of the same amount.

PAUSE AND CHECK 9.P1 A shop has revenue of Tk 800 and expenses of Tk 500. After steps 1 and 2, what balance does Income Summary hold, and is it a debit or a credit?

The figure shows the four closing entries as arrows that end in Owner's Capital.

The figure below shows the same four entries as arrows. Revenue, Expenses and Drawings sit on the left. Income Summary sits in the middle. Owner's Capital sits on the right, and every path ends there. Each arrow is labelled with its step number. Where an arrow points shows where a balance moves; the line style tells the kind of balance: a solid line for a credit and a dashed line for a debit.

Revenue has a credit balance; expenses and drawings have debit balances.

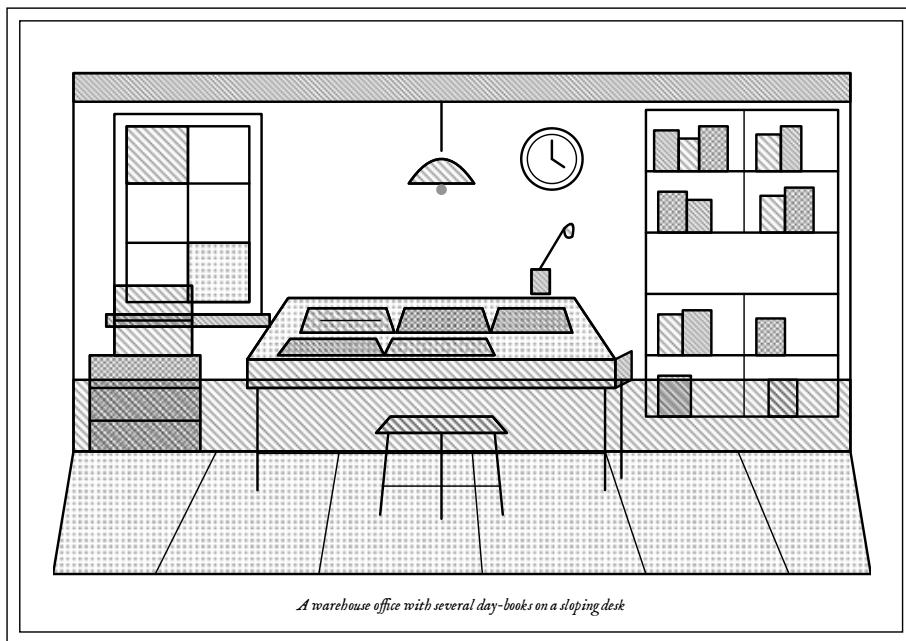


Solid line: credit balance closed. Dashed line: debit balance closed.

FIG. 9.2. Closing entries as arrows into capital

Three points often cause confusion. First, closing does not erase history. The journal keeps every entry, and the income statement keeps every figure. Second, drawings are not an expense. They do not pass through the income summary, because they are the owner's withdrawal and not a cost of earning revenue. Third, some

Closing never erases records; it only resets the accounts that measure one period.



CHAPTER XI

Special Journals

How can a shop record hundreds of credit sales without writing the same words hundreds of times?



This chapter shows how a business with many repeated transactions saves time and avoids error. You will meet the sales journal and the purchases journal, learn how their totals reach the ledger, and see how a control account checks the individual accounts of customers.

After both entries, the expense account holds only the wages that belong to 2027.

Reverse accruals that routine payments will settle; most deferrals need no reversal.

Reversing is a convenience, not a requirement, and the result is the same either way.

The reversal changes nothing in the old period's statements and corrects the new period's expense.

Before moving on: reverse accruals on the first day of the new period; the payment then follows the usual routine.

Here is the data for the workshop's first year, used in every step below.

REVERSAL AND PAYMENT OF WAGES, MEGHNA BINDING SERVICE, 2027

DATE	ACCOUNT	DEBIT (TK)	CREDIT (TK)
1 January 2027	Wages Payable	14,000	
1 January 2027	Wages Expense		14,000
5 January 2027	Wages Expense	28,000	
5 January 2027	Cash		28,000

Now examine the accounts. Wages Payable holds a credit of Tk 14,000 from December and a debit of Tk 14,000 from the reversal, so it stands at zero. Wages Expense holds a credit of Tk 14,000 from the reversal and a debit of Tk 28,000 from the payment, so its balance is a debit of Tk 14,000. That is exactly the wages for work done in 2027. The old debt is gone, the new expense is correct, and the payment was recorded without any splitting.

The next question is when to reverse. The usual answer is: reverse an accrual that a later routine payment or receipt will settle. Wages payable is the classic case. Interest owed or interest earned but not yet received is another. Deferrals are different. In the deferral entries of Chapter 7, section 7.2, the original payment or receipt was first recorded in an asset or liability account, such as Prepaid Insurance or Unearned Revenue. The adjusting entry already left each balance correct, so a reversal is not needed.

Reversing is a convenience and never a requirement. A bookkeeper who splits the payment correctly by hand reaches the same balances. Two rules keep the method safe. Date the reversal on the first day of the new period, after the old period has been closed and its statements prepared. And never reverse a closing entry, because closing entries must stay in place for the period that ended.

The effect on the next period is easy to state. The balance sheet at 31 December 2026 still shows wages payable of Tk 14,000, because the reversal is dated 2027. The income statement for 2027 shows only the wages that belong to 2027. If the bookkeeper forgets the reversal and records the full payment, the 2027 wages are overstated by Tk 14,000.

Before moving on. A reversing entry cancels an earlier accrual on the first day of a new period. It is used for accruals that routine payments will settle, and it is optional. After the reversal, the full payment is recorded in the usual way, and the expense account shows only the new period's share.

* * *

A WORKED EXAMPLE: STATEMENTS AND CLOSING FOR MEGHNA BINDING SERVICE

Case file, illustrative. Meghna Binding Service, statements for 2026. Revenue Tk 788,000. Expenses Tk 604,000. Capital at 1 January 2026 Tk 400,000. Drawings for the year Tk 80,000. Assets at 31 December 2026: cash Tk 150,000, accounts receivable Tk 110,000,

+ Tk 600,000 = Tk 4,860,000. Liabilities: Tk 1,270,000 + Tk 480,000 = Tk 1,750,000; with capital, Tk 1,750,000 + Tk 3,110,000 = Tk 4,860,000. *Answer:* ending capital is Tk 3,110,000, and both sides total Tk 4,860,000. *Check:* Tk 4,860,000 – Tk 1,750,000 = Tk 3,110,000.

Now change one figure to see how the method responds. Suppose purchase returns had been Tk 300,000 instead of Tk 200,000. Then cost of goods sold would be Tk 3,300,000 + Tk 10,200,000 – Tk 300,000 – Tk 3,500,000 = Tk 9,700,000, and gross profit would be Tk 14,400,000 – Tk 9,700,000 = Tk 4,700,000. Profit would rise by Tk 100,000 to Tk 850,000, and capital would rise by the same amount. A larger return to suppliers lowers the cost of goods sold.

Three slips are common. The first is forgetting the closing inventory, which would give a cost of goods sold of Tk 13,300,000 and a gross profit of only Tk 1,100,000. The second is placing the loan due within one year among the non-current items. The third is adding drawings to capital instead of subtracting them.

* * *

SUMMARY

A merchandising business records sales, returns, purchases and inventory. Net sales are sales less returns and allowances. The cost of goods sold is opening inventory plus purchases less purchase returns less closing inventory. Gross profit is net sales less cost of goods sold, and profit is gross profit less operating expenses. The balance sheet separates current from non-current items. Four kinds of error are corrected by new entries.

THINK IT THROUGH

10.T1 Explain why a compensating error does not stop the trial balance from agreeing, and give an example of your own.

10.T2 A bookshop forgets to count its closing inventory of Tk 2,000 when it calculates its cost of goods sold. What happens to the cost of goods sold, to gross profit and to the balance sheet?

WHERE PEOPLE GO WRONG

Some students treat everything bought as the cost of goods sold. Only the goods that were sold count. Goods that remain are inventory, an asset, until they are sold.

Some students start from sales and forget the returns. Net sales are always smaller than sales when goods come back or prices are reduced.

Some students place a whole long loan below the current items. The instalment due within one year is current, and only the rest is non-current.

Some students stop checking when the trial balance agrees.

Changing one figure shows how the statements respond.

Three slips are common in these calculations.

The summary gathers the chapter in a few lines.

Mistake one: treating all purchases as the cost of goods sold.

Mistake two: ignoring returns and allowances.

Mistake three: treating a long loan as wholly non-current.

Mistake four: believing that an agreeing trial balance

CLOSING ENTRIES OF MEGHNA BINDING SERVICE, 31 DECEMBER 2026

STEP	ACCOUNT	DEBIT (TK)	CREDIT (TK)
1	Revenue	788,000	
1	Income Summary		788,000
2	Income Summary	604,000	
2	Expenses (all accounts)		604,000
3	Income Summary	184,000	
3	Owner's Capital		184,000
4	Owner's Capital	80,000	
4	Drawings		80,000

Check the closing. After step 1, Income Summary holds a credit of Tk 788,000. After step 2 it holds Tk 788,000 – Tk 604,000 = a credit of Tk 184,000. Step 3 removes it, so the account is at zero. Capital is Tk 400,000 + Tk 184,000 = Tk 584,000 after step 3 and Tk 584,000 – Tk 80,000 = Tk 504,000 after step 4. This equals the balance sheet figure. On 1 January 2027, the reversal and the payment of 5 January from section 9.3 leave Wages Expense with a debit of Tk 14,000, the share that belongs to 2027.

Four slips are common. The first is a wrong sign: adding drawings to capital instead of subtracting them. The second is a mixed-up period: heading the balance sheet *For the year ended* instead of *At*. The third is a forgotten liability, such as Tk 12,000 of unearned revenue, which makes the two sides disagree. The fourth is closing Drawings through the income summary, which understates net income in the report.

* * *

SUMMARY

The income statement, the statement of owner's equity and the balance sheet are prepared in that order, each feeding the next. After they are prepared, closing entries move the balances of temporary accounts into Owner's Capital through the income summary, and permanent accounts carry their balances forward. On the first day of the new period, a reversing entry cancels an accrual so that the routine payment is recorded in the usual way. For Meghna Binding Service, the year produced net income of Tk 184,000 and ending capital of Tk 504,000.

THINK IT THROUGH

9.T1 Suppose Meghna Binding Service did not close its Revenue account on 31 December 2026, and then recorded Tk 900,000 of revenue during 2027. What balance would the account show at the end of 2027, and why would that figure mislead a reader?

9.T2 Explain in your own words why the balance sheet cannot be prepared before the statement of owner's equity.

500,000. The three statements of Chapter 9 therefore link here to the classified layout.

PAUSE AND CHECK 10.P2 A loan of Tk 90,000 is repayable in four equal yearly instalments of Tk 22,500, the first due in six months. Which amount is a current liability?

Different readers use these classes. A bank that lends money to Meghna and Sons compares current assets of Tk 4,260,000 with current liabilities of Tk 1,270,000. The current assets are about three times as large, which suggests that the shop can pay its debts of the coming year. The owner reads the same figures to decide how much to withdraw. One limit must be remembered: inventory is a current asset, but it becomes cash only if customers buy it.

Before moving on. A current asset is turned into cash or used up within one year. A current liability must be paid within one year. Items beyond one year are non-current, and owner's equity forms its own group.

* * *

10.4 CORRECTION OF ERRORS

The trial balance is the list that tests whether total debits equal total credits. Many students believe that when it agrees, the books must be right. They are not always right. Some mistakes leave the totals equal. Accountants name four kinds, and each has its own correction.

An **error of omission** is a mistake in which a transaction is left out of the books completely. A credit sale of Tk 800 that is never entered is an example. Neither side is recorded, so the trial balance still agrees.

An **error of commission** is a mistake in which an entry is made in the right class of account but the wrong account, or with a wrong amount. Tk 700 paid for electricity and debited to Telephone Expense is an example. Both are expenses, so the class is right and the account is wrong.

An **error of principle** is a mistake in which an entry is made in the wrong class of account, such as an expense recorded as an asset. Tk 40,000 spent on a delivery van and debited to Repairs Expense is an example. A van is an asset, and the entry placed it among the expenses.

A **compensating error** is two or more mistakes that cancel each other so that the trial balance still agrees. Suppose Wages is debited Tk 300 too much and Rent is debited Tk 300 too little. The debit total is correct, though both accounts are wrong.

A lender compares what is due soon with what can be turned into cash soon.

Before moving on: current items fall within a year; non-current items last longer.

An agreeing trial balance does not prove that the books are free of error.

An error of omission leaves a transaction out of the books completely.

Error of omission. A mistake in which a transaction is left out of the books completely.

An error of commission uses the wrong account of the right class, or a wrong amount.

Error of commission. A mistake in which an entry is made in the right class of account but the wrong account, or with a wrong amount.

An error of principle puts an entry in the wrong class of account.

Error of principle. A mistake in which an entry is made in the wrong class of account, such as an expense recorded as an asset.

A compensating error is two mistakes that cancel each other in the trial balance.

Compensating error. Two or more mistakes that cancel each other so that the trial balance still agrees.

Percentages make a comparison possible. The stationery seller's gross profit is Tk 3,300 on net sales of Tk 9,300, which is about 35 per cent. A shop with a higher percentage keeps more from each taka of sales. Appendix A1 shows how to calculate a percentage. Two shops can have the same gross profit and different profit, because one of them spends more on running costs. That is the reason the statement separates the two classes.

The next figure shows the layout. The six boxes are the lines of the statement from top to bottom. The two heavy boxes are the results: gross profit after the first three lines, and profit at the foot. The brackets on the right show the two parts: the goods, and the running costs.

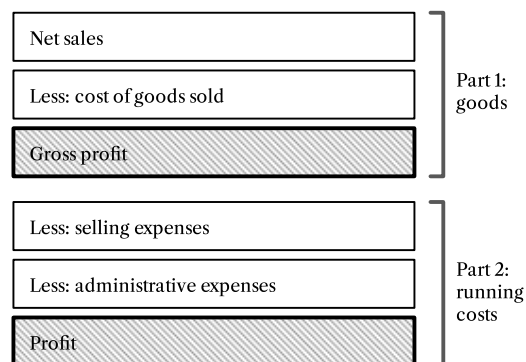


FIG. 10.2. The layout of a classified income statement

Case file, illustrative. The table below shows the 2025 income statement of Meghna and Sons. Read it from top to bottom: net sales, the build-up of the cost of goods sold, gross profit, the two groups of operating expenses, and profit.

CLASSIFIED INCOME STATEMENT OF MEGHNA AND SONS, YEAR ENDED 31 DECEMBER 2025 (TK)

ITEM	TK
Net sales	14,400,000
Opening inventory	3,300,000
Add: purchases	10,200,000
Less: purchase returns	(200,000)
Less: closing inventory	(3,500,000)
Cost of goods sold	9,800,000
Gross profit	4,600,000
Selling expenses	1,990,000
Administrative expenses	1,860,000
Total operating expenses	3,850,000
Profit	750,000

Before moving on: Gross profit is net sales less cost of goods sold

A percentage of net sales lets the reader compare gross profit across years or shops.

The figure shows the order of the lines in a classified income statement.

Meghna and Sons' 2025 Statement puts its figures in this layout.

Before moving on: the

This chapter builds on two sections of Chapter 9.

Recall: the three statements, Chapter 9, 'Financial Statements and Closing Entries', section 9.1 'The Three Statements'.

Recall: closing the books, Chapter 9, 'Financial Statements and Closing Entries', section 9.2 'Closing the Books'.

A bookseller who still holds unsold books cannot find profit by subtracting all purchases.

A business that buys goods to sell again must separate goods sold from goods still held.

The running case shows the same gap between the goods bought and the cost of the goods sold.

The chapter has four sections and one worked example.

The word sales means revenue from goods sold, not a price reduction event.

Sales. Word in use. In daily life, the act of selling, or a time when prices are lowered. In accounting, the revenue earned from goods sold to customers.

A sales return is goods sent back; an allowance is a price reduction for a faulty item.

Net sales is the revenue the

You will need two sections of Chapter 9, 'Financial Statements and Closing Entries': section 9.1 'The Three Statements' and section 9.2 'Closing the Books'. You will also use Appendix A1, 'Maths You Will Use', for percentages.

A bookseller buys 100 books at Tk 100 each and sells 80 of them at Tk 150 each. What profit did the bookseller make? A hasty answer subtracts the cost of all 100 books from the takings: Tk 12,000 – Tk 10,000 = Tk 2,000. That answer is wrong, and it is wrong in a useful way. Twenty books are still on the shelf. They have not been lost or used, and they are still worth Tk 2,000 to the bookseller. Only the cost of the 80 books that were sold, Tk 8,000, belongs against the takings, so the profit is Tk 4,000.

A business that buys goods in order to sell them again is called a merchandising business. Its main difficulty is the one in the puzzle: at any date, some of the goods bought have been sold, and some have not. The accounts of such a business must separate the two.

Case file, illustrative. Meghna and Sons, the family bookshop in Dhanmondi, Dhaka, bought books costing Tk 10,200,000 during 2025, before returns. Yet its cost of goods sold for 2025 is Tk 9,800,000. The difference is explained by opening inventory, purchase returns and closing inventory, as this chapter shows.

This chapter has four sections. Section 10.1 teaches sales, purchases, returns and the cost of goods sold. Section 10.2 arranges these in a classified income statement. Section 10.3 arranges the balance sheet in the same way. Section 10.4 teaches the kinds of error that can occur in the books and how each is corrected. A worked example with Meghna and Sons follows.

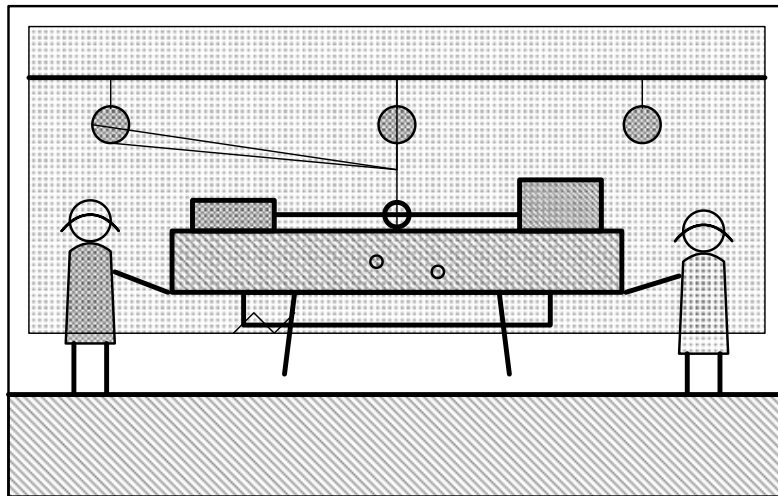
* * *

10.1 SALES, PURCHASES AND RETURNS

Think of a stationery seller with a small stall. In one month she sells goods for Tk 10,000. That figure is her sales. Then a customer returns notebooks worth Tk 500 because the pages are loose, and she gives another customer a price reduction of Tk 200 because a pen is scratched. Does she still have Tk 10,000 of sales? She does not, because she did not keep all of that money. Each of these cases has a name, and the accounts record them separately.

A sales return is goods that a customer sends back, and the business refunds or cancels the amount. A sales allowance is a reduction in the price that a customer agrees to pay because the goods are faulty. Both are recorded in their own accounts, and not by changing the original sale. The business can then see how often goods come back.

The amount that remains is called net sales. Net sales means



A machine shop of the iron age, with lathe, leather belt and workmen

CHAPTER XV

Plant Assets

Two bills of the same size, one for a new machine and one for a repair: why do they live in different parts of the accounts?



This chapter explains the long-lived assets a business uses to do its work, such as machines and buildings. You learn what counts as the cost of such an asset, which payments belong in that cost and which do not, and how to tell spending that builds an asset from spending that only keeps it working. The rules decide how large the profit and the balance sheet appear.

* * *

SUMMARY

Goods bought at different prices form layers, and a cost flow assumption decides which layers are treated as sold. First-in first-out sells the oldest costs first. Last-in first-out sells the newest costs first. Weighted average cost gives every unit the same average cost. For Meghna and Sons, with rising prices, gross profit was Tk 12,600, Tk 11,600 and Tk 12,180 under the three methods, and ending inventory was Tk 10,200, Tk 9,200 and Tk 9,780. In every method, cost of goods sold plus ending inventory equals the cost available, Tk 32,600.

The summary restates the whole chapter in a few lines.

THINK IT THROUGH

14.T1 Prices rise during the period. Rank the three methods from the highest to the lowest gross profit, and explain the ranking in a few sentences.

14.T2 Prices fall during the period. How does the ranking change? What stays the same for all three methods?

WHERE PEOPLE GO WRONG

Some students believe that FIFO describes which goods leave the shelf. It describes only the order of costs, and the goods may leave in any order. Others believe that LIFO always gives the lowest profit. That is true when prices rise, but when prices fall LIFO gives the highest profit. A third belief is that the weighted average is the plain average of the unit costs. It is the total cost divided by the total units, which gives larger layers more weight. A fourth belief is that a method can create profit. A method only moves cost between periods; the total cost over the life of the business does not change. Last, students sometimes change the method each year to suit the result. A business should use one method consistently, so that profits can be compared.

Five common misunderstandings are stated here, each with its correction.

IN HISTORY

The choice of cost flow assumption is a matter of law and standards as well as of arithmetic. In the United States, the Revenue Act of 1939 allowed LIFO for income tax. A rule known as LIFO conformity requires a business that uses LIFO for its tax return to use it in its financial reports as well.² In a time of rising prices, LIFO gives a higher cost of goods sold, a lower profit and a lower tax bill, which explains why many American businesses chose it.

Other countries follow the standards of the International Accounting Standards Board, a body based in London that issues International Financial Reporting Standards. Its International
² The Revenue Act of 1939 and the conformity rule are described in articles in The Tax Adviser, a journal of the American Institute of Certified Public Accountants.

The United States allowed LIFO for tax in 1939, while international standards now bar it.

Under international rules the position differs, because International Accounting

A small puzzle: two careful people record the same money and reach different totals.

Neither person is careless, so the difference needs an explanation, not a search for blame.

The same difference appears in the running case, and the chapter will explain all of it.

Imagine that you keep a small notebook for your savings account. Every time you put money in, you write the amount. Every time you take money out, you write that too. At the end of the month you add everything up and your notebook says Tk 5,000. Then the bank sends you its own list of what happened in the account. Its final figure is Tk 5,700. The difference is Tk 700.

Who is wrong? Perhaps the bank made a mistake. Perhaps you did. Many students expect this answer, and it is usually false. In most months both records are correct. They differ because the bank and you learn about each event at different times, and because some events reach one record before the other. Your task is not to decide which record is right. Your task is to explain, item by item, how one total becomes the other. If you can do that, you have proved that both records are sound, and you have found the true balance as well.

Case file, illustrative. On 31 October 2026 the cash book of Meghna and Sons showed a bank balance of Tk 612,000. The bank statement for the same date showed a balance of Tk 655,000. The difference is Tk 43,000. The case file lists seven items that explain this difference; by the end of the chapter each one has a place.

YOU WILL NEED

This chapter uses the idea of a special journal from Chapter 11 'Special Journals', section 11.1 'Why Special Journals'. It also uses the ledger and posting from Chapter 5 'The Journal and the Ledger', section 5.3 'The Ledger and Posting'. The arithmetic is addition, subtraction and checking two totals against each other; Appendix A1 'Maths You Will Use' holds a short reference.

* * *

12.1 THE CASH BOOK

A business handles money in two places. Some money is in the drawer: notes and coins from customers, kept for small payments. The rest is in the bank account: cheques deposited, transfers received, and payments made by cheque. Think of your own purse and your own bank card. You know the amount in each, and you add them to know what you have. A business needs the same habit, kept with more care. It records both kinds of money in one book.

This book is called the **cash book**. A cash book is a book that serves as both journal and ledger for cash and bank receipts and payments. For example, when a customer pays Tk 500 in notes, you write the amount once in the cash book. Many students find this confusing, because they have learned that a journal and a ledger are two different books. Usually an entry is written in the journal first and posted to the ledger afterwards, which is two steps. For cash and bank, the cash book joins the two steps into one. This saves work, because a cash book is also the ledger account for Cash and the ledger account for Bank.

Every business receives and pays money in two ways, and it needs one book that records both.

The cash book is the first record and the ledger account for cash and bank, in one place.

Recall: journal, Chapter 5 'The Journal and the Ledger', section 5.3 'The Ledger and Posting'.

Cash book. A book that serves as both journal and ledger for cash and bank receipts and payments.

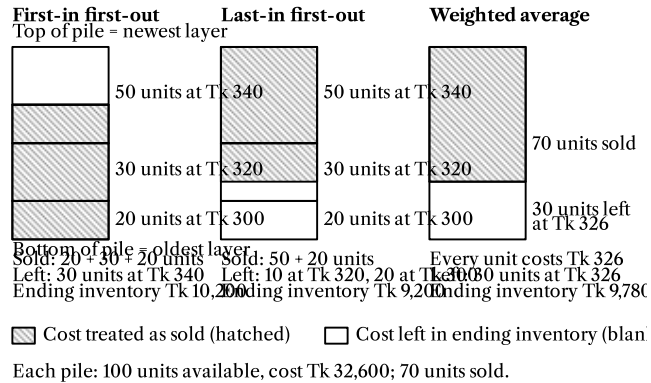


FIG. 14.1. Three methods as layers of cost leaving a pile

The table collects the results. Read each column down for one method, and each row across to compare the methods. In every column the cost of goods sold plus the ending inventory equals Tk 32,600. Notice also that the weighted average figures lie between the FIFO and LIFO figures. This is no accident, because an average of the costs must lie between the lowest and the highest.

RESULTS OF THE THREE METHODS FOR THE DICTIONARY SETS

MEASURE	FIRST-IN FIRST-OUT	LAST-IN FIRST-OUT	WEIGHTED AVERAGE
Cost of goods sold (Tk)	22,400	23,400	22,820
Ending inventory (Tk)	10,200	9,200	9,780
Gross profit (Tk)	12,600	11,600	12,180

The second figure is a bar chart of the results for Meghna and Sons. Each group of bars belongs to one method. The vertical axis shows taka in thousands, and the height of a bar shows the size of that measure. In each group the three bars are cost of goods sold, ending inventory and gross profit, in that order. The bars are marked by different hatching, so the chart reads in grey as well as in colour. Notice how little the bars differ between methods: the choice moves a small share of the cost.

Comparing the three methods shows that weighted average cost lies between the other two.

The second figure shows the results of the three methods side by side.

A bank statement is a list of the account's movements, not a claim that someone has made.

The principle: a difference arises only when one record learns of an event before the other.

A deposit in transit is a receipt the business has recorded but the bank has not.

Deposit in transit. Money recorded as deposited in the books but not yet shown on the bank statement.

An outstanding cheque is a payment the business has recorded but the bank has not yet paid.

Outstanding cheque. A cheque written and recorded but not yet paid by the bank.

Clear. Word in use. In daily life, easy to understand or free of obstacles. In banking, a cheque clears when the bank has paid it and moved the money out of the writer's account.

A credit memo is the bank telling the business that it added money to the account.

Credit memo. A bank's notice that it has added money to the account, for example a collection or interest.

A debit memo is the bank telling the business that it took money out of the account.

Debit memo. A bank's notice that it has deducted money, for example a service charge.

A bank statement lists each receipt and payment the bank has recorded, with a balance after each. It is a source document, because it is written evidence that an event took place. It is also the bank's version of the account, so it is the natural thing to compare with the cash book.

Here is the principle that explains every difference. Each event in the account is recorded by two parties. If both record it in the same period, the records agree. If one records it earlier, the records differ until the other catches up. We can therefore sort the differences into two groups. In the first group, the business has recorded the event and the bank has not yet done so. The bank will catch up without any action from us, so no entry is needed in our books. In the second group, the bank has recorded the event and the business has not. The business must make an entry to catch up. Keep these two groups in mind, because they decide what we do with each item.

The first item in group one is the **deposit in transit**. A deposit in transit is money recorded as deposited in the books but not yet shown on the bank statement. For example, a shop banks Tk 400 after the bank has closed on the last day of the month. The shop records the receipt at once. The bank records it on the next working day. The statement therefore omits it, and the bank balance looks lower than the cash book balance by Tk 400.

The second item in group one is the **outstanding cheque**. An outstanding cheque is a cheque written and recorded but not yet paid by the bank. For example, you write a cheque for Tk 900 to a supplier and record the payment the same day. The supplier needs a few days to deposit it, and the banks need more days to clear it. Until it clears, the bank statement still shows the money in your account, so the bank balance looks higher than the cash book balance by Tk 900.

Now we move to group two, the events that the bank records first. The first is the **credit memo**. A credit memo is a bank's notice that it has added money to the account, for example a collection or interest. Suppose the bank collects a Tk 300 payment on the business's behalf, or pays Tk 300 of interest. The bank adds the money and informs the business with a credit memo. Why is it called a credit? The bank looks at the account from its own side. The money in your account is money the bank owes you, a liability of the bank. When the bank owes you more, it records a credit. For you, the memo increases the balance.

The opposite item is the **debit memo**. A debit memo is a bank's notice that it has deducted money, for example a service charge. Suppose the bank charges Tk 100 for running the account. It deducts the money and sends a debit memo. The bank owes you less, so it records a debit. For you, the memo decreases the balance. Remember the pairing: a credit memo adds to your balance and a debit memo subtracts from it. The names follow the bank's

Under FIFO, ending inventory is Tk 10,200, cost of goods sold is Tk 22,400 and gross profit is Tk 12,600. *Check.* Directly: $20 \times 300 + 30 \times 320 + 20 \times 340 = 6,000 + 9,600 + 6,800 = \text{Tk } 22,400$.

PAUSE AND CHECK 14.P1 Suppose Meghna and Sons had sold only 60 sets. Using FIFO, find the ending inventory, the cost of goods sold, and check that they add to Tk 32,600.

Exceptions and limits need attention. FIFO is an assumption about costs, not a statement about which dictionary left the shelf. In our case the prices rise from Tk 300 to Tk 340. FIFO charges the cheap, old costs to the sale, so cost of goods sold is low and profit is high. The ending inventory carries the newest, dearest costs, so the asset in the balance sheet is close to what the goods cost now. When prices fall, the effects reverse. The worked case in this chapter shows this.

Before moving on. First-in first-out takes the oldest layers for cost of goods sold and keeps the newest layers as ending inventory. For the dictionary sets, it gives Tk 22,400 and Tk 10,200. You should now be able to say which layers remain after a sale under FIFO.

* * *

14.3 LAST-IN FIRST-OUT

Think of a stack of plates in a kitchen. A cook adds clean plates on top, and takes the top plate when one is needed. The plate placed last is taken first. The bottom plate may stay in the stack for months. A pile of coal or sand in a yard behaves in the same way, because the newest load lies on top and is taken first.

The **last-in first-out** method, often written LIFO, is a method that treats the newest costs as sold first, leaving the oldest costs in inventory. The method reverses the order of FIFO. In our pile, the newest layer is the second purchase, so its cost leaves first. Then the first purchase, and then the opening layer, if more units must be taken.

We apply the method to the same case. *Given.* The same layers and the same sale of 70 units for Tk 35,000. *Find.* The ending inventory, the cost of goods sold and the gross profit. *Formula.* The formulas are the same as for FIFO: $C = A - E$ and $GP = S - C$, but E is now built from the oldest layers. *Steps.* The newest 70 units are 50 from the second purchase and 20 from the first purchase. The 30 units left are 20 from the opening layer and 10 from the first purchase. So $E = 20 \times \text{Tk } 300 + 10 \times \text{Tk } 320 = \text{Tk } 6,000 + \text{Tk } 3,200 = \text{Tk } 9,200$. Then $C = \text{Tk } 32,600 - \text{Tk } 9,200 = \text{Tk } 23,400$. Then $GP = \text{Tk } 35,000 - \text{Tk } 23,400 = \text{Tk } 11,600$. *Answer.* Under LIFO, ending inventory is Tk 9,200, cost of goods sold is Tk 23,400 and gross profit is Tk 11,600. *Check.* Directly: $50 \times 340 + 20 \times 320 = 17,000 + 6,400 = \text{Tk } 23,400$.

Again we must state the limits. LIFO charges the newest, dearest costs to the sale, so cost of goods sold is high and gross profit is low. Ending inventory is valued at old prices, so the asset in the balance sheet may be far below what the goods would cost today. A

Pause and check that you can take costs from the oldest layers first.

When prices rise, FIFO gives the lowest cost of goods sold and the highest ending inventory.

Recap: FIFO sells the oldest costs first and leaves the newest costs in inventory.

GO SLOWLY HERE.

A stack of plates shows the idea: the plate placed last is the plate taken first.

Last-in first-out treats the newest costs as sold first.

Last-in first-out. A method that treats the newest costs as sold first, leaving the oldest costs in inventory.

We take 70 units from the newest layers, and 30 units of older cost remain.

When prices rise, LIFO gives the highest cost of goods sold and the lowest profit.

recorded. Each half ends with a figure, and we call these two figures the adjusted balances. If the work is correct, the adjusted balances are equal, and that common figure is the true amount in the bank.

The rules follow from what you learned in section 12.2. In words: the adjusted bank balance is the bank statement balance plus deposits in transit, less outstanding cheques. The adjusted book balance is the cash book bank balance plus credit memos, less debit memos and dishonoured cheques. In symbols, let S be the bank statement balance, D the deposits in transit and C the outstanding cheques. Let B be the cash book bank balance, M the credit memos, N the debit memos and H the dishonoured cheques. Then the adjusted bank balance is $S + D - C$, and the adjusted book balance is $B + M - N - H$. A deposit in transit is added to the bank side because the bank will soon receive it. An outstanding cheque is subtracted because the bank will soon pay it.

IN SHORT

Adjusted bank balance = $S + D - C$. Adjusted book balance = $B + M - N - H$. The two must be equal. If they are not, an item is missing or wrongly placed.

The figure shows the two halves side by side, ending in a single agreed balance.

The figure below shows the layout. Read it from top to bottom in each column. The left column begins with the bank statement balance and shows the two timing items. The right column begins with the cash book balance and shows the three items from the bank. Each column ends in a box, and a dashed line joins the two boxes to show that they must be equal.

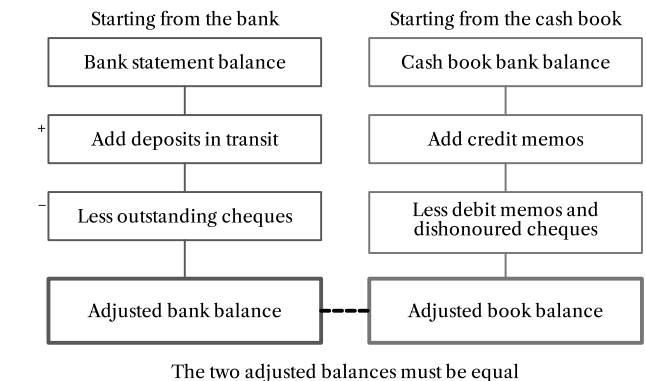


FIG. 12.1. Two-sided layout of a bank reconciliation

Follow this method. First, compare the cash book with the statement, line by line, and tick every pair that agrees. Second, list the unticked items from the cash book: these are deposits in transit and outstanding cheques. Third, list the unticked items from the statement: these are credit memos, debit memos, dishonoured cheques and direct payments. Fourth, calculate the two adjusted balances

Four steps give a method that works for any month.

Imagine the goods as a pile, built up as purchases arrive. Each purchase adds a **layer** of its own: a group of units that were bought at one time and at one unit cost. The opening inventory, which is the goods held when the period began, is the first layer. The sack bought for Tk 100 is one layer. The sack bought for Tk 120 is a second layer, placed on top of the first. When goods are sold, we must decide which layer the sold units are taken from.

A **cost flow assumption** is a rule that decides which purchase costs are treated as sold and which remain in inventory. Suppose a shop sells goods from a pile with two layers. The rule says which layer's cost goes to cost of goods sold first. The word flow refers to the order of costs, not to the movement of goods on the shelf. A shop may hand out its newest dictionary first, yet still assume that its oldest cost was sold first. We call it an assumption because we choose it, and no one proves it.

We now test the idea with the smallest numbers possible. *Given.* One unit bought at Tk 10 and one unit bought at Tk 12, both on hand. One unit is sold for Tk 20. *Find.* The cost of goods sold, the cost of the unit left, and the gross profit, under each of two rules: the oldest cost is sold first, or the newest cost is sold first. *Formula.* Gross profit equals net sales minus cost of goods sold. In symbols, $GP = S - C$, where GP is gross profit, S is net sales and C is cost of goods sold. *Steps.* If the oldest cost is sold first, $C = Tk 10$, the unit left costs Tk 12, and $GP = Tk 20 - Tk 10 = Tk 10$. If the newest cost is sold first, $C = Tk 12$, the unit left costs Tk 10, and $GP = Tk 20 - Tk 12 = Tk 8$. *Answer.* The profit is Tk 10 under the first rule and Tk 8 under the second. *Check.* In both cases the cost of goods sold plus the cost of the unit left equals Tk 22, the cost of both units.

The unit left in the tiny calculation has a name. **Ending inventory** is the cost of goods still on hand at the end of the period. The second sack in our opening puzzle, if the customer took the first, is the ending inventory, valued at Tk 120. Ending inventory appears as an asset in the balance sheet. It is also the opening inventory of the next period, so an error in one period carries into the next.

The tiny calculation shows a rule that never changes. The cost of goods available for sale is the opening inventory plus the purchases of the period. Each unit of that cost is either sold or left. In words: cost of goods available equals cost of goods sold plus ending inventory. In symbols, $A = C + E$, where A is the cost of goods available, C is the cost of goods sold and E is the ending inventory. A cost flow assumption does not change A. It decides only how A is divided between C and E.

Consider why the choice matters, from four viewpoints. The owner sees a higher or lower profit, and so a higher or lower reward for the year. The tax collector sees a larger or smaller amount on which tax is due, in countries where tax follows profit. The bank sees a higher or lower value for the goods that support

The word layer helps us picture goods bought at different times and prices.

Layer. Word in use. In daily life, a thickness of material lying over another. In accounting, a group of units bought at one time at one cost.

A cost flow assumption is a rule that names which layers are treated as sold.

Flow. Word in use. In daily life, water or traffic moving along a path. In accounting, the order in which costs are treated as leaving inventory, which need not match the movement of the goods.

Cost flow assumption. A rule that decides which purchase costs are treated as sold and which remain in inventory.

A tiny calculation with two sacks shows that each rule gives a different profit.

Ending inventory is the cost of the goods still on hand when the period closes.

Ending inventory. The cost of goods still on hand at the end of the period.

Every cost the business could sell ends up either in cost of goods sold or in ending inventory.

The method matters because it moves cost between the income statement and the balance sheet.

GO SLOWLY HERE.

The task is to find the adjusted balance on each side and to check that they agree.

The arithmetic is shown line by line so that you can repeat it by hand.

The answer is a single balance, and a quick test confirms the work.

Four items need journal entries, because the books of Meghna and Sons are missing them.

After posting, the Bank column of the cash book equals the adjusted balance.

Common slips: a wrong sign, a missed item, a misplaced half

bank Tk 2,500; school payment direct to the bank not yet in the books Tk 40,000.

Find. The adjusted bank balance and the adjusted book balance, and the entries that follow. The formula is the same as before: $S + D - C$ for the bank side, and $B + M - N - H$ for the book side. Here M is the interest of Tk 2,500 plus the school payment of Tk 40,000. N is the service charge of Tk 1,500. H is the dishonoured cheque of Tk 8,000.

OCTOBER RECONCILIATION, 31 OCTOBER 2026

BANK STATEMENT SIDE	TK	CASH BOOK SIDE	TK
---	---	---	---
Statement balance	655,000	Cash book balance	612,000
Add deposit in transit	48,000	Add interest credited	2,500
Less cheque no. 4417	36,000	Add school payment	40,000
Less cheque no. 4421	22,000	Less service charge	1,500
		Less dishonoured cheque	8,000
Adjusted bank balance	645,000	Adjusted book balance	645,000

Steps. Bank side: $655,000 + 48,000 = 703,000$. The cheques total $36,000 + 22,000 = 58,000$. Then $703,000 - 58,000 = 645,000$. Book side: $612,000 + 2,500 = 614,500$. Then $614,500 + 40,000 = 654,500$. Then $654,500 - 1,500 = 653,000$. Then $653,000 - 8,000 = 645,000$.

Answer. Both adjusted balances are Tk 645,000. This is the true bank balance, and it is the figure for the balance sheet. *Check.* The original difference was $655,000 - 612,000 = 43,000$. The bank side moved by $48,000 - 58,000 = -10,000$. The book side moved by $2,500 + 40,000 - 1,500 - 8,000 = +33,000$. And $10,000 + 33,000 = 43,000$, so every taka is explained.

Now the entries. Four items came from the bank, so four entries are needed. Suppose the school owed Meghna and Sons Tk 40,000 for books supplied earlier. The table shows each entry.

ENTRIES AFTER THE OCTOBER RECONCILIATION

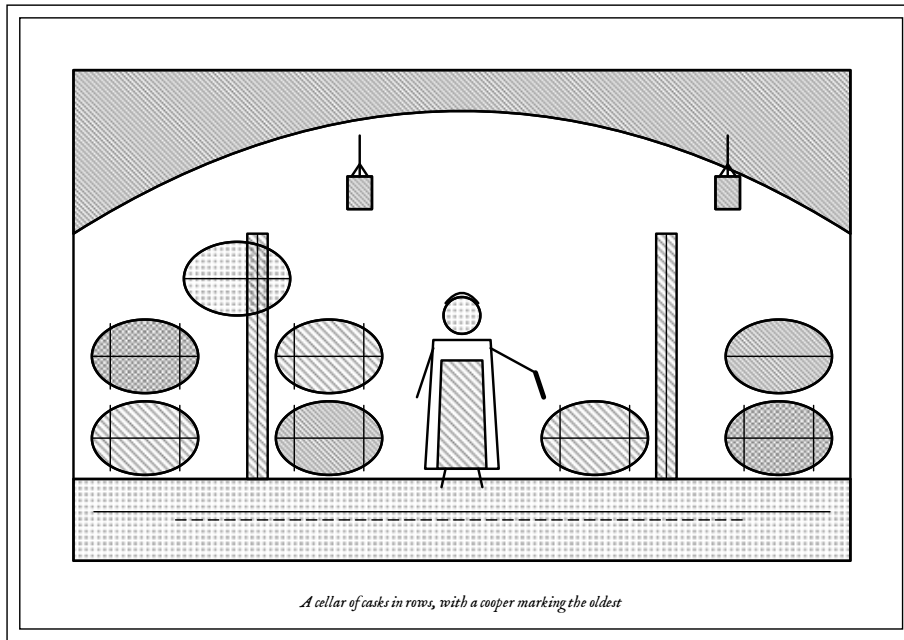
ITEM	DEBIT	CREDIT	TK
---	---	---	---
Interest credited	Bank	Interest Revenue	2,500
School payment	Bank	Account Receivable (the school)	40,000
Service charge	Bank Charges	Bank	1,500
Dishonoured cheque	Account Receivable (the customer)	Bank	8,000

The net effect on the Bank account is $+2,500 + 40,000 - 1,500 - 8,000 = +33,000$. So the cash book balance changes from Tk 612,000 to Tk 645,000. The deposit in transit and the two cheques have no entry; they will appear on the statement in November, and the bank side will then agree without help.

Three slips are common. The first is a wrong sign: a dishonoured cheque is subtracted from the cash book side, not added. The

REVIEW QUESTIONS

- 12.1 The cash book shows a bank balance of Tk 3,000. The statement shows Tk 3,400. A deposit of Tk 700 is in transit. A cheque for Tk 300 is outstanding. The bank collected Tk 900 for the business (a credit memo) and charged Tk 100 (a debit memo). Prepare the reconciliation statement and state the adjusted balance.
- 12.2 The cash book shows Tk 95,050 and the statement shows Tk 91,250. Deposits in transit are Tk 12,000 and Tk 6,500. Outstanding cheques are no. 0215 for Tk 9,800 and no. 0219 for Tk 4,150. A credit memo shows a collection of Tk 5,000. A debit memo shows a service charge of Tk 750. A customer's cheque for Tk 3,500 was dishonoured. Prepare the reconciliation statement.
- 12.3 In a later month the statement shows Tk 18,400 and the cash book shows Tk 20,150. A deposit of Tk 5,000 is in transit. Cheques of Tk 3,200 and Tk 1,100 are outstanding. A service charge of Tk 250 and a dishonoured cheque of Tk 800 are not in the books. Say which balance is higher before adjusting, then reconcile.
- 12.4 The cash book shows Tk 40,000 and the statement shows Tk 47,000. A deposit of Tk 4,000 is in transit and a cheque for Tk 9,000 is outstanding. A customer paid Tk 3,000 directly into the bank. The bank credited interest of Tk 500 and charged Tk 1,500. Reconcile, write the journal entries, and state the figure for Bank on the balance sheet.



CHAPTER XIV

Costing Inventory: FIFO, LIFO and Weighted Average

Two identical items were bought at different prices. Which cost did the customer take away?



This chapter teaches three rules for deciding which purchase costs leave inventory when goods are sold. You will apply each rule to one set of figures, compare the results, and see how the choice changes profit and the value of the goods still on hand.

The lesson fits this chapter. Goods that cannot be sold at cost must be reported at a lower value, and the loss falls in the period when the fall is recognised. Cisco's write-down was large in size, but the same rule applies to the 120 old copies of Meghna and Sons.

REVIEW QUESTIONS

13.1 Classify each item as raw materials, work in process, finished goods, or not inventory: (a) reels of paper in a binding workshop; (b) sewn sheets with no cover; (c) a stapler on the owner's desk; (d) bound volumes ready for collection; (e) a supplier's books stored in the shop but still owned by the supplier.

13.2 A shop holds 200 copies of a guide book. Each cost Tk 450. Each can now be sold for Tk 500, and the cost of selling is Tk 90 a copy. Find the net realisable value per copy, and the write-down, if any, for all 200 copies.

13.3 A shop begins the year with inventory of Tk 40,000 at cost and buys goods for Tk 65,000. A count at year end shows goods with a cost of Tk 30,000. Find the cost of goods sold under a periodic system.

13.4 A perpetual record shows 500 units of a book. A count finds 470. Cost is Tk 60 a unit. Find the cost of the difference and say what the shop should do first.

13.5 An owner holds goods with a total cost of Tk 600,000 and a holding cost of 12% a year. Find the yearly holding cost. In the next year, 150 units that cost Tk 80 each have a net realisable value of Tk 55. Find the write-down, and state the new reported value of those units.

The case teaches that a count and a rule on value matter most when demand changes quickly.

A small puzzle: two bad mornings in a bookshop, one with too few goods and one with too many.

The puzzle is to find a sensible amount between none and too many, and then to record it honestly.

The running case gives a first number: the goods held by the bookshop at the end of 2025.

Imagine two bad mornings in a bookshop. On the first morning a student walks in and asks for a particular textbook. The shelf is empty. The student shrugs and walks to another shop, and the sale is lost for ever. On the second morning the owner opens the store room at the back and finds a thousand copies of a book that nobody has asked for in a year. The money that paid for those copies is not in the bank. It is locked in paper and dust.

Both mornings are mistakes, and they pull in opposite directions. If the owner buys more to avoid the first morning, the second morning comes closer. If the owner buys less to avoid the second, the first comes closer. How much should a business hold? And once it holds something, how should it count the goods and say what they are worth? Accountants answer the second question with care, because the figure appears twice: in the income statement, through cost of goods sold, and in the balance sheet, as an asset. A mistake in the count is a mistake in the profit.

Case file, illustrative. On 31 December 2025 Meghna and Sons held goods for sale with a total cost of Tk 3,500,000. The year's net sales were Tk 14,400,000 and its cost of goods sold was Tk 9,800,000. Two groups within the Tk 3,500,000 need special attention: 800 titles that have not sold for 18 months, and 120 copies of an old edition.

YOU WILL NEED

This chapter uses cost of goods sold from Chapter 10 'Accounts of a Merchandising Company', section 10.1 'Sales, Purchases and Returns'. It also uses the idea of current assets from Chapter 10 'Accounts of a Merchandising Company', section 10.3 'Classified Balance Sheet'. The arithmetic is multiplication, percentages and subtraction; Appendix A1 'Maths You Will Use' holds a short reference.

* * *

13.1 WHAT INVENTORY IS

Begin at home. Most families keep a supply of rice, oil and soap in the kitchen. They do not eat the rice the moment they buy it. They keep it until it is needed, and when it runs low they buy more. A shopkeeper behaves in the same way, with one difference: the shopkeeper keeps the goods in order to sell them, not to use them.

In accounting this supply has a name: **inventory**. Inventory means goods held for sale, or materials held to make goods for sale, in the ordinary course of business. For example, the books on the shelves of Meghna and Sons are inventory, because the shop intends to sell them to customers. Inventory is an asset, because the business owns it and expects to receive money from it. It is a current asset, because the business expects to sell it within a short time, usually within one year.

A shop and a household both keep a supply of things for later use, and the business version has a name.

Inventory is the accounting name for goods held for sale and for materials held to make them.

Recall: current assets, Chapter 10, section 10.3 'Classified Balance Sheet'.

Inventory. Goods held for sale, or materials held to make goods for sale, in the ordinary course of business.

and its quantity, ideally with a second person who checks the first. Goods that belong to others are left out. Goods that have been sold but not yet collected are also left out, because they no longer belong to the business.

Under the perpetual system, suppose the records say 700 and the count finds 660. The difference of 40 may come from theft, damage, a mistake in recording, or a sale not entered. The business investigates first. If no cause is found, it reduces the Inventory account to the counted figure and charges the loss as an expense. The count is the final authority, because goods that cannot be found cannot be sold.

The last idea in this chapter is about value. A business usually reports inventory at its cost. But suppose an item can no longer be sold for as much as it cost. A business should not report an asset at more than the money it can bring in. The money it can bring in is the **net realisable value**. Net realisable value is the expected selling price of an item less the cost of selling it. For example, a textbook that cost Tk 100 may now sell for only Tk 150 after a discount, and selling it costs Tk 20 in packing and delivery. Its net realisable value is $150 - 20 = \text{Tk } 130$.

The rule is plain. For each item or group of the same item, compare the cost with the net realisable value. If cost is lower, report cost. If net realisable value is lower, report that figure and treat the difference as a loss in the period when the value fell. In the example above, cost Tk 100 is lower than Tk 130, so the book stays at Tk 100. If the discount were bigger and the net realisable value fell to Tk 70, the book would be reported at Tk 70, and a loss of Tk 30 would be recorded. We never raise an item above its cost because its value rises. Standards on this point differ between countries, so a business must follow the standards of the place where it reports.

Before moving on. A perpetual system updates Inventory after every purchase and sale. A periodic system finds it only by a count at period end. Both give the same cost of goods sold when the records are right. Inventory is reported at the lower of cost and net realisable value. You should now be able to say what each system shows during the year and how to test whether an item must be written down.

WORKED EXAMPLE: EXCESS TITLES AND AN OLD EDITION

Case file, illustrative. Given for 31 December 2025: inventory at cost Tk 3,500,000. Excess inventory: 800 titles at cost Tk 380,000, unsold for 18 months, holding cost 10% a year. Old edition: 120 copies, cost Tk 300 each, net realisable value Tk 180 each.

Find. (a) the share of inventory in the excess titles, and the yearly holding cost; (b) the write-down for the old edition, and the value at which it is reported.

Part (a), Steps. The excess titles are $380,000 \div 3,500,000 = 0.1086$, which is about 10.9% of the inventory. The holding cost is the cost times the rate: $380,000 \times 10\% = 38,000$. The value at which it is reported is $380,000 - 38,000 = 342,000$.

After a count, any difference between the records and the shelves must be explained and corrected.

Cost is not the only measure, because some goods will sell for less than they cost.

Net realisable value. The expected selling price of an item less the cost of selling it.

The rule: report the lower of cost and net realisable value, item by item.

Before moving on, we collect the ideas of this section.

The two groups from the case file are examined in full, with every line of arithmetic shown.

The task has two parts: the cost of holding the titles and the write-down of the old edition.

Part (a) uses a percentage of a number, which is the same step twice.

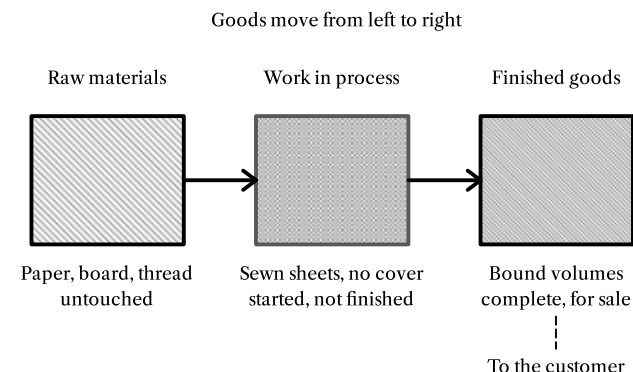


FIG. 13.1. Three kinds of inventory in a factory chain

From four viewpoints the kinds matter in different ways.

Before moving on, we collect the ideas of this section.

A business could try to hold nothing at all, and the reasons it does not form a framework.

Four reasons explain why every trading business holds some inventory.

The kinds matter to different people. The owner wants to know how much money is tied up in each stage. The manager of the workshop watches the middle stage, because goods that stay long in the middle are a sign of delay. The employee who handles the materials needs a count of what is waiting. A lender looking at the balance sheet sees the total and asks how quickly it turns into cash. Finished goods are closest to cash, and raw materials are furthest from it.

Before moving on. Inventory is owned goods held for sale, or owned materials held to make goods for sale. It is a current asset until the sale, when its cost becomes cost of goods sold. A shop holds finished goods only. A factory holds raw materials, work in process and finished goods. You should now be able to say, for any item in a business, whether it is inventory and, if so, which kind.

13.2 WHY HOLD INVENTORY

Why does a business not hold nothing? It would save storage, it would never have unsold goods, and it would never have a loss from old books. The answer is that customers do not wait, and suppliers do not deliver at the moment of need. Section 13.1 asked what inventory is. This section asks why a business accepts the burden of holding it. There are four main reasons.

The first reason is the customer. A customer who finds the shelf empty usually goes elsewhere, so goods must be ready when the customer arrives. The second reason is delay in supply. A publisher may need two weeks to deliver, so the shop needs enough goods to last until the delivery comes. The third reason is price. Buying a large quantity at one time often earns a lower price per item, and fewer deliveries mean lower transport costs. The fourth reason is demand that changes with the season. A bookshop sells many more textbooks in the weeks before classes start, so it must prepare before that time. A factory has a fifth reason: the stages of making need materials and part-made goods to keep the workers busy.

paid in. Partners who contribute unequally usually want a different rule, which is the reason deeds exist.

Now consider the books. When one person owns a business, one account records the owner's equity. When several people own it, the accountant keeps one account for each person. This account is called the partner's capital. It shows the amount that partner has put into the firm and left there, as a credit balance, which is a balance on the right-hand side of the account. When a partner pays in cash, we debit Cash and credit that partner's capital. When a partner brings an asset other than cash, such as a delivery van, we debit that asset instead of Cash. The capital of all the partners together makes up the owners' equity of the firm.

Different readers use the same records in different ways. A partner reads the capital accounts to see what share of the firm is his or her own. A bank that lends to the firm reads them to learn how much owners' money stands behind the loan. An employee has less interest in them, but may still ask whether the owners are able to pay wages in a poor year.

Case file, illustrative. A partnership deed is dated 1 July 2026. It records capital of Tk 2,500,000 from Nurul, Tk 1,000,000 from Rafiq and Tk 600,000 from Tariq. The three capital accounts together total Tk 4,100,000. The deed also fixes how profit is shared; section 18.2, 'Sharing Profit', explains the rule. On the balance sheet each of the three amounts appears on its own line.

Before moving on. A partnership is a business owned by two or more people who share profit and loss. The partnership deed records their agreement; without a deed, the law of the place decides. Each partner has a separate capital account, and the accounts together form the owners' equity of the firm.

* * *

18.2 SHARING PROFIT

The three friends of the opening can settle their puzzle in two layers. In the first layer, each friend is paid a reward for the money put in, so the largest lender gains most. In the second layer, the profit that remains is divided in a fixed proportion that reflects the work each friend does. Accountants use the same two layers, and each layer has a name.

We begin with the second layer. A **profit-sharing ratio** is the fixed proportion in which partners divide profit and loss. A ratio of 5:3:2 cuts the amount into ten equal parts: five go to the first partner, three to the second and two to the third. Of Tk 100, those partners receive Tk 50, Tk 30 and Tk 20. In symbols, a partner's share is $S = (p \div P) \times R$, where p is that partner's number of parts, P is the total number of parts, and R is the amount to be divided.

Capital. Word in use. In daily life, a sum of money, or the main city of a country. In accounting, the amount owners have put into the business and left there.

Each partner's capital account records what that one partner has put into the firm.

A lender reads the deed to learn who owns the firm and how much capital stands behind it.

Recall: balance sheet, Chapter 9, section 9.1 'The Three Statements'.

The case file records three partners and the capital each one contributed.

Three ideas carry forward: the partnership, its deed and the capital accounts.

The two friends' puzzle is solved by dividing profit in two layers.

Share. Word in use. In daily life, a part of something divided among people. In accounting, a partner's part of the profit; in a company, a unit of ownership, taught in Chapter 19.

Profit-sharing ratio. The fixed proportion in which partners divide profit and loss.

A ratio cuts profit into equal parts and gives each partner a fixed number of them.

A small puzzle: two payments for the same machine, one that stays in the asset and one that does not.

The puzzle asks which payments become part of the machine in the accounts and which do not.

The running case supplies the payments for one machine, which the chapter follows to the end.

Imagine that the owner of a bookshop buys a machine to cut the edges of books. The seller sends an invoice. A lorry brings the machine, and the lorry driver wants payment. An electrician connects it, and a technician tests it on a few sheets of paper. Four months later a blade breaks, and a mechanic repairs the machine. Each of these events costs money, and the owner pays for all of them. Yet the accountant will treat them in two very different ways.

Why does the difference matter? Suppose the accountant adds every payment to the machine. The machine looks more valuable, and the profit for the year looks larger, because fewer costs are charged against it. Suppose instead that the accountant charges every payment as an expense at once. The profit for the year looks smaller, and the machine looks cheaper than it really was. Neither extreme is honest. The accountant needs a rule that decides, payment by payment, where each one belongs. This chapter builds that rule.

Case file, illustrative. Meghna and Sons bought a book-cutting machine on 1 April 2026. The invoice was Tk 300,000. Freight was Tk 10,000, installation was Tk 15,000 and a trial run cost Tk 5,000. On 14 August 2026 a breakdown required a repair that cost Tk 8,000.

YOU WILL NEED

This chapter uses the balance sheet and the income statement from Chapter 9 'Financial Statements and Closing Entries', section 9.1 'The Three Statements'. It also uses the idea of a cost paid now that benefits later periods, from Chapter 7 'Adjusting the Accounts', section 7.2 'Deferrals'. The arithmetic is addition and subtraction; Appendix A1 'Maths You Will Use' holds a short reference.

* * *

15.1 WHAT PLANT ASSETS ARE

Think about a family that runs a small food stall. The rice and vegetables on the table are for sale, and they are gone in a day. The stove, the large pot and the table itself are different. The family does not sell them. It uses them every day for years. Accountants separate these two groups, because they behave differently in the accounts. The first group is inventory, which Chapter 13 explains. The second group is the subject of this section.

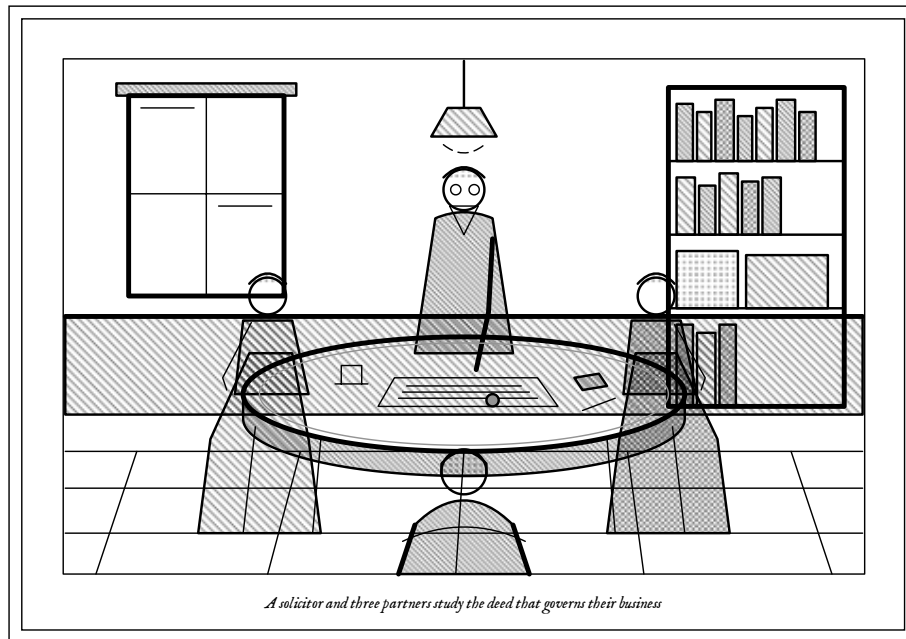
The accounting name for the stove, the pot and the table is **plant asset**. A plant asset is a long-lived physical asset used in running a business, such as a machine or a building. For example, the book-cutting machine of Meghna and Sons is a plant asset. The word *plant* may mislead you. In daily speech it means a living plant or a factory. In accounting it applies to the equipment of any business, even a small shop with no factory at all.

A business owns some things to sell and other things to use, and this chapter is about the second group.

Plant asset is the name for long-lived physical things used in the business.

Plant asset. A long-lived physical asset used in running a business, such as a machine or a building.

Plant. Word in use. In daily life, a growing thing such as a tree, or a factory. In



CHAPTER XVIII

Partnership Accounts

Who receives what when three people share one business?



This chapter shows how the books of a business with several owners are kept. It explains each partner's capital, the rules for sharing profit, the admission of a new partner, and the ending of the business by liquidation. Each method is worked by hand, from small numbers to a realistic case.

different business. You should now be able to say, for any item, whether it is a plant asset, and give the reason.

15.2 COST OF A PLANT ASSET

How much did the machine cost? Many students say: the amount on the invoice. This answer is tempting, and it is incomplete. The owner paid for the lorry, the electrician and the test as well. The machine was not usable until all of these were done. An honest cost must include them.

The accounting name for this honest cost is **acquisition cost**. Acquisition cost is all the money spent to buy an asset and make it ready for use. For example, if a machine costs Tk 200,000 and the owner pays another Tk 10,000 to bring it and install it, the acquisition cost is Tk 210,000. The word *acquisition* means "getting". The test for each payment is simple to say: was it necessary to get the asset and to make it ready for use? If yes, it belongs in the cost.

Let us look at the usual items. The first is the price on the invoice, less any trade discount the seller gives. The second is freight, the cost of bringing the asset to the business, because the asset is of no use at the seller's door. The third is installation, which includes the electrician and any special base or wiring. The fourth is testing, because a machine is not ready until it has been tested. The fifth is any tax or duty on the purchase that the business cannot recover. Where a tax can be reclaimed from the government, it is not part of the cost; rules on this differ between countries, so the business checks the law of its own place.

Other payments are not part of the cost. Repairs after the asset is working are not part of it, because the asset was already ready. Training the staff to use the machine is not part of it, because the training does not make the machine itself ready. Insurance for the first year of use, and the running costs such as electricity, are not part of it either, because they relate to using the asset, not to getting it. Damage caused by careless handling during delivery is also excluded. It is a loss for the period, not a part of the asset. Many students find this list confusing at first. The test makes it easy: a payment up to the moment the asset is ready for use is in the cost, and a payment after that moment is out.

Case file, illustrative. For the book-cutting machine bought on 1 April 2026: invoice Tk 300,000; freight Tk 10,000; installation Tk 15,000; trial run Tk 5,000. These four payments were all made to bring the machine to a working state. The repair of Tk 8,000 on 14 August 2026 came after the machine was working.

The figure below shows the cost as a stack. Read it from the bottom upward. The widest block at the bottom is the invoice. Each block above it is one more payment added to the cost. The top block is the total, the acquisition cost. The blocks are not drawn to scale, because the smaller payments would be too thin to see.

The first task is to decide how much the asset cost, and the answer is larger than the invoice.

The rule says the cost covers everything spent to buy the asset and make it ready.

Acquisition cost. All the money spent to buy an asset and make it ready for use.

The items that belong in the cost can be listed, and each has a reason.

The items that do not belong are those that come after the asset is ready, or that do not add to it.

The case file lists four payments to the moment of readiness and one payment afterwards.

The figure shows how the cost builds up from its parts, one on top of the other.

17.3 A trader had assets of Tk 60,000 and liabilities of Tk 15,000 at the start of a year, and assets of Tk 72,000 and liabilities of Tk 18,000 at the end. She withdrew Tk 20,000 for private use and brought in Tk 8,000 of savings. Find the profit.

17.4 A club bought a piano for Tk 30,000 in the year. Explain where this appears in the receipts and payments account, and in the income and expenditure account and balance sheet.

repairs and training. You should now be able to list the payments for one asset and state which belong in the cost.

15.3 CAPITAL AND REVENUE SPENDING

Sections 15.1 and 15.2 dealt with the cost of one asset. We now widen the question. Every day the business pays for things: wages, rent, paper, repairs, new machines. Some of these payments create something that lasts for years. Others are used up in the period. If the accountant puts a payment in the wrong group, the profit and the balance sheet are both wrong. We need two names for the two groups.

The first group is **capital expenditure**. Capital expenditure is spending that creates or improves an asset lasting more than one period. For example, the purchase of the book-cutting machine, at Tk 330,000, is capital expenditure. It is recorded as an asset on the balance sheet. The word *capital* here has nothing to do with the owner's money. It points to the long life of the thing that the spending creates.

The second group is **revenue expenditure**. Revenue expenditure is spending that keeps an asset in working order and is treated as an expense of the period. For example, the repair of Tk 8,000 on 14 August 2026 is revenue expenditure. It is recorded as an expense in the income statement of 2026. The name can confuse a new reader. It does not mean that the spending produces revenue. It means that the spending is matched against the revenue of the period in which it is made.

Why is the line drawn here? Think of the matching principle and the accrual basis of accounting, which you have met. An expense belongs to the period whose revenue it helps to earn. A repair keeps the machine running this year, so its cost belongs to this year. A new machine will help earn revenue for many years, so its cost is spread across those years. If the accountant charged the whole machine to the first year, that year would look poor and the later years would look rich, which is a false picture.

Some cases are not obvious. A mechanic works on a machine. Is that a repair or an improvement? The test asks what the spending does. If it restores the asset to the condition in which it was first bought, it is a repair: revenue expenditure. If it makes the asset last longer, work faster or do more than before, it is an improvement: capital expenditure. For example, replacing a broken blade with a new blade of the same kind is a repair. Fitting a motor that makes the machine cut twice as fast is an improvement, and its cost is added to the cost of the machine.

The table below sets the two kinds side by side. Read each row as a question and each column as an answer.

Every payment a business makes either builds an asset or is used up, and the two kinds need names.

The first group is spending that creates or improves a lasting asset.

Capital expenditure. Spending that creates or improves an asset lasting more than one period.

Improvement. Word in use. In daily life, any change for the better. In accounting, a change that makes an asset last longer, work faster or do more, and so is added to its cost.

The second group is spending that keeps an asset in order and is used up in the period.

Revenue expenditure. Spending that keeps an asset in working order and is treated as an expense of the period.

The matching principle gives the reason for the split.

The hard cases are the repairs and improvements, and a single test sorts them.

A table compares the two kinds of spending question by question.

January 2026, and a bill of Tk 500 for newspapers was unpaid on 31 December 2025.

Find. The income and expenditure account for 2025, the surplus, and the balance sheet at 31 December 2025.

Formula. In words: subscriptions for the year equal cash received, less arrears of the earlier year, less advance receipts for the next year, plus arrears at the end of this year. In symbols, with R for cash received, A₁ for earlier arrears, P for advance receipts and A₂ for closing arrears: income = R - A₁ - P + A₂.

Steps. Subscriptions: 30,000 - 1,500 - 2,000 + 3,000 = Tk 29,500. Rent: 12,000 - 1,000 = Tk 11,000. Newspapers: 9,000 + 500 = Tk 9,500. Dinner surplus: 8,000 - 6,000 = Tk 2,000. The table of Tk 10,000 is an asset and is not an expense. Repairs stay at Tk 2,000.

INCOME AND EXPENDITURE ACCOUNT FOR THE YEAR ENDED 31 DECEMBER 2025

INCOME	TK	EXPENDITURE	TK
---	---	---	---
Subscriptions	29,500	Rent	11,000
Donations	5,000	Newspapers	9,500
Surplus on annual dinner	2,000	Repairs	2,000
		Surplus for the year	14,000
Total	36,500	Total	36,500

At 1 January 2025 the club held cash of Tk 4,000 and was owed Tk 1,500, and it owed nothing, so its accumulated fund was Tk 5,500. The year added a surplus of Tk 14,000, so the fund at the end is 5,500 + 14,000 = Tk 19,500. The assets are the table, Tk 10,000; cash, Tk 8,000; fees owed by members, Tk 3,000; and rent paid in advance, Tk 1,000. The total is Tk 22,000. The liabilities are fees paid in advance, Tk 2,000, and the unpaid newspaper bill, Tk 500, a total of Tk 2,500.

Answer. The surplus for 2025 is Tk 14,000 and the accumulated fund at the end is Tk 19,500. *Check.* Assets less liabilities are 22,000 - 2,500 = Tk 19,500, which equals the fund. The agreement proves that the adjustments are consistent. The surplus of Tk 14,000 is larger than the rise in cash of Tk 4,000 because of the new table (Tk 10,000 of cash that is not an expense) and the other adjustments.

Three slips are common. The first is to use the cash figure for subscriptions without adjusting. The second is to charge the table as an expense, which would turn a healthy surplus into a small one. The third is to add arrears of the earlier year instead of subtracting them. The formula protects you: earlier arrears and advance receipts are subtracted, and closing arrears are added.

SUMMARY

A non-trading organisation exists to serve members or a cause. Its first summary is the receipts and payments account, which shows cash only. The income and expenditure account applies the accrual

The task is to find the income and expenditure for the year and the accumulated fund at the end.

The subscriptions come first, because they need the longest calculation.

The Steps are shown one line at a time.

The balance sheet shows what the club owns and owes at the end of the year.

The answer and the check show that the two statements agree.

Common slips: using cash for subscriptions, treating the table as an expense, and the wrong sign on an adjustment.

The chapter's ideas in a few lines.

The entries record each kind of spending in its proper place.

Common slips: adding after-readiness payments, leaving out freight or installation, and confusing a repair with an improvement.

The chapter's ideas in a few lines.

Belief: the cost of an asset is the invoice price.

Belief: any payment connected with a machine is part of its cost.

Belief: capital expenditure means spending the owner's capital.

Belief: revenue expenditure produces revenue.

Belief: an improvement and a repair are the same thing.

A famous fraud began with the wrong choice between

Check. The two errors are equal, because one payment is placed in the wrong statement.

The correct entries, if both were paid by bank, are these. For the purchase: debit Machinery Tk 330,000 and credit Bank Tk 330,000. For the repair: debit Repairs Expense Tk 8,000 and credit Bank Tk 8,000.

Three slips are common. The first is to add the repair to the cost because it concerns the machine. The test is timing and purpose, not the name of the asset. The second is to leave out freight or the trial run because they are not on the invoice. The third is to treat a restoring repair as an improvement, or the other way round.

SUMMARY

A plant asset is a long-lived physical asset used in running a business. An intangible asset has no physical form but lasts as well. The acquisition cost of a plant asset is all the money spent to buy it and make it ready for use. Capital expenditure creates or improves an asset lasting more than one period; revenue expenditure keeps an asset working and is an expense of the period. Wrong classification changes profit and the balance sheet together.

THINK IT THROUGH

- 15.T1 Suppose a business wrongly charges the Tk 15,000 installation of the machine as an expense in 2026. What is the effect on 2026 profit and on the balance sheet figure for the machine?
- 15.T2 A manager wishes to show a high profit this year. Explain why she might be tempted to capitalise repairs, and why this is dishonest even if it is not noticed at once.

WHERE PEOPLE GO WRONG

The cost includes freight, installation, testing and any tax that cannot be recovered. The invoice is only the first part.

Only payments up to the moment the asset is ready for use are part of the cost. Repairs, training and running costs come later and are expenses.

The word *capital* in this phrase refers to the lasting nature of the asset that the spending creates. It has no link to the owner's own money.

Revenue expenditure is an expense that is matched against the revenue of the same period. The name comes from that matching, not from producing revenue.

A repair restores the asset to its first condition and is an expense. An improvement raises its life, speed or output and is added to the asset.

IN HISTORY

In June 2002 the American telephone company WorldCom, one of the largest in the country, told the public that it had wrongly

amounts still owed. This is exactly the method of Chapter 7, now applied to a club.

The second change deals with the table. A table bought for Tk 10,000 will serve for years, so its cost is not an expense of this year alone. It is an asset, and it appears on the balance sheet. Chapter 15 'Plant Assets' deals with such assets, and Chapter 16 'Depreciation' teaches how a share of the cost is charged to each year of use. In this chapter we do not charge depreciation, so that the principle of conversion stands out.

A third matter concerns events. When the club holds a dinner, it receives money from tickets and pays for food. It is clearer to show only the difference, the surplus or deficit on the dinner, than to list both amounts among the other items. For the dinner of the example, tickets of Tk 8,000 less costs of Tk 6,000 give a surplus of Tk 2,000.

The club also prepares a balance sheet at the year end. It lists the assets: the table, the cash, the fees still owed by members and the rent paid in advance. It lists the liabilities: the fees paid in advance by members and the unpaid newspaper bill. It ends with the accumulated fund, which equals assets less liabilities. The fund at the end equals the fund at the start plus the surplus for the year, or less the deficit.

PAUSE AND CHECK 17.P1 A club receives Tk 1,000 in January for the year that begins in the following April. Is it income of the year in which it is received? Give a reason.

Before moving on. The income and expenditure account applies the accrual basis to a non-trading organisation. It adjusts the cash record for amounts owed and amounts paid early, removes the purchase of lasting assets and shows the result as a surplus or a deficit. The balance sheet ends with the accumulated fund. You should now be able to say how cash received for subscriptions becomes income for the year.

The second change removes the purchase of a lasting asset.

Events that earn money are shown by their net result.

The club also draws up a balance sheet, which uses the same form as any other.

Before moving on, we collect the ideas of this section.

17.3 THE SOLE TRADER FROM INCOMPLETE RECORDS

Now we turn to the second puzzle. A sole trader is one person who owns and runs a business alone. Many small traders do not keep a journal and a ledger. They keep a cash box, a few bills and a memory. At the year end they, or a bank, or the tax office, want to know the profit. Records are incomplete, but two facts can be found by counting: what the business owns and owes at the start, and what it owns and owes at the end.

That list is called the **statement of affairs**. A statement of affairs is a list of a business's assets and liabilities at one date, used to find capital when records are incomplete. For example, the trader counts her tools, her goods, the money customers owe her and the cash in her box, and then lists what she owes to suppliers. The assets less the liabilities give the owner's equity at that date. The statement of affairs looks like a balance sheet, but it is built from counting and memory, not from ledger accounts.

How can profit come from two lists? Think of the owner's equity

Some small traders keep no full books, yet they still need to know their profit.

A list of assets and liabilities at one date is the tool, and it has a name.

Statement of affairs. A list of a business's assets and liabilities at one date, used to find capital when records are incomplete.

The profit is found from the

The main income of most clubs is the **subscription**. A subscription is a regular payment made by a member for belonging to an organisation. For example, each member of the reading room pays Tk 500 a year. Other income comes from donations, from the entry fees of new members and from events such as an annual dinner. A subscription needs special care because members may pay late, on time or early. We shall return to this in section 17.2.

Small clubs usually keep little more than a cash book, and the treasurer's first report is a summary of it. This summary is called the **receipts and payments account**. A receipts and payments account is a summary of all cash received and paid in a period, with opening and closing balances. For example, the account of the reading room lists every taka that came into the cash box and every taka that went out during the year. The cash balance at the start is added on the receipts side, and the balance at the end is found by subtraction.

The account has two sides. The left side lists the receipts, beginning with the opening cash balance. The right side lists the payments, ending with the closing cash balance, which makes both sides equal. We now build a small account with clean numbers, so that the method is the only thing to learn. *Given*. The reading room had cash of Tk 4,000 on 1 January 2025. During 2025 it received subscriptions of Tk 30,000, donations of Tk 5,000 and Tk 8,000 from sales of tickets to an annual dinner. It paid rent of Tk 12,000, newspapers of Tk 9,000, dinner costs of Tk 6,000, repairs of Tk 2,000 and Tk 10,000 for a new table. *Find*. The closing cash balance.

Steps. Receipts: $4,000 + 30,000 + 5,000 + 8,000 = \text{Tk } 47,000$. Payments: $12,000 + 9,000 + 6,000 + 2,000 + 10,000 = \text{Tk } 39,000$. Closing cash: $47,000 - 39,000 = \text{Tk } 8,000$. *Answer*. The cash on 31 December 2025 is Tk 8,000. *Check*. The treasurer said Tk 8,000, so the account agrees with the box.

RECEIPTS AND PAYMENTS ACCOUNT FOR THE YEAR ENDED 31 DECEMBER 2025

RECEIPTS	TK	PAYMENTS	TK
---	---	---	---
Opening cash	4,000	Rent	12,000
Subscriptions	30,000	Newspapers	9,000
Donations	5,000	Dinner costs	6,000
Dinner tickets	8,000	Repairs	2,000
		New table	10,000
		Closing cash	8,000
Total	47,000	Total	47,000

The receipts and payments account is useful. It is easy to prepare, and it proves that the cash is in the box. But it has three limits, and

The main source of money is the regular fee paid by members.

Subscription. A regular payment made by a member for belonging to an organisation.

Subscription. Word in use. In daily life, a payment to receive a newspaper or a service for a period. In a club, the yearly or monthly fee that a member pays for belonging to it.

A club keeps simple records, and its first summary is of cash only.

Receipts and payments account. A summary of all cash received and paid in a period, with opening and closing balances.

The account has the same two-sided form as a cash book.

The steps add each side and subtract to find the closing balance.

The account has strengths, and also limits that explain the rest of the chapter.

YOU WILL NEED

You will need the cost of a plant asset from Chapter 15 'Plant Assets', section 15.2 'Cost of a Plant Asset', and the difference between capital and revenue spending from Chapter 15 'Plant Assets', section 15.3 'Capital and Revenue Spending'. Appendix A1 'Maths You Will Use' covers percentages and rounding, which this chapter uses throughout.

A bookshop buys a costly machine, and the question is which years should bear its cost.

Charging the whole cost to one year gives an unfair picture of every year.

The remedy is to charge a share of the cost to each year of use, and this chapter teaches how.

On 1 April 2026, Meghna and Sons pays Tk 330,000 for a book-cutting machine. The machine will cut books for about five years. In the year of purchase the shop spends Tk 330,000, and its profit for that year is far smaller than usual. In the next four years the machine cuts thousands of books, yet the shop records no cost at all for it. Those years look more profitable than they really are.

Consider what the owner would conclude. In the first year, the bookshop seems to lose money, and a lender may refuse a loan. In the later years, it seems to earn large profits, and the owner may spend too much. Both pictures are false, because the machine helps to earn revenue in all five years. Many students find this point difficult at first, because they remember that the cash left the business on one day. The accounts, however, must follow the use of the machine and not the day of payment.

The remedy has a name, and this chapter teaches it. Section 16.1 explains the idea and the words that belong to it. Section 16.2 and section 16.3 teach three methods of choosing the share for each year. Section 16.4 shows how to record the share and what happens when the asset is sold. Go slowly. The same machine returns in every section, so only the method changes.

* * *

16.1 WHY DEPRECIATE

GO SLOWLY HERE.

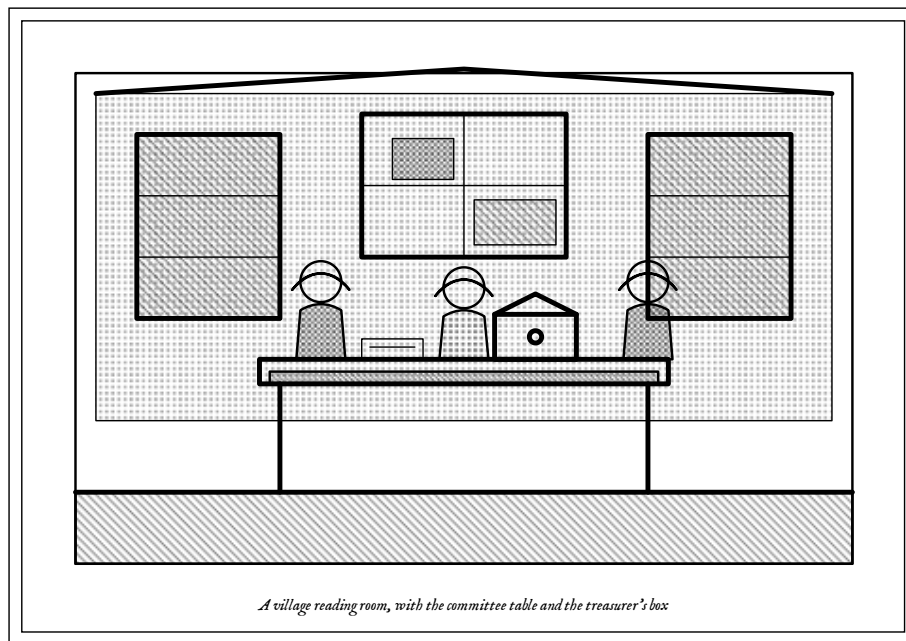
A bicycle bought for a delivery job shows why cost should be shared over the years of use.

Depreciation is the spreading of an asset's cost over the years of its use.

Depreciation. The spreading of a plant asset's cost over the years it is expected to be useful.

Suppose you begin a part-time delivery job and buy a bicycle for Tk 10,000. You expect to use it for four years, and then to sell it for about Tk 2,000. Would it be fair to say that all your earnings in the first year were reduced by Tk 10,000? The bicycle helps you earn in each of the four years. A fair view charges a share of its cost to each year. You pay once, but you use it four times.

Depreciation is the spreading of a plant asset's cost over the years it is expected to be useful. The bicycle shows the idea: a share of Tk 10,000 is charged in each of the four years. Depreciation is a method of allocation. It does not measure the fall in the market price of the asset, and it does not mean that cash is paid each year. The cash was paid on the day of purchase.



CHAPTER XVII

Accounts of Clubs and Sole Traders

The treasurer says the box holds more money than last year: is the club better off?



This chapter shows how organisations that do not aim at profit keep their accounts, and how their yearly cash record is turned into a fair statement of surplus or deficit. It then turns to a small trader who has kept no full books, and shows how to find the year's profit from the change in owner's equity. The same principles of Chapters 7 and 9 apply throughout.

GO SLOWLY HERE.

Equal shares are the easiest rule, and the case file gives the figures for the machine.

The straight-line method charges the same amount in every year.

Straight-line method. A depreciation method that charges an equal amount each year.

The machine's yearly charge is Tk 60,000, and the nine months of 2026 carry Tk 45,000.

Pause and check that you can calculate the straight-line charge.

The units-of-production method links the charge to how much the machine is used.

Units-of-production method. A depreciation method that charges an amount based on how much the asset is used in the period.

The machine's charge is Tk 0.60 for each book, so 90,000 books give Tk 54,000.

why a machine bought in one year should not reduce the profit of that year alone.

* * *

16.2 STRAIGHT-LINE AND UNITS-OF-PRODUCTION

Case file, illustrative. Meghna and Sons bought a book-cutting machine on 1 April 2026 at a cost of Tk 330,000. The residual value is Tk 30,000, the useful life is five years, and the capacity is 500,000 books. The business closes its books on 31 December each year.

The **straight-line method** is a depreciation method that charges an equal amount each year. The bicycle in section 16.1 used it, at Tk 2,000 a year. The word straight describes the picture of book value falling by equal steps, which forms a straight line on a graph. The method is the most common because it is easy to calculate and easy to explain.

Given. Cost Tk 330,000, residual value Tk 30,000, useful life five years, bought on 1 April 2026. *Find.* The charge for a full year, and the charge for 2026. *Formula.* Yearly charge equals cost minus residual value, divided by useful life in years. In symbols, $D = (C - R) \div n$. For a part of a year, multiply by the fraction of the year, $m \div 12$, where m is the number of months of use. *Steps.* $D = (Tk\ 330,000 - Tk\ 30,000) \div 5 = Tk\ 300,000 \div 5 = Tk\ 60,000$. From 1 April to 31 December is 9 months, so the 2026 charge is $Tk\ 60,000 \times 9 \div 12 = Tk\ 45,000$. *Answer.* The full-year charge is Tk 60,000, and the charge for 2026 is Tk 45,000. *Check.* $Tk\ 60,000 \times 5 = Tk\ 300,000$, which is the cost less the residual value.

PAUSE AND CHECK 16.P1 A van costs Tk 120,000, has a residual value of Tk 20,000 and a useful life of four years. What is the straight-line charge for a full year?

Some assets wear out by use more than by time. The **units-of-production method** is a depreciation method that charges an amount based on how much the asset is used in the period. A cutting machine is used more in a busy year than in a quiet year, and its wear follows the number of books cut. The method first finds a charge for each unit of use, and then multiplies by the units used.

Given. The machine can cut 500,000 books in its life, and it cut 90,000 books in 2026. *Find.* The charge for each book, and the charge for 2026. *Formula.* The charge for each unit equals cost minus residual value, divided by the total units expected. In symbols, $k = (C - R) \div U$ and $D = k \times u$, where U is the total units expected, k is the charge for each unit and u is the units used in the period. *Steps.* $k = Tk\ 300,000 \div 500,000 = Tk\ 0.60$ a book. $D = 90,000 \times Tk\ 0.60 = Tk\ 54,000$. *Answer.* The 2026 charge is Tk 54,000. *Check.* If all 500,000 books were cut, the total would be $500,000 \times Tk\ 0.60 = Tk\ 300,000$, which equals the amount to be

the proceeds. *Check.* Under straight-line, the debits are Tk 250,000 + Tk 140,000 + Tk 90,000 = Tk 480,000, which equals the credit to the van. Under declining balance, the debits are Tk 250,000 + Tk 244,800 = Tk 494,800, and the credits are Tk 480,000 + Tk 14,800 = Tk 494,800.

Common slips deserve a warning. The first is to forget the residual value in the straight-line formula, which makes every charge too large. The second is to subtract the residual value from the cost before applying the declining-balance rate, instead of applying the rate to the full remaining cost. The third is to forget the fraction of the year in the year of purchase. Test every answer by asking whether the total of the charges equals the cost less the residual value.

* * *

SUMMARY

Depreciation spreads the cost of a plant asset over its useful life, after subtracting the residual value. The straight-line method charges an equal amount each year, the units-of-production method charges by use, and the declining-balance method charges a fixed percentage of the remaining cost. The charge is recorded as an expense and as accumulated depreciation. Book value is cost less accumulated depreciation, and the gain or loss on a sale is the proceeds minus the book value. For Meghna and Sons, the machine's straight-line charge is Tk 60,000 a year, and a sale after three years for Tk 120,000 gives a loss of Tk 30,000.

THINK IT THROUGH

16.T1 A manager raises the estimated life of the book-cutting machine from five years to eight years, with the same residual value. What happens to the yearly straight-line charge, to the profit of each year, and to the total charged over the whole life?

16.T2 A student says: "Depreciation sets aside cash for buying a new machine." Explain what is wrong with this statement.

WHERE PEOPLE GO WRONG

Some students believe that depreciation measures the fall in the market price of an asset. It does not: it spreads the cost over the years of use, by a rule. Others believe that depreciation is a payment of cash. It is not: the cash was paid when the asset was bought, and the yearly entry moves no cash. A third belief is that book value equals the price at which the asset could be sold. Book value is a figure in the records, and the market price may be higher or lower. A fourth belief is that a faster method gives a larger total charge. Over the whole life all methods charge the same total, and only the timing differs. Last, students sometimes depreciate land. Land is usually not depreciated, because it does not wear out in

Three slips occur often in this work, and each one can be prevented.

The summary restates the whole chapter in a few lines.

Five common misunderstandings are stated here, each with its correction.

YEAR	STRAIGHT-LINE	DECLINING-BALANCE
1	60,000	132,000
2	60,000	79,200
3	60,000	47,520
4	60,000	28,512
5	60,000	12,768
Total	300,000	300,000

The figure shows the remaining cost falling faster under declining-balance than under straight-line.

The first figure draws the remaining cost of the machine against the years. The horizontal axis shows the years since purchase, and the vertical axis shows taka. A point shows the remaining cost at the end of that year. The solid line, for straight-line, falls by equal steps. The dashed line, for declining-balance, falls steeply at first and then flattens. Both lines end at the residual value of Tk 30,000.

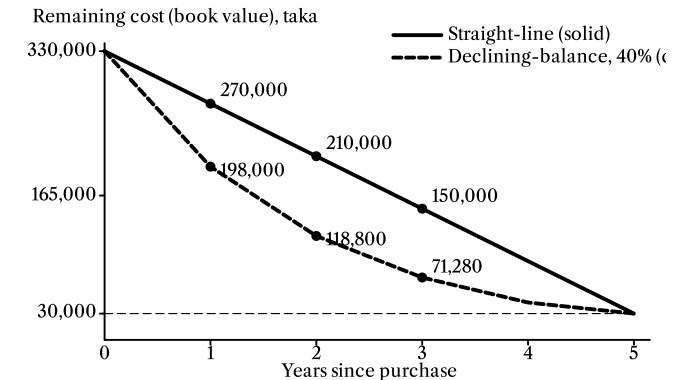


FIG. 16.1. *Straight-line against declining-balance as two descending lines*

Recap: declining-balance charges more early, less later, and stops at the residual value.

Before moving on. The declining-balance method applies a fixed rate to the remaining cost. For the machine at 40%, the charges are Tk 132,000 and Tk 79,200 in the first two years. You should now be able to say why the charge falls each year, and why the last charge may be cut.

* * *

16.4 RECORDING AND DISPOSAL

At the end of each year the business records the charge in an adjusting entry, of the kind taught in Chapter 7 'Adjusting the Accounts'. The entry debits an expense account, called depreciation expense, which reduces profit. It credits a second account, which collects the charges and is shown as a deduction from the cost of the asset. The cost account itself is left unchanged.

Each year the charge is recorded by debiting an expense and crediting a separate account.

Compensating error. Two or more mistakes that cancel each other so that the trial balance still agrees. Example: Wages debited Tk 300 too much and Rent debited Tk 300 too little, so the debit total is still correct. (*Chapter X*)

Compound entry. A journal entry that has more than one debit or more than one credit. Total debits still equal total credits. Buying a press for Tk 30,000, part paid in cash and part owed, debits one account and credits two. (*Chapter V*)

Control account. A main ledger account that sums up the balances of all the accounts in a subsidiary ledger. If both are correct, the control account balance equals the total of the individual balances. A difference shows that an error exists somewhere. (*Chapter XI*)

Cost flow assumption. A rule that decides which purchase costs are treated as sold and which remain in inventory. It concerns the order of costs, not the movement of goods. For example, a shop may sell its newest dictionary first but still assume that the oldest cost was sold first. (*Chapter XIV*)

Cost of goods sold. The cost of the goods that were sold during the period. It equals opening inventory plus purchases less purchase returns less closing inventory. Example: Tk 2,000 + Tk 6,000 – Tk 300 – Tk 1,700 = Tk 6,000. (*Chapter X*)

Credit. An entry on the right side of an account. It increases liabilities, owner's capital and revenue, and it decreases assets, drawings and expenses. Like debit, the word names a side only and carries no judgement. (*Chapter IV*)

Credit memo. A bank's notice that it has added money to the account, for example a collection made for the business or interest paid. The name follows the bank's books, where more owed to the customer is a credit. It is added on the book side of the reconciliation. (*Chapter XII*)

Current asset. An asset expected to be turned into cash or used up within one year. Examples are cash, accounts receivable, inventory and prepaid expenses. Current assets are listed in the order in which they come closest to cash. (*Chapter X*)

Current liability. A debt that must be paid within one year. Accounts payable are the usual example. The part of a long loan that falls due within one year is also current. Example: a loan instalment due in six months. (*Chapter X*)

Debit. An entry on the left side of an account. It increases assets, drawings and expenses, and it decreases liabilities, owner's capital and revenue. The word names a side; it does not mean good, bad, more or less. (*Chapter IV*)

Debit memo. A bank's notice that it has deducted money from the account, for example a service charge. The name follows the bank's books, where less owed to the customer is a debit. It is subtracted on the book side of the reconciliation. (*Chapter XII*)

Declining-balance method. A depreciation method that charges a fixed percentage of the asset's remaining book value each year. The charge is largest in the first year and falls after that. For example,

Interest on capital. A payment credited to partners for their capital before the remaining profit is divided.

Interest on capital rewards money in proportion to the capital each partner contributed.

Interest on capital divides profit; it is not a cost paid to an outsider.

The appropriation account shows profit being divided, layer by layer, in one place.

Example 1 uses small round numbers so that the method is the only thing to learn.

Three common slips are a wrong period, a wrong order of layers and a wrong divisor.

Two layers, interest first and ratio second, are the core of this section.

The first layer rewards money. **Interest on capital** is a payment credited to partners for their capital before the remaining profit is divided. A partner with capital of Tk 100,000 and a rate of 10% a year is credited Tk 10,000 for a full year. In words, interest equals capital times rate times time. In symbols, $I = C \times r \times t$, where C is the capital, r is the yearly rate and t is the time in years. Six months is half a year, so $t = 6 \div 12$.

Interest on capital is a way of dividing profit. It is not a cost paid to an outsider, and no cash need leave the firm. It exists only if the deed provides for it. We also assume that the profit is large enough to cover all the interest; a deed must say what happens if it is not.

PAUSE AND CHECK 18.P1 A partner has capital of Tk 200,000. The deed allows interest at 5% a year. How much interest is credited for six months?

Both layers are recorded in an appropriation account. The word *appropriation* means setting money aside for a stated use. This account begins with the profit from the income statement. It takes out interest on capital, divides the remainder by the profit-sharing ratio, and ends with the total credited to each partner. In this way the account shows who received what, and why, in one place.

Example 1 uses small, round numbers. Work through each row of the table in order; every line of arithmetic is shown.

EXAMPLE 1: DIVIDING ONE YEAR'S PROFIT IN TWO LAYERS

PART	WORKING
Given	Capital: A Tk 100,000, B Tk 50,000. Interest 10% a year. Ratio 2:1. Profit for the year Tk 30,000.
Find	The total profit credited to A and to B.
Formula	In words: interest = capital $\times$ rate $\times$ time; share = (parts $\div$ total parts) $\times$ remainder. In symbols: $I = C \times r \times t$ and $S = (p \div P) \times R$.
Step 1	Interest for A = $100,000 \times 10\% \times 1 =$ Tk 10,000.
Step 2	Interest for B = $50,000 \times 10\% \times 1 =$ Tk 5,000.
Step 3	Remainder R = $30,000 - (10,000 + 5,000) =$ Tk 15,000.
Step 4	A's share = $(2 \div 3) \times 15,000 =$ Tk 10,000. B's share = $(1 \div 3) \times 15,000 =$ Tk 5,000.
Answer	A is credited Tk 20,000 (10,000 + 10,000). B is credited Tk 10,000 (5,000 + 5,000).
Check	$20,000 + 10,000 =$ Tk 30,000, which equals the profit.

Three slips are common. First, students forget that the rate is yearly and credit a full year of interest for a shorter period. Second, they divide the profit by the ratio before taking out interest, which reverses the layers. Third, they divide by one part of the ratio instead of its total: with a ratio of 2:1 the divisor is 3, not 2.

Before moving on. A profit-sharing ratio divides profit and loss in fixed parts. Interest on capital pays each partner for money put in, and it comes out of profit first. The appropriation account records both layers and ends with each partner's total.

* * *

GLOSSARY

Account. A record that collects all the increases and decreases of one kind of item. The Cash account, for example, shows every receipt and payment of cash, so its balance tells the business how much cash it holds. (*Chapter IV*)

Accounting. The process of recording, sorting and reporting the money effects of a business's events to people who need them. When a shop sells a novel, accounting records the sale, sorts it with other sales, and reports the total to the owner, the bank and others. (*Chapter II*)

Accounting cycle. The series of steps repeated each period, from recording transactions to preparing closing records. It begins with source documents, passes through the journal and the ledger, and ends with reports and closing records, after which the next period starts the same steps again. (*Chapter V*)

Accounting equation. The rule that assets always equal liabilities plus owner's equity, written $A = L + E$. It holds because every asset was provided either by a lender or by the owner. (*Chapter III*)

Accounting period. The span of time, such as a month, a quarter or a year, for which a business measures and reports its results. For example, Meghna Binding Service reports for the calendar year 2026. (*Chapter VII*)

Accounting standard. A written rule about how to record and report items, so that different businesses report in the same way. For example, a standard may say how goods held for sale must be measured. In daily life, a standard means a level of quality. (*Chapter II*)

Accrual basis of accounting. A method that records revenue when it is earned and an expense when it is incurred, whether or not cash has moved. For example, wages earned in December belong to December even if paid in January. (*Chapter VII*)

Accrued expense. A cost that has already been used up but not yet paid, shown as a liability. For example, wages earned by workers in the last days of December but paid in January. (*Chapter VII*)

Accrued revenue. Revenue that has already been earned but not yet received or billed, shown as an asset. For example, binding work finished in December for which the invoice is sent in January. (*Chapter VII*)

Accumulated depreciation. The total depreciation charged to date on an asset, shown as a deduction from its cost. It grows by one charge each year. For example, after three years of Tk 60,000 it equals Tk 180,000. (*Chapter XVI*)

Acquisition cost. All the money spent to buy an asset and make it ready for use. It includes the invoice price less discounts, freight, installation and testing. Payments after the asset is ready, such as repairs, are not part of it. (*Chapter XV*)

Adjusted trial balance. A trial balance prepared after the adjusting entries have been taken into account. It shows the correct balance

Liquidation. The process of selling a business's assets, paying its debts and sharing what remains among the owners.

Liquidation is the whole process of closing a business and paying everyone.

Realisation. The sale of assets, in liquidation, and the gain or loss on that sale.

Realisation is the sale of the assets and the gain or loss that results.

The case file records the sale of all assets at a loss and the payment of debts and partners.

Paying the partners follows four steps in a fixed order: sell, share the loss, pay debts, pay partners.

Example 2 works the liquidation arithmetic in the standard order.

Liquidation is the process of selling a business's assets, paying its debts and sharing what remains among the owners. A shop that sells its books and fittings, repays its suppliers and bank, and gives the balance to its owners is in liquidation. Debts to outsiders come first. The owners receive only what is left, because their claim comes after every other claim.

The first step has its own name. **Realisation** is the sale of assets, in liquidation, and the gain or loss on that sale. If an asset with a book amount of Tk 100 sells for Tk 80, the realisation loss is Tk 20. The book amount is the amount at which the asset stands in the accounts. The partners share a loss on realisation in the profit-sharing ratio, as section 18.2 explained, unless the deed says otherwise.

Case file, illustrative. The liquidation takes place on 31 December 2027. The assets, at a book amount of Tk 5,300,000, are sold for Tk 4,400,000. The liabilities are Tk 1,200,000. The loss of Tk 900,000 is shared as Tk 450,000, Tk 270,000 and Tk 180,000. Cash of Tk 3,200,000 remains after the debts and is paid to the partners as Tk 2,050,000, Tk 730,000 and Tk 420,000.

Paying the partners comes last, and the order matters. First, sell the assets and find the loss. Second, share the loss in the ratio 5:3:2 by reducing each partner's capital. Third, pay the liabilities to outsiders from the cash received. Fourth, pay each partner the capital that remains. A realisation account collects the assets sold and the cash received, so that its balance shows the loss. The table records the four steps.

ENTRIES FOR THE LIQUIDATION, 31 DECEMBER 2027

STEP	DEBIT	CREDIT	TK
1	Cash	Realisation	4,400,000
1	Realisation	Assets	5,300,000
2	Capital: Nurul 450,000, Rafiq 270,000, Tariq 180,000	Realisation	900,000
3	Liabilities	Cash	1,200,000
4	Capital: Nurul 2,050,000, Rafiq 730,000, Tariq 420,000	Cash	3,200,000

Example 2 works the arithmetic. The case file implies that each partner's capital on 31 December 2027 equals the amount in the deed, and the working uses those amounts.

IN HISTORY

Until 1976, the United States federal government counted its fiscal year from 1 July to 30 June. The Congressional Budget and Impoundment Control Act of 1974, signed into law on 12 July 1974, changed this. Under section 501, the fiscal year began on 1 October from 1976. To make the change, the period from 1 July to 30 September 1976 was treated as a separate three-month transition quarter, between fiscal years 1976 and 1977. The case shows that a fiscal year is a choice, and that a change of choice needs a bridging period.

A true account of a government that chose a new fiscal year.

REVIEW QUESTIONS

20.1 Classify each cost of a bookshop as a product cost or a period cost: books bought, freight in, shop rent, wages of the shop assistants, delivery to a customer.

20.2 A business records net sales of Tk 600,000, product costs of Tk 380,000 and period costs of Tk 150,000 for a quarter. Find the gross profit and the profit.

20.3 Trace the cycle from one transaction to the post-closing check. Name each step in order and say which steps change account balances.

20.4 Explain in two sentences the difference between an account payable and an account receivable, and say where each appears on the balance sheet.

20.5 A business has a fiscal year from 1 April to 31 March. Name the dates of its second quarter, and say which of these is an interim period: the year, the second quarter, or the full cycle.

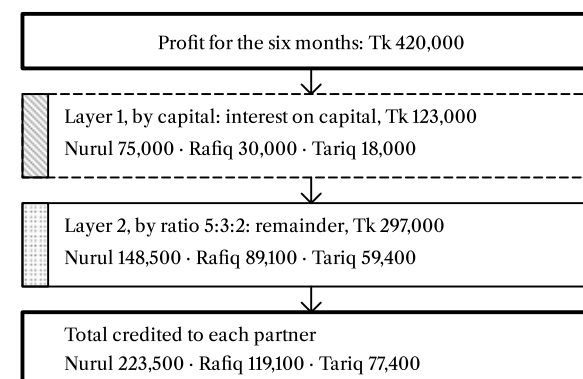


FIG. 18.1. Profit shared in two layers: interest on capital first, then the ratio

Case file, illustrative. The deed is dated 1 July 2026. Capital: Nurul Tk 2,500,000, Rafiq Tk 1,000,000 and Tariq Tk 600,000. The profit-sharing ratio is 5:3:2 and interest on capital is 6% a year. The profit from 1 July to 31 December 2026 is Tk 420,000.

Example 3 divides this profit. Notice the period: 1 July to 31 December is six months, so $t = 6 \div 12 = 0.5$ year. Many students find this step confusing, because the rate is stated for a year while the profit covers half of one. The table keeps both facts in view.

EXAMPLE 3: DIVIDING SIX MONTHS OF PROFIT

PART	WORKING
Given	Capital: Nurul Tk 2,500,000, Rafiq Tk 1,000,000, Tariq Tk 600,000. Interest 6% a year. Ratio 5:3:2. Profit for six months Tk 420,000.
Find	The total profit credited to each partner.
Formula	In words: interest = capital $\times$ rate $\times$ time; share = (parts $\div$ total parts) $\times$ remainder. In symbols: $I = C \times r \times t$ and $S = (p \div P) \times R$.
Step 1	Interest: Nurul $2,500,000 \times 6\% \times 0.5 = \text{Tk } 75,000$. Rafiq $1,000,000 \times 6\% \times 0.5 = \text{Tk } 30,000$. Tariq $600,000 \times 6\% \times 0.5 = \text{Tk } 18,000$.
Step 2	Total interest = $75,000 + 30,000 + 18,000 = \text{Tk } 123,000$.
Step 3	Remainder $R = 420,000 - 123,000 = \text{Tk } 297,000$.
Step 4	Shares of R: Nurul $(5 \div 10) \times 297,000 = \text{Tk } 148,500$. Rafiq $(3 \div 10) \times 297,000 = \text{Tk } 89,100$. Tariq $(2 \div 10) \times 297,000 = \text{Tk } 59,400$.
Step 5	Totals: Nurul $75,000 + 148,500 = \text{Tk } 223,500$. Rafiq $30,000 + 89,100 = \text{Tk } 119,100$. Tariq $18,000 + 59,400 = \text{Tk } 77,400$.
Answer	Nurul is credited Tk 223,500, Rafiq Tk 119,100 and Tariq Tk 77,400.
Check	$223,500 + 119,100 + 77,400 = \text{Tk } 420,000$, which equals the profit.

The case file records the profit for the first six months and the rules that divide it.

Example 3 divides the case-file profit in standard order, including the change of period.

The appropriation account sets out the same figures in the form accountants use.

The appropriation account shows the same figures in the form that accountants use. Read each column downward: a partner's interest, then the share of the remainder, then the total.

below. At the end of a full year, closing entries move the revenue and expense balances into owner's equity. For an interim report, many businesses prepare the statements without closing the accounts, because the year has not ended.

IN SHORT

The cycle has six verbs: record, post, test, adjust, report and close. Record and post fill the ledger. Test and adjust prepare it. Report tells the story. Close resets the temporary accounts for the next period.

PAUSE AND CHECK 20.P2 A business pays Tk 5,000 in cash to settle an account payable of the same amount. Which two totals on the balance sheet change, and does the balance sheet still balance?

Before moving on. The ledger feeds the trial balance; adjustments correct it; the statements report it; closing entries reset the temporary accounts. A payable is owed by us; a receivable is owed to us. The trial balance tests; the balance sheet reports.

You should now be able to say: "The statements are linked, so profit flows from the income statement through owner's equity to the balance sheet."

WORKED EXAMPLE: ONE QUARTER OF MEGHNA AND SONS

This example applies the chapter to the case file. We first sort the costs, and then we find the profit for the interim period. Take each step slowly and notice where each figure comes from.

Given. Net sales for the quarter ended 30 September 2026 are Tk 3,900,000. Product costs are books bought Tk 2,450,000 and freight in Tk 40,000. Period costs are rent Tk 180,000 and wages Tk 735,000. To keep the arithmetic clear, assume that all the books bought were sold in the quarter and that inventory did not change. *Find.* The total product cost, the total period cost, the gross profit and the profit for the quarter.

Formula. Product cost total equals books bought plus freight in, or $P = B + F$. Period cost total equals rent plus wages, or $C = R + W$. Gross profit equals net sales minus product cost total, or $G = S - P$. Profit equals gross profit minus period cost total, or $N = G - C$. Here S stands for net sales, and N stands for profit.

Steps. First, $P = \text{Tk } 2,450,000 + \text{Tk } 40,000 = \text{Tk } 2,490,000$. Second, $C = \text{Tk } 180,000 + \text{Tk } 735,000 = \text{Tk } 915,000$. Third, $G = \text{Tk } 3,900,000 - \text{Tk } 2,490,000 = \text{Tk } 1,410,000$. Fourth, $N = \text{Tk } 1,410,000 - \text{Tk } 915,000 = \text{Tk } 495,000$.

INCOME STATEMENT FOR THE QUARTER ENDED 30 SEPTEMBER 2026 (TK)

LINE	AMOUNT
Net sales	3,900,000
Less product costs (books bought and freight in)	2,490,000
Gross profit	1,410,000

Recap of the linked Statements.

The reader should now be able to state how the Statements are linked.

A worked example sorts the costs and finds the profit for the quarter ended 30 September 2026.

The given figures and the question to be answered.

The formulas are stated in words first and then in symbols.

The steps show every line of arithmetic with units.

IN HISTORY

Goldman Sachs, a New York firm founded in 1869, was owned by its partners for 130 years.

On 4 May 1999 the firm sold shares to the public and its partners became shareholders.

Marcus Goldman, a German immigrant, founded Goldman Sachs in New York City in 1869. The firm grew into one of the best-known financial firms in the United States, and for 130 years its owners were partners. In the 1990s the partners debated for years whether to change this. Some feared the loss of the partnership's special culture; others wanted permanent capital to fund the firm's growth.

The partners voted to make the change. On 4 May 1999 the firm's shares began to trade, in what was then the second-largest first public sale of shares in United States history. The lesson for this chapter is that a partner's capital belongs to named people, who may take it back when they leave, while money raised from shares stays in the business. Chapter 19, 'Company Accounts: Shares', section 19.1 'Share Capital', teaches that second form.

REVIEW QUESTIONS

18.1 Two partners share profit in the ratio 3:1 and the deed gives no interest on capital. The profit is Tk 80,000. How much does each partner receive?

18.2 A new deed gives Mira capital of Tk 450,000 and Naveed capital of Tk 300,000. Interest on capital is 8% a year, and the ratio is 3:2. The profit for the year is Tk 211,000. Find the total credited to each partner.

18.3 Under the same deed as question 18.2, the profit of Tk 180,000 is for nine months. Find the total credited to each partner.

18.4 On 1 April 2028 a new partner, Priya, pays Tk 300,000 by cheque into the firm's bank account, and the amount is credited as her capital. Journalise the admission.

18.5 Partners P and Q have capital of Tk 400,000 and Tk 200,000 and share profit and loss 2:1. Their assets, at a book amount of Tk 850,000, are sold for Tk 700,000, and the liabilities are Tk 250,000. Find the loss, each partner's share of it, and the payment to each partner.

PRODUCT COSTS AND PERIOD COSTS COMPARED

FEATURE	PRODUCT COST	PERIOD COST
Tied to	Goods held for sale	A stretch of time
First appears	In inventory, on the balance sheet	Straight in the income statement
Reaches income statement	When the goods are sold	In the period incurred
Examples from the case	Books bought, freight in	Rent, wages

PAUSE AND CHECK 20.P1 Meghna and Sons pays Tk 40,000 to a carrier to bring books into the shop, and Tk 180,000 as rent. Which payment is a product cost, and which is a period cost?

Chapter 10 shows how a classified income statement groups costs under headings.³ The product costs sit inside cost of goods sold, which is deducted from net sales to find gross profit. The period costs sit below gross profit as operating expenses. You can think of the income statement as a staircase: product costs come off first, then period costs.

Some costs are easy to misplace. Delivery to a customer is freight out, and it is tied to the act of selling, not to the buying of goods, so it is a period cost. Freight in, which brings goods to the shop, is a product cost. The name "freight" appears in both, but the question that sorts them is the same: is the cost tied to goods held for sale, or to time and selling? A second exception concerns unsold goods. If the shop buys books in September and sells none before the quarter ends, the whole product cost is still in inventory on 30 September, and none of it has reached the income statement.

Each reader of the reports uses this split differently. The owner wants to know whether the books bought are selling, which the product costs show. The manager wants to control rent and wages, which belong to the period and do not change with each sale. A lender wants to know whether sales cover both classes of cost. If the two classes were mixed, none of these readers could separate the cost of the goods from the cost of running the shop.

Before moving on. A product cost belongs to goods held for sale and waits in inventory until the sale. A period cost belongs to the time in which it is incurred and reaches the income statement at once. Freight in is a product cost; rent and wages are period costs.

You should now be able to say: "A product cost follows the goods, and a period cost follows the calendar."

20.3 THE LINKED STATEMENTS

Think of a water pipe that feeds three taps in a house. If the pipe is blocked, all three taps fail. The accounts work the same way. Every figure in the ledger feeds the reports, and a change at the source changes every tap it feeds. Learn the pipe first and the taps follow.

³ Chapter 10, 'Accounts of a Merchandising Company', section 10.2 'Classified Income Statement'.

A classified income statement groups these costs under separate headings.

A cost can look the same on paper yet belong to the other class, so ask what it is tied to.

The viewpoint of the owner, the manager and the lender explains why the split matters.

Recap of product cost and period cost.

The reader should now be able to state what separates the two classes of cost.

GO SLOWLY HERE.

A single number in the ledger flows through every report, much as one water pipe feeds several taps.

YOU WILL NEED

Chapter 9, 'Financial Statements and Closing Entries', section 9.3 'Reversing Entries'; Chapter 18, 'Partnership Accounts', section 18.1 'Forming a Partnership'; Appendix A1, 'Maths You Will Use'.

A bookshop needs far more money than its four partners own, so it looks to many small contributors.

The puzzle is to give each contributor a small, equal piece of the business and a firm limit on the risk.

This chapter takes three steps: what share capital is, which kinds of share exist, and how shares are issued.

A company is owned by many people through shares, and a share is one equal part of that ownership.

Recall: balance sheet, Chapter 9, section 9.3 'Reversing Entries'.

Stock. Word in use. In everyday English, a supply of goods kept ready for sale. In the United States and some other places, stock also means a company's shares. This book uses inventory for goods and stock only for company shares.

Suppose a bookshop wishes to open ten new branches. The cost is Tk 5,000,000. The four partners who own it have already put in all the money they can spare. A thousand neighbours would each be willing to give Tk 5,000, if they could be sure of a fair return. Yet the partners cannot ask a thousand people to become partners. A deed with a thousand names would be impossible to write, and in many countries every partner is personally responsible for the firm's debts. No sensible neighbour would accept that risk for Tk 5,000.

What arrangement would be attractive to the neighbours and safe for the partners? Each neighbour needs a small piece of the business, a paper to prove it, a claim on profit, and a promise that the loss can never be more than the money paid. The partners need a way to keep control of the shop while they raise the money. Merchants found an answer long ago, and it is the subject of this chapter. The money raised becomes the capital of a company, and each small piece of ownership is called a share.

We shall proceed in three steps. First, we see what the money raised is called and how its size is limited. Second, we meet the two chief kinds of share. Third, we record an issue of shares in the books, from the first application to the last entry. Chapter 18 described a business with a few owners who know each other. Here the owners may be many, and the records change to suit them.

* * *

19.1 SHARE CAPITAL

Ten families in a village decide to dig a well. The well will cost Tk 100,000, so each family pays Tk 10,000 and receives a written note of its part. Nobody owns the well alone, and nobody's part is larger than another's. A company works in a similar way, on a larger scale. Its owners are called shareholders. A shareholder is a person who owns one or more shares, and a share is one of the equal parts into which the ownership of a company is divided.

The accountant needs a name for the money that the shareholders pay in. **Share capital** is the money a company receives from shareholders in exchange for shares. If a shareholder pays Tk 1,000 for 100 shares, then Tk 1,000 enters the share capital. In a partnership, each partner has a separate capital account, as Chapter 18 showed. A company keeps one share capital account for all its shareholders together. On the balance sheet it appears among the owners' equity. In the United States, share capital is often called capital stock.

December. Bangladesh's government fiscal year is different: it runs from 1 July to 30 June.

The case file shows an important point. A government and a business in the same country can use different twelve-month periods, and neither one is wrong. Each chooses the period that suits its own work. Many students find this confusing, because they expect one year for everyone. The remedy is to ask of every report the same question: from which date to which date does it run?

Why would a business choose a year that is not the calendar year? One common reason is the pattern of trade. A seller of school uniforms might sell most items in the weeks before term begins. If the year ended in the middle of that rush, the shelves would be in constant movement and inventory would be hard to count. Ending the year when trade is quiet gives a calmer count. Rules differ from country to country, and tax law or company law can limit the choice, so a business must check the rules of its own country.

A fiscal year answers the question "How did we do?" only once a year. A lender who must decide this week whether to lend more money cannot wait. An owner who sees losses building would like to know in the third month, and not after the twelfth. For this reason businesses also prepare reports for shorter slices of time.

An **interim period** is a reporting period shorter than a year, such as a quarter. A quarter is three months; four quarters make a year.¹ Months, quarters and half-years are all interim periods.

Case file, illustrative. Meghna and Sons prepared a report for the interim period of the quarter ended 30 September 2026. Net sales in that quarter were Tk 3,900,000. The quarter began on 1 July 2026 and is the third quarter of the shop's calendar year.

The table below sets four periods side by side. Read each row from left to right: the name, the dates, and the length. The dates for Meghna and Sons come from the case file; the other two come from the facts you have already read.

FOUR REPORTING PERIODS COMPARED

PERIOD	DATES	LENGTH
Meghna and Sons, calendar year	1 January to 31 December	12 months
Meghna and Sons, interim period	1 July to 30 September 2026	3 months
Bangladesh government fiscal year	1 July to 30 June	12 months
United States federal fiscal year	1 October to 30 September	12 months

¹ A quarter may begin in any month. For a business with a calendar year, the quarters end on 31 March, 30 June, 30 September and 31 December.

Two bodies in one country may use two different twelve-month periods.

Some businesses end their year when trade is quiet, because counting inventory is then easier.

A year gives news only once, so lenders and owners also want reports for shorter slices.

Interim period. A reporting period shorter than a year, such as a quarter.

An interim period is a reporting period shorter than a year, such as a quarter.

Period. Word in use. In daily life, a period is any stretch of time, such as a school lesson. In accounting, a period is the span of time that one set of reports covers, such as a month, quarter or year.

The case file records one interim period of three months.

Different bodies measure different periods, and a table lays the dates side by side.

The case file gives a proposed company a ceiling of Tk 10,000,000 and a first issue to its partners.

Shares of face value Tk 10 given for net assets of Tk 4,100,000 are recorded as Tk 4,100,000 of share capital.

The entry is shown in short form; the full entry debits each asset and credits each liability.

Different readers want different things from the same figures: owners, managers, lenders and employees.

Authorised capital is a ceiling and issued capital is what has been sold; the two are not interchangeable.

Three terms carry forward: share capital, authorised capital and issued capital.

Investors differ: one wants steady income and another wants a voice and a chance of higher reward.

Case file, illustrative. Meghna and Sons Limited is a proposal to turn the business into a company. Its authorised capital is Tk 10,000,000, in 1,000,000 shares of face value Tk 10. The first issue is 410,000 shares to the partners, in exchange for net assets of Tk 4,100,000. Net assets are the assets of the business less its liabilities. The partners' own capital accounts, which Chapter 18 described, are replaced by the shares they receive.

Now we record the first issue. The partners hand over net assets, and the company gives them shares in return. The company debits the net assets it receives. Debit means the left-hand side of an account. It credits Share Capital with the face value of the shares given. Here 410,000 shares × Tk 10 = Tk 4,100,000, which equals the value of the net assets, so no further entry is needed. The table shows the entry.

JOURNAL ENTRY FOR THE FIRST ISSUE OF SHARES

ACCOUNT	DEBIT (TK)	CREDIT (TK)
Net assets taken over	4,100,000	
Share Capital		4,100,000
Shares issued to the partners in exchange for net assets		

This entry is shown in short form. In the full entry, every asset is debited by its own amount and every liability is credited by its own amount, and the difference is Tk 4,100,000. After this entry, the issued capital is Tk 4,100,000. The capital not yet issued is Tk 10,000,000 – Tk 4,100,000 = Tk 5,900,000.

Each reader looks at share capital for a different reason. A shareholder asks what part of the company he or she owns. A manager asks how much more may be raised without changing the founding documents. A bank asks how much owners' money stands behind its loan. An employee, a supplier or a customer asks mainly whether the company is stable.

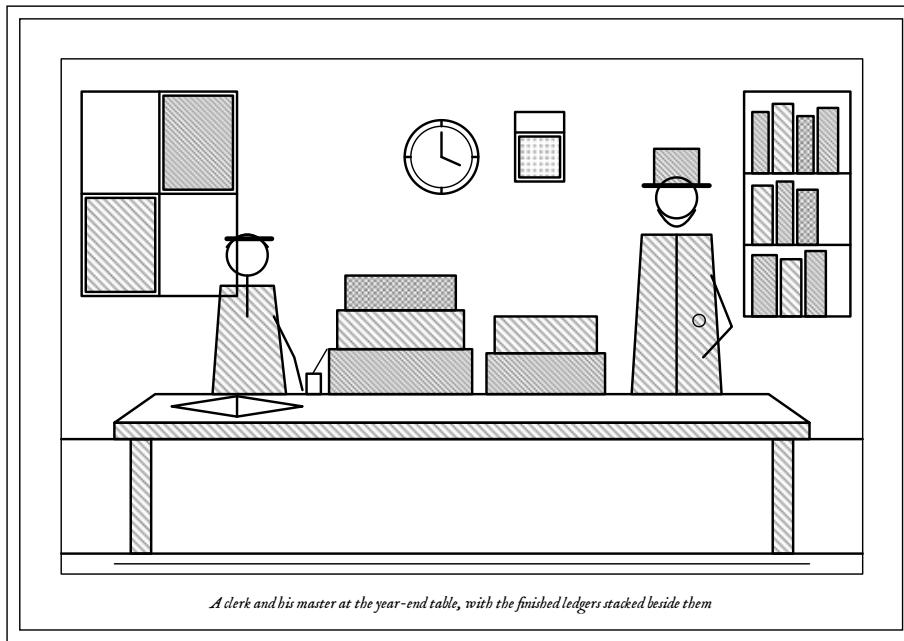
Many students find these two terms confusing, because both are amounts of capital in the founding documents' language. The test is simple. If shareholders have paid for it, it is issued. If the company has only permission to sell it, it is authorised and not yet issued.

Before moving on. Share capital is the money a company receives from shareholders in exchange for shares. Authorised capital is the ceiling set by the founding documents. Issued capital is the part of the ceiling that has been sold, and it is never larger than the ceiling.

* * *

19.2 TYPES OF SHARE

Two neighbours each have Tk 100,000 to invest in the bookshop. Anwar is retired and wants a steady, predictable income. Babita is young and is willing to wait for a larger reward, and she wishes to have a say in how the shop is run. One kind of share cannot suit both of them equally. For this reason, companies create more than



CHAPTER XX

The Whole Cycle and Reporting Periods

Which twelve months are we reporting, and which costs belong to them?



This chapter completes your first pass through the accounting cycle for a trading business. It explains why a business reports for a chosen year and for shorter periods inside it, and why some costs wait in inventory while others reach the income statement at once. It then links every report you have met into one connected whole.

EXAMPLE 2: PAYING A DIVIDEND TO THE TWO KINDS OF SHARE

PART	WORKING
Given	Preference capital Tk 1,000,000 at a fixed rate of 8% a year. Ordinary capital Tk 2,000,000. The company decides to pay Tk 400,000 as dividends for the year.
Find	The dividend to the preference shareholders, the dividend to the ordinary shareholders, and the amount for each ordinary share.
Formula	In words: preference dividend = preference capital $\times$ rate; ordinary dividend = total dividend - preference dividend. In symbols: $P = C \times r$ and $O = T - P$, where C is preference capital, r is the rate, and T is the total dividend.
Step 1	$P = 1,000,000 \times 8\% = \text{Tk } 80,000.$
Step 2	$O = 400,000 - 80,000 = \text{Tk } 320,000.$
Step 3	There are $2,000,000 \div 10 = 200,000$ ordinary shares, so each receives $320,000 \div 200,000 = \text{Tk } 1.60.$
Answer	The preference shareholders receive Tk 80,000. The ordinary shareholders receive Tk 320,000, which is Tk 1.60 for each share.
Check	$80,000 + 320,000 = \text{Tk } 400,000$, which equals the total dividend.

A dividend is paid out of profit, so it is a division of the company's profit and not an expense like rent.

The case file names no preference shares, so every share in the case has the same rights.

Rights vary by country and by company; the founding documents of each company state its own rules.

Two kinds of share, ordinary and preference, trade vote and reward against fixed income and priority.

A company that wants outsiders' money must invite applications, choose among them and then deliver the shares.

A common slip is to treat a dividend as an expense. It is a division of profit among the owners, as the appropriation account of Chapter 18 was. Another slip is to pay the ordinary shareholders first. The preference dividend always comes first, as the table showed.

Case file, illustrative. The proposal for Meghna and Sons Limited states one face value, Tk 10, for all its shares. The case file names no preference shares, so this chapter treats every share in the case as an ordinary share.

The rights described here are the common pattern. The exact rights of each kind of share depend on the law of the country and on the founding documents of each company. Some preference shares carry a vote in certain circumstances, and some fixed dividends may be carried forward when profit is too small to pay them. Read the documents of the company concerned.

Before moving on. An ordinary share carries a vote and a claim on profit that remains after other claims are paid. A preference share carries a fixed dividend, paid before the ordinary dividend, usually without a vote. The right choice for an investor depends on whether safety or reward matters more.

* * *

19.3 ISSUING SHARES

Suppose a company offers shares to the public. How does it know who wants them, how much money has arrived, and who has actually become a shareholder? The process has two formal steps, and each step has its own name and its own entry in the books.

19.2 A company issues 50,000 shares of face value Tk 10 at Tk 16. Applicants pay Tk 8 a share with the application and Tk 8 a share on allotment. Find the share capital and the share premium, and journalise the application and the allotment.

19.3 A company has authorised capital of Tk 2,000,000 in shares of face value Tk 5. It has issued 120,000 shares. Find the number of shares authorised, the issued capital, and the number of shares still unissued.

19.4 Mira and Naveed, former partners with capital of Tk 450,000 and Tk 300,000, form a company. They receive shares of face value Tk 10, equal in value to their capital, in exchange for net assets of Tk 750,000. Find the number of shares each receives, and journalise the issue.

Three slips are common: credit capital with the whole price, forget the premium, or mix the two steps.

Application comes first, then allotment, then premium: this is the order of the technique.

The case file offers 200,000 shares at Tk 25, so the same method is now applied to a realistic issue.

The case file records the public issue, the two money stages, and the totals credited.

Example 4 works the public issue in the standard order, with the journal entries at the end.

Three slips are common. The first is to credit Share Capital with the whole issue price, which hides the premium. The second is to record the issue before the allotment, so that Share Capital appears before anybody owns a share. The third is to credit the application money to Share Capital directly, which skips the temporary account.

Before moving on. A share application is a request to buy shares, sent with money. Allotment is the company's decision to give shares to applicants. Share premium is the amount paid above face value, and it is credited to its own account. The entries follow the order of events: application, allotment, premium.

* * *

WORKED CASE: THE PUBLIC ISSUE OF MEGHNA AND SONS LIMITED

Example 3 used small numbers. The case file now gives a realistic issue. Read the figure below from left to right. The bar is the issue price of Tk 25 for one share. The bar is the issue price of Tk 25 for one share. The hatched part is the face value of Tk 10, and the dotted part is the premium of Tk 15. The words under the bar name the money asked at each step.

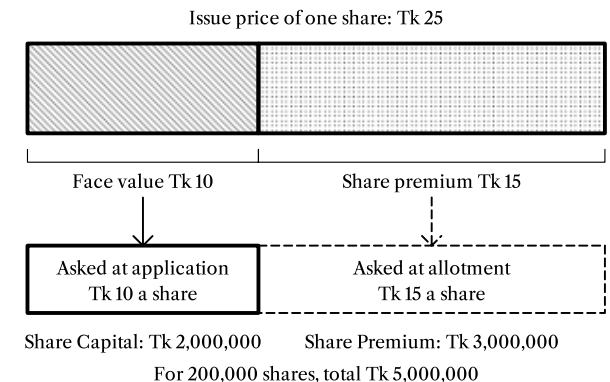


FIG. 19.1. The issue price of one share, split into face value and share premium

Case file, illustrative. Meghna and Sons Limited makes a public issue of 200,000 shares at Tk 25, which is a face value of Tk 10 and a premium of Tk 15. Application money is Tk 10 a share, which is Tk 2,000,000. Allotment money is Tk 15 a share, which is Tk 3,000,000. The total is Tk 5,000,000. Share capital credited is Tk 2,000,000, and share premium credited is Tk 3,000,000.

Example 4 works the case in full. Many students find this example confusing, because the application money equals the face value and the allotment money equals the premium. This is a feature of the case, and it is not a rule: other issues divide the money in other ways, as question 19.2 shows.

What is accountancy, and how are the books of a business kept? This treatise answers from the beginning. It sets out the nature and purpose of accounting, then follows the work through recording transactions, classifying them in ledgers, summarising them in a trial balance, and reporting the result in final accounts. Each chapter opens with a puzzle, closes with review questions, and needs no earlier knowledge.

Depreciation. The spreading of a plant asset's cost over the years it is expected to be useful. It is a method of allocation, not a measure of the fall in market price and not a payment of cash. For example, a Tk 10,000 bicycle may carry a charge of Tk 2,000 a year. (*Chapter XVI*)

Dishonoured cheque. A cheque that the bank refuses to pay, usually because the writer's account lacks funds. The bank takes back any amount it credited, so the business subtracts it on the book side and restores the customer's account receivable. For example, a customer's cheque for Tk 800 returned unpaid. (*Chapter XII*)

Double entry. The rule that every transaction is recorded twice, once as a debit and once as a credit of equal amount. It keeps the accounting equation in balance and exposes many mistakes, though equal totals do not prove that every entry is right. (*Chapter IV*)

Drawings. Money or goods that the owner withdraws from the business for personal use. Drawings lower the owner's equity, but they are not an expense, because they do not help to earn revenue. (*Chapter III*)

Ending inventory. The cost of goods still on hand at the end of the period. It appears as an asset in the balance sheet and becomes the opening inventory of the next period. Cost of goods sold plus ending inventory equals the cost of goods available. (*Chapter XIV*)

Error of commission. A mistake in which an entry is made in the right class of account but the wrong account, or with a wrong amount. Example: electricity of Tk 700 debited to Telephone Expense, when both are expenses. (*Chapter X*)

Error of omission. A mistake in which a transaction is left out of the books completely. Neither side of the entry is made, so the trial balance still agrees. Example: a credit sale of Tk 800 that is never entered. (*Chapter X*)

Error of principle. A mistake in which an entry is made in the wrong class of account, such as an expense recorded as an asset. Example: Tk 40,000 spent on a delivery van and debited to Repairs Expense. (*Chapter X*)

Expense. A cost used up in earning revenue, such as rent or wages. For example, rent paid for a month in which a workshop has used the room is an expense; buying shelving is not. (*Chapter III*)

Financial accounting. Accounting that prepares reports for outsiders, such as lenders, owners and tax authorities. Its reports follow written rules, so that outsiders can compare one business with another. A bank's yearly report on a borrower is an example. (*Chapter II*)

Financial statement. A formal report of a business's results or position, prepared at the end of a period. A period may be a month, a quarter or a year. The income statement, which shows profit, and the balance sheet are examples taught in a later chapter. (*Chapter II*)

Finished goods. Completed products held for sale. They are closest to being turned into cash. Bound volumes ready for collec-

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Journal. The book of first entry, in which transactions are recorded in date order with debits and credits. Each entry shows the date, the debited account, the credited account, the amounts and a narration. A purchase on credit is written here before anywhere else. (*Chapter V*)

Last-in first-out. A method that treats the newest costs as sold first, leaving the oldest costs in inventory. Often written LIFO. When prices rise it gives the highest cost of goods sold and the lowest gross profit. International Accounting Standard 2 bars it. (*Chapter XIV*)

Ledger. The book containing all of a business's accounts, where the effects of journal entries are collected. Each account has its own page, so all the Cash entries or all the Rent entries sit together and can be balanced. (*Chapter V*)

Liability. An amount that a business owes to others, such as a bank loan or an unpaid supplier bill. For example, Tk 200,000 owed to a publisher for books is a liability until it is paid. (*Chapter III*)

Liquidation. The process of selling a business's assets, paying its debts and sharing what remains among the owners. Outsiders are paid first. Each partner then receives capital less his or her share of any loss, so the partners are the last to be paid. (*Chapter XVIII*)

Management accounting. Accounting that prepares reports inside the business to help managers plan and control. No outside rule governs these reports, and the manager designs them to answer a question, such as which books sell fastest each week. (*Chapter II*)

Manufacturing business. A business that makes goods from materials and then sells them. It holds materials, goods half finished and goods ready to sell. A bakery that turns flour and fuel into bread is an example. (*Chapter I*)

Matching principle. The rule that expenses are recorded in the same period as the revenue they helped to earn. For example, insurance for October to December is placed beside the revenue of those months. (*Chapter VII*)

Merchandising business. A business that buys finished goods and sells them again at a higher price. It does not change the goods. A bookshop that buys books from publishers and sells the same books to readers is an example. (*Chapter I*)

Narration. A short note under a journal entry explaining the transaction. A good narration names the event, not the accounts. For example, "Bought shelving for cash" tells a later reader why Shelving was debited and Cash credited. (*Chapter V*)

Net income. The amount by which total revenue exceeds total expenses for a period. It is found by subtracting total expenses from total revenue, and it increases owner's equity. Example: revenue of Tk 6,000 less expenses of Tk 3,400 gives net income of Tk 2,600. (*Chapter VIII*)

Net loss. The amount by which total expenses exceed total revenue for a period. It is found by subtracting total revenue from total expenses, and it reduces owner's equity. Example: expenses of Tk 6,000 less revenue of Tk 3,400 gives net loss of Tk 2,600. (*Chapter VIII*)

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Plant asset. A long-lived physical asset used in running a business, such as a machine or a building. It is owned, used in the work of the business, and lasts more than one period. A book-cutting machine in a bookshop is a plant asset. (*Chapter XV*)

Posting. Copying the debits and credits from the journal to the right accounts in the ledger. Posting changes no amount. Each journal line is copied once, and the folio columns record that it has been done. (*Chapter V*)

Preference share. A share that gets a fixed dividend before ordinary shareholders, usually without a vote. For example, a holder of preference shares at 8% receives 8% of their face value before any ordinary dividend, if the company has enough profit. (*Chapter XIX*)

Prepaid expense. A cost paid in advance that is an asset until it is used up, after which it becomes an expense. For example, a year of insurance paid on 1 October is mostly still an asset on 31 December. (*Chapter VII*)

Product cost. A cost of buying or making goods held for sale. It waits in inventory until the goods are sold and then moves to cost of goods sold. The price paid for books and freight in are examples. (*Chapter XX*)

Profit-sharing ratio. The fixed proportion in which partners divide profit and loss. A ratio of 5:3:2 cuts the amount into ten equal parts, giving five to the first partner, three to the second and two to the third. Unless the deed says otherwise, it also divides losses. (*Chapter XVIII*)

Purchases journal. A special journal that records only purchases of goods on credit. Each line comes from an invoice received from a supplier and shows the date, supplier and amount. Its total is posted once to the main ledger. Cash purchases and purchases of equipment are not recorded in it. (*Chapter XI*)

Raw materials. Materials a manufacturer holds to turn into products. They have not yet been changed by the factory's work. In a binding workshop, reels of paper, board, thread and glue are raw materials. (*Chapter XIII*)

Realisation. The sale of assets, in liquidation, and the gain or loss on that sale. The loss is the book amount less the amount received. For example, an asset with a book amount of Tk 100 sold for Tk 80 gives a realisation loss of Tk 20. (*Chapter XVIII*)

Receipts and payments account. A summary of all cash received and paid in a period, with opening and closing balances. It has the form of a cash book. It shows cash only, so it ignores amounts owed, amounts paid early and the difference between lasting assets and everyday costs. (*Chapter XVII*)

Residual value. The amount a business expects to receive when it disposes of an asset at the end of its useful life. It may be zero. Only the cost less the residual value is spread over the years. For example, a machine costing Tk 330,000 with a residual value of Tk 30,000 has Tk 300,000 to spread. (*Chapter XVI*)

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early, so the cash received in a year is adjusted to find the income that belongs to the year. (*Chapter XVII*)

Subsidiary ledger. A ledger holding the individual accounts of customers or suppliers, supporting a main ledger account. A business may keep one for customers and another for suppliers. Each line of a special journal is posted to the matching account in it. (*Chapter XI*)

Suspense account. A temporary account that holds a difference until its cause is found. It takes the side that makes the two column totals agree, and it must be cleared by a correcting entry before reports are prepared. (*Chapter VI*)

T-account. A simple account drawn as a letter T, with the name of the account across the top. The left side holds the debit entries and the right side holds the credit entries. It is a quick way to show an account's entries and balance. (*Chapter IV*)

Temporary account. An account for revenue, expense or drawings that measures one period only. It is closed to zero at the end of the period, so the next period starts afresh. Example: the Revenue account of one year. (*Chapter IX*)

Transaction. An event that changes the money position of a business and must be recorded. Buying shelving for cash is a transaction; a customer saying she may buy next month is not, because no money has yet changed. (*Chapter III*)

Transposition error. A mistake in which two digits of a number are written in the wrong order, such as 53 for 35. Posted on one side only, it makes the trial balance totals differ by a multiple of 9, here 18. (*Chapter VI*)

Trial balance. A list of all ledger account balances, with debits and credits totalled, to test that they are equal. It is taken at one date. If the two totals agree, the test is passed; if they differ, a slip must be found. (*Chapter VI*)

Unadjusted trial balance. A trial balance taken before the period-end adjustments have been recorded. The January list of Meghna Binding Service is unadjusted, because no adjustment has yet been made to the accounts. (*Chapter VI*)

Unearned revenue. Money received before the work is done, held as a liability until the work is performed. For example, a school pays Tk 24,000 in advance; the part not yet worked remains a liability. (*Chapter VII*)

Units-of-production method. A depreciation method that charges an amount based on how much the asset is used in the period. A charge for each unit of use is found first and then multiplied by the units used. For example, Tk 0.60 a book times 90,000 books gives Tk 54,000. (*Chapter XVI*)

Useful life. The period, or the amount of use, over which an asset is expected to serve the business. It may be given in years, kilometres or units made. It is an estimate and may be shorter than the time the asset would physically last. (*Chapter XVI*)

Weighted average cost. A method that gives every unit the aver-

19.2 Share capital = $50,000 \times 10 = \text{Tk } 500,000$. Share premium = $50,000 \times 6 = \text{Tk } 300,000$. Application: debit Bank 400,000; credit Share Application 400,000. Allotment: debit Share Application 400,000 and Bank 400,000; credit Share Capital 500,000 and Share Premium 300,000. Check: $500,000 + 300,000 = 800,000 = 400,000 + 400,000$.

19.3 Shares authorised = $2,000,000 \div 5 = 400,000$. Issued capital = $120,000 \times 5 = \text{Tk } 600,000$. Shares still unissued = $400,000 - 120,000 = 280,000$, worth Tk 1,400,000.

19.4 Total shares = $750,000 \div 10 = 75,000$. Mira receives $450,000 \div 10 = 45,000$ shares and Naveed receives $300,000 \div 10 = 30,000$ shares. Entry: debit Net assets taken over 750,000; credit Share Capital 750,000.

CHAPTER XX · THE WHOLE CYCLE AND REPORTING PERIODS

20.p1 Freight in (Tk 40,000) is a product cost, because it is part of bringing goods to the shop. Rent (Tk 180,000) is a period cost, because it belongs to the time the building is used.

20.p2 Cash, an asset, falls by Tk 5,000, and accounts payable, a liability, falls by Tk 5,000. Both sides fall by the same amount, so the balance sheet still balances.

20.t1 Two changes: every report would cover 1 July to 30 June, and the quarters would end on 30 September, 31 December, 31 March and 30 June, so the quarter ended 30 September 2026 would become a first quarter. Moving from a December year-end also needs a bridging period of six months, 1 January to 30 June. The accounts, the debit and credit rules and the split between product costs and period costs stay the same.

20.t2 A year ending when trade is quiet, such as soon after the term-start rush, is a good choice. Inventory is low and easy to count then, and the whole busy season falls inside one year.

20.1 Product costs: books bought and freight in. Period costs: shop rent, wages of the shop assistants and delivery to a customer (freight out).

20.2 Gross profit is $\text{Tk } 600,000 - \text{Tk } 380,000 = \text{Tk } 220,000$. Profit is $\text{Tk } 220,000 - \text{Tk } 150,000 = \text{Tk } 70,000$.

20.3 The order is: journal, posting to the ledger, trial balance, adjusting entries, adjusted trial balance, the three statements, closing entries, post-closing trial balance. The adjusting entries and the closing entries change account balances.

20.4 An account payable is an amount the business owes, and it appears among the liabilities. An account receivable is an amount owed to the business, and it appears among the assets.

20.5 The second quarter runs from 1 July to 30 September. The second quarter is an interim period; the year is the fiscal year, and the cycle is a set of steps and not a period of time.

15.2 (a) revenue; (b) capital; (c) capital; (d) revenue; (e) capital, because it is done before first use.

15.3 Profit is Tk 12,000 too high and the plant asset is Tk 12,000 too high on the balance sheet.

15.4 Acquisition cost = 600,000 + 30,000 + 18,000 = Tk 648,000. Revenue expenditure for the year = 22,000 + 9,000 = Tk 31,000.

CHAPTER XVI · DEPRECIATION

16.p1 (Tk 120,000 – Tk 20,000) ÷ 4 = Tk 25,000 a year.

16.p2 Year 1: 25% × Tk 80,000 = Tk 20,000, leaving Tk 60,000. Year 2: 25% × Tk 60,000 = Tk 15,000, leaving Tk 45,000.

16.t1 The yearly charge falls from Tk 60,000 to (Tk 330,000 – Tk 30,000) ÷ 8 = Tk 37,500, so each year's profit rises by Tk 22,500. The total charged over the whole life is unchanged at Tk 300,000, because only the timing changes.

16.t2 Depreciation is an allocation of cost, not a transfer of cash. The cash was paid on the day of purchase, and the yearly entry moves no cash. Saving for a new machine is a separate decision about cash.

16.1 Charge: (Tk 50,000 – Tk 10,000) ÷ 4 = Tk 10,000 a year. After two years the accumulated depreciation is Tk 20,000, so the book value is Tk 30,000.

16.2 Straight-line: (Tk 275,000 – Tk 25,000) ÷ 8 = Tk 31,250 a year; for 3 months, Tk 31,250 × 3 ÷ 12 = Tk 7,812.50. Units-of-production: Tk 250,000 ÷ 400,000 = Tk 0.625 a unit; 36,000 × Tk 0.625 = Tk 22,500.

16.3 Year 1: Tk 30,000, remaining Tk 120,000. Year 2: Tk 24,000, remaining Tk 96,000. Year 3: Tk 19,200, remaining Tk 76,800. Gain = Tk 80,000 – Tk 76,800 = Tk 3,200.

16.4 Yearly charge: (Tk 240,000 – Tk 40,000) ÷ 5 = Tk 40,000. Entry for 2026: debit depreciation expense Tk 40,000, credit accumulated depreciation Tk 40,000. After three years the accumulated depreciation is Tk 120,000 and the book value is Tk 120,000, so the sale gives a loss of Tk 20,000. Sale entry: debit cash Tk 100,000, debit accumulated depreciation Tk 120,000, debit loss on sale Tk 20,000; credit press at cost Tk 240,000.

CHAPTER XVII · ACCOUNTS OF CLUBS AND SOLE TRADERS

17.p1 No. The money is for a later year, so it belongs to the period that it pays for. It is received early and is shown as a liability, fees paid in advance, until that period begins.

17.t1 Cash can rise because of money not charged against the year, such as a loan or the sale of an asset, or because the club bought no lasting assets. A deficit arises when expenditure on the accrual basis, including amounts owed but unpaid, exceeds income that belongs to the year, such as when many fees are still owed.

17.t2 Profit = 46,000 – 50,000 + 9,000 – 0 = Tk 5,000. The equity fell by Tk 4,000 only because she withdrew Tk 9,000, so the business was profitable.

2.2 The weekly book-sales report is management accounting, because it is made for the manager inside the business and follows no outside rule. It is not enough for the bank, which is an external user and needs a formal report that follows written standards.

2.3 The business entity rule is broken, because family costs are mixed with the business. Anyone who reads her records, such as a lender or a tax authority, is misled about how well the business is doing.

2.4 The going concern assumption no longer holds, because the shop will not use its shelves for years. The records must then show what the shelves would fetch when sold, instead of the price paid under historical cost.

CHAPTER III · THE ACCOUNTING EQUATION

3.p1 Owner's equity = 80,000 – 25,000 = Tk 55,000. Check: liabilities plus equity, 25,000 + 55,000 = 80,000, which equals the assets.

3.p2 Owner's equity = 40,000 + 15,000 – 9,000 – 2,000 = Tk 44,000.

3.t1 Buying shelving for cash: one asset (shelving) rises by Tk 120,000 and another (cash) falls by Tk 120,000. Total assets are unchanged, so liabilities and owner's equity need no change and the equation stays equal.

3.t2 She is not right. Paying the publisher lowered cash by Tk 80,000 and lowered the account payable by Tk 80,000. Owner's equity did not change, so no expense arose.

3.1 (a) asset; (b) liability; (c) asset; (d) liability; (e) owner's equity.

3.2 Liabilities = 240,000 – 150,000 = Tk 90,000. Check: 90,000 + 150,000 = 240,000.

3.3 Owner's equity at the end = 300,000 + 410,000 – 365,000 – 50,000 = Tk 295,000.

3.4 Equity = 1,250 – 480 = 770 thousand, which is Tk 770,000.

3.5 Assets Tk 252,000 (cash 162,000 and equipment 90,000); liabilities Tk 50,000 (90,000 – 40,000); owner's equity Tk 202,000 (200,000 + 30,000 – 18,000 – 10,000). Check: 50,000 + 202,000 = 252,000.

CHAPTER IV · ACCOUNTS, DEBITS AND CREDITS

4.p1 Debit. Rent is an expense, and expenses increase by debit, so its balance usually falls on the debit side.

4.p2 Shelving is an asset that increases, so it is debited. Cash is an asset that decreases, so it is credited.

4.t1 Cash is an asset, and assets stand on the left of the equation A + D + E = L + C + R. Increases of left-side items are recorded on the left, which is the debit side.

4.t2 Debit Drawings and credit Cash, both for Tk 10,000. Drawings is its own type because the owner's personal use is not a cost of earning revenue, so it is not an expense.

4.1 Debit Cash 2,000 and credit Owner's Capital 2,000; debit Shelf 500 and credit Cash 500; debit Cash 300 and credit Sales Revenue

426,400. The difference is Tk 10,000, which suggests a posting is missing or wrong.

11.4 Total: Tk 36,000 + Tk 54,500 + Tk 29,500 = Tk 120,000. Debit Purchases Tk 120,000 and credit the accounts payable control account Tk 120,000. Closing balance: Tk 200,000 + Tk 120,000 = Tk 320,000.

CHAPTER XII · THE CASH BOOK AND BANK RECONCILIATION

12.p1 A difference of Tk 1,000 means an item is missing, wrongly placed or wrongly added. Tick the items again and check the arithmetic; make no entry until the two adjusted balances agree.

12.t1 Entries are needed for the service charge (Tk 1,500), the dishonoured cheque (Tk 8,000), the interest (Tk 2,500) and the school payment (Tk 40,000), because the bank recorded them first. No entry is needed for the deposit in transit or the two cheques, because the business recorded them first and the bank will catch up.

12.t2 The business should contact the payee to learn why the cheque has not been presented, because rules on old cheques differ between banks and countries. Until it clears, the cheque stays on the reconciliation as an outstanding cheque and is subtracted on the bank side each month.

12.1 Bank side: $3,400 + 700 - 300 = 3,800$. Book side: $3,000 + 900 - 100 = 3,800$. The adjusted balance is Tk 3,800 on both sides.

12.2 Bank side: $91,250 + 12,000 + 6,500 - 9,800 - 4,150 = 95,800$. Book side: $95,050 + 5,000 - 750 - 3,500 = 95,800$. The adjusted balance is Tk 95,800.

12.3 The cash book is higher by Tk 1,750 before adjusting. Bank side: $18,400 + 5,000 - 3,200 - 1,100 = 19,100$. Book side: $20,150 - 250 - 800 = 19,100$. The adjusted balance is Tk 19,100.

12.4 Bank side: $47,000 + 4,000 - 9,000 = 42,000$. Book side: $40,000 + 3,000 + 500 - 1,500 = 42,000$. Entries: debit Bank, credit Account Receivable Tk 3,000; debit Bank, credit Interest Revenue Tk 500; debit Bank Charges, credit Bank Tk 1,500. Bank on the balance sheet is Tk 42,000.

CHAPTER XIII · INVENTORY: MEANING AND MEASUREMENT

13.p1 It will be hidden. A periodic system calculates cost of goods sold by subtraction, so the cost of the 3 stolen books is included in cost of goods sold with no separate sign.

13.t1 The extra holding cost is $380,000 \times 10\% = \text{Tk } 38,000$. The shop may also face a write-down if the titles can only be sold for less than their cost.

13.t2 Two possible costs: lost sales when demand rises or a delivery is late, and the loss of customers who do not return. More frequent orders may also raise transport costs and remove bulk-buying discounts.

13.1 (a) raw materials; (b) work in process; (c) not inventory, because it is not held for sale; (d) finished goods; (e) not inventory, because the shop does not own it

6.4 The debits fall short by Tk 7,200, so a suspense account with a debit balance of Tk 7,200 is added in the Debit column.

6.5 Difference = $81,000 - 18,000 = \text{Tk } 63,000$, with debits larger ($63,000 \div 9 = 7,000$). The credits are short, so the suspense balance of Tk 63,000 goes on the credit side. Correcting entry: debit Suspense 63,000, credit Rent 63,000. Narration: Corrected rent posted as 81,000 instead of 18,000, and cleared the suspense account.

CHAPTER VII · ADJUSTING THE ACCOUNTS

7.p1 Monthly rent = $24,000 \div 12 = \text{Tk } 2,000$. November and December are two months, so Tk 4,000 belongs to the year. The matching principle (with the accrual basis) decides it: the cost is placed in the period in which it is used.

7.p2 The Tk 6,500 owed to workers is an accrued expense, a liability. The Tk 11,000 of unbilled work is accrued revenue, an asset.

7.t1 The records would show a profit of Tk 280,000 instead of Tk 184,000, which is Tk 96,000 too high. The owner might withdraw too much, and a lender or tax authority would judge the business on a wrong figure. Assets would also be overstated and liabilities understated.

7.t2 The Tk 36,000 was paid in cash on 1 October. The adjustment on 31 December only moves the used part, Tk 9,000, from the asset Prepaid Insurance to the expense Insurance Expense. No money moves on that date.

7.1 $6,000 \div 6 = \text{Tk } 1,000$ a month. March to June is 4 months, so the expense is Tk 4,000 and the prepaid expense is $6,000 - 4,000 = \text{Tk } 2,000$.

7.2 Supplies expense = $52,500 - 18,200 = \text{Tk } 34,300$. Entry: debit Supplies Expense 34,300; credit Supplies 34,300.

7.3 $18,000 \div 6 = \text{Tk } 3,000$ a month. November and December are 2 months, so revenue earned is Tk 6,000 and unearned revenue is $18,000 - 6,000 = \text{Tk } 12,000$.

7.4 Depreciation = $240,000 \div 8 = \text{Tk } 30,000$: debit Depreciation Expense, credit Accumulated Depreciation. Wages: debit Wages 9,000, credit Wages Payable 9,000. Unbilled work: debit Accounts Receivable 15,000, credit Service Revenue 15,000. Expenses rise by 39,000 and revenue by 15,000, so profit falls by Tk 24,000.

CHAPTER VIII · THE WORKSHEET

8.p1 Tk 400, a debit: the debit of Tk 1,000 less the credit of Tk 600 leaves a debit of Tk 400.

8.p2 The debit side, because revenue (Tk 6,000) exceeds expenses (Tk 3,400), so the debit column is short by Tk 2,600 and the balancing figure makes the two totals equal.

8.t1 The credit total is $\text{Tk } 8,500 + \text{Tk } 2,600 = \text{Tk } 11,100$, so the difference is Tk 300 (debit larger). Wages Payable is Tk 300, so check first whether that balance was extended to the balance sheet credit column.